Draft version 22 July 2026
Typeset using LaTeX **twocolumn** style in AASTeX631

arXiv:2607.19293v1 [gr-qc] 21 Jul 2026

GWTC-5.0: Tests of General Relativity

The LIGO Scientific Collaboration, the Virgo Collaboration, and the KAGRA Collaboration

(Compiled: 22 July 2026)

ABSTRACT

The signals from the LIGO–Virgo–KAGRA network of gravitational-wave (GW) detectors allow us to perform sensitive tests of general relativity (GR) in the dynamical and strong-field regime of gravity. We present the results of seven tests of GR using the observed binary signals in the fifth GW Transient Catalog (GWTC-5.0), i.e., up to and including the second part of the fourth observing run (O4b). We restrict our analysis to the confident signals, henceforth called *events*, observed by at least two detectors that have estimated false alarm rates $\leq 10^{-3}$ yr^{-1}. These include 72 events from O4b and five events from the first part of the fourth observing run that are now analyzed due to their increased significance from updated search results, bringing the total number of events for tests of GR in the cumulative GWTC to 168. After subtracting the best-fit waveforms, we find the residuals are consistent with detector noise for all events considered. We also find no strong evidence for additional polarizations beyond those predicted by GR. We perform tests of GW generation, improving the constraints on deviations from the GR post-Newtonian coefficients by factors of 1.2 to 2.6. Finally, we find overall consistency of the remnants with GR using both time- and frequency-domain methods. For GW240621_195059, postmerger data are consistent with the dominant quadrupolar ($\ell = |m| = 2$) mode of a Kerr black hole and its first overtone, with spurious high-frequency content preventing a spectroscopic constraint of GR. In the frequency-domain ringdown analysis, the GR prediction lies in the tails of the combined results, possibly due to the limited catalog size. However, the combined results indicate improved consistency with GR over GWTC-4.0, owing to the contribution of GW250114 with a network matched-filter signal-to-noise ratio of 76.9. Overall, we find no evidence for physics beyond GR.

1. INTRODUCTION

Gravitational wave (GW) observations of compact binary coalescences (CBCs) have opened a new regime for testing gravity in the strong-field, highly dynamical sector. Since the first detection of a merger involving two black holes (BHs) (Abbott et al. 2016a), followed by observations of systems containing neutron stars (NSs) (Abbott et al. 2017a, 2021a), the population of detected GW signals (hereafter referred to as *events*) has grown rapidly, enabling increasingly precise tests of Einstein's theory of general relativity (GR). These observations of binary BH (BBH), binary NS (BNS), and NS–BH binary (NSBH) mergers probe gravitational dynamics in regimes that were previously inaccessible, complementing precision tests of gravity performed in the Solar System (Will 2014), with binary pulsars (Kramer et al. 2021), through observations of stars orbiting Sgr A* (Abuter et al. 2018; Do et al. 2019; Abd El Dayem et al. 2025), using BH shadow measurements (Akiyama et al. 2019, 2022a,b), and from cosmological observations (Ishak 2019; Ferreira 2019).

The full author list is given at the end.
For correspondence: LSC P&P Committee, for LVK Publications, via lvc.publications@ligo.org

Analyses of the GW signals detected during the first three observing runs of the Laser Interferometer Gravitational-Wave Observatory (LIGO) (Aasi et al. 2015) and Virgo observatory (Acernese et al. 2015) have already provided a wide range of tests of GR, including consistency checks of the waveform morphology, constraints on parameterized deviations from GR predictions, and studies of the properties of the remnant BH (Abbott et al. 2016b, 2017b,c, 2019a,b, 2020c, 2021b, 2025). To date, these tests have shown overall consistency with the predictions of GR, simultaneously improving constraints on a wide range of possible deviations from the theory.

The fourth observing run of the LIGO–Virgo–KAGRA Collaboration (LVK) has further expanded the number and diversity of observed compact binaries. The first part of this run already yielded a rich sample of binaries (Abac et al. 2025a, 2026a), including several events whose properties are particularly relevant for testing gravitational dynamics. These events were analyzed in detail in the GWTC-4.0 tests of GR catalog papers (Abac et al. 2026b,c,d). For example, the likely NSBH system GW230518_125908 provided strong constraints on deviations in the inspiral dynamics (Abac et al. 2026c), while GW231226_101520 offered improved measurements of the remnant BH properties and the post-merger signal (Abac et al. 2026d). The tests of GR on the two single-detector events GW230529_181500 (Abac et al. 2024) and GW230814_230901 (Abac et al. 2026e) are reported else-

where (Sänger et al. 2026; Abac et al. 2026e). Analyses of the first part of the fourth observing run (O4a) observations have strengthened the constraints obtained in earlier catalog studies for an expanded range of tests to probe any deviations from GR.

The second part of the fourth observing run (O4b), from 2024 April 10 to 2025 January 28, has continued to expand this sample and includes several events that provide especially informative probes of strong-field gravity. The asymmetric, high-spin binary merger GW241011_233834 is among the first detections allowing us to probe a region of the binary parameter space that has so far been sparsely populated by observations, providing a valuable opportunity to test gravitational dynamics in unequal-mass systems (Abac et al. 2025b). Similarly, the high signal-to-noise ratio (SNR) observation of BBH coalescence GW250114_082203 (henceforth shortened to GW250114; Abac et al. 2025c) has enabled particularly strong constraints on both post-Newtonian (PN) coefficients and quasi-normal mode (QNM) deviations (Abac et al. 2026f). Detailed analyses of this event have yielded stringent tests of the Kerr nature of BHs as well as Hawking's area theorem (Abac et al. 2025c, 2026f). We also include GW240925_005809, a loud event observed for which the original detector calibration was subsequently revised in the C01 recalibration (Abac et al. 2026g). Comparisons with astrophysical calibration measurements show that the recalibrated strain data are consistent with the expected detector response, while remaining calibration uncertainties can be incorporated through joint inference of calibration and source parameters. These studies indicate that any potential biases are adequately controlled, supporting the inclusion of this event in our analyses. These notable events illustrate the increasing scientific reach of GW observations as the detector network continues to become more sensitive.

In this work, we analyze the GW signals that constitute the fifth Gravitational-Wave Transient Catalog (GWTC-5.0; Abac et al. 2026h,i), combining the new detections from O4b with previously observed events. We restrict ourselves to events that pass the same selection criteria described in the previous tests of GR papers (Abbott et al. 2025; Abac et al. 2026b) of having an estimated false-alarm rate (FAR) $\leq 10^{-3}\mathrm{yr}^{-1}$ and having been observed by at least two detectors. Thus, of the O4b candidates, this paper includes 72 new events. Of the O4a candidates, this paper covers five additional events that only passed the FAR threshold in the reanalyses carried out with the PyCBC and GstLAL search pipelines using updated configurations (Abac et al. 2026i), and thus were not analyzed in the GWTC-4.0 tests of GR papers (Abac et al. 2026b,c,d). We also include the events from the previous observing runs: O1 (the first observing run, from 2015 September 12 to 2016 January 19; Abbott et al. 2016c), O2 (the second observing run, from 2016 November 30 to 2017 August 25; Abbott et al. 2019c), O3a (the first half of the third observing run, from 2019 April 1 to 2019 October 1; Abbott et al. 2021c, 2024), O3b (the second half of the third observing run, from 2019 November 1 to 2020 March 27; Abbott et al. 2023), and O4a (from 2023 May 24 to 2024 January 16; Abac et al. 2026a). Together, these observations provide the largest sample of compact binary mergers yet used to test gravitational dynamics.

Table 1 gives an overview of the testing GR pipelines and sections where they are described, as well as main results and highlights of the improvements with respect to GWTC-4.0. Many of the tests covered have additional, narrower selection criteria for choosing which events are relevant for them, based on the supported physical parameters; these are described in the respective sections.

While GWTC-4.0 tests of GR papers (Abac et al. 2026b,c,d) reported results from a broader range of pipelines, the present paper prioritizes tests that provide broad coverage of potential deviations from GR, have well understood systematics, and are expected to yield the most informative constraints with the current set of events. Several analyses reported in earlier catalogs are therefore not included in this paper due to technical reasons or to limit the scope of the paper, as detailed in the following.

The inspiral–merger–ringdown (IMR) consistency test (Hughes & Menou 2005; Ghosh et al. 2016, 2018) is not included because Narayan et al. (2023, 2026) showed that this test can exhibit spurious biases even for unlensed, quasicircular binary signals in certain portions of the parameter space. This is in addition to finding the expected biases when applying this test and other ones to lensed or eccentric binary signals using an unlensed, quasicircular GR baseline. The subdominant multipole amplitudes test (Puecher et al. 2022; Gupta et al. 2025) provides comparatively weak constraints for current data and can only be applied to a limited subset of events, with the best constraint so far coming from GW241011_233834, whose source was a asymmetric BBH system (Abac et al. 2025b). The Test Infrastructure for GEneral Relativity (TIGER; Li et al. 2012a; Agathos et al. 2014; Meidam et al. 2018; Roy et al. 2026) post-inspiral analysis is less straightforward to interpret than tests based on deviations in individual PN coefficients. The multiparameter PN coefficients test using principal component analysis (Saleem et al. 2022; Mahapatra et al. 2025) is excluded to simplify the scope of the paper, since it probes deviations in the inspiral dynamics that are partially redundant with those constrained by the other included parameterized PN deviation analyses. The spin-induced moment tests (Krishnendu et al. 2017, 2019; Mehta et al. 2023; Divyajyoti et al. 2024) only analyze a relatively small subset of the events and also obtained the most stringent constraints to date from GW241011_233834 (Abac et al. 2025b) due to the event's high SNR, large primary spin, and unequal mass ratio. The line-of-sight acceleration (Vijaykumar et al. 2023; Tiwari et al. 2026) analysis does not directly probe deviations from GR and only a few events satisfy its selection criteria. The test for dispersive propagation (Mirshekari et al. 2012; Baka et al. 2026) is currently investigating correlations between the binary parameters and the modifications in the dispersion relation, as well as refining its event selection criterion to select a subset of events that contribute most significantly to the constraints. The spacetime symmetry breaking test (Haegel et al. 2023) for anisotropic birefringent

propagation effects is not considered to simplify the scope of the paper. The pyRing (Carullo et al. 2019) ringdown analysis is currently undergoing development to improve its computational efficiency and scalability for large catalog analyses. Finally, the searches for post-ringdown echoes (Lo et al. 2019; Tsang et al. 2018; Miani et al. 2023; Uchikata et al. 2023) are discontinued as these analyses are not expected to provide significantly stronger constraints than what have been obtained already.

The rest of this paper is organized as follows. Section 2 discusses the events included, and details which tests were performed for each. Section 3 describes the general methods common between many of the tests. Section 4 describes the residual test. Section 5 describes the tests for possible non-GR polarizations. Section 6 describes the tests of modifications in the GW generation from CBC systems, specifically tests of parameterized deviations in individual PN coefficients with TIGER, using the IMRPhenomXPHM_SpinTaylor waveform model, and the Flexible Theory-Independent (FTI) test, using the SEOBNRv5HM_ROM waveform model. In Section 7, we discuss three ringdown tests of remnants of BBH mergers: the ringdown analysis (RD) considering the post-inspiral signal in the time domain (Section 7.1), pSEOBNR analyzing the entire signal in the frequency domain (Section 7.2), and the QNM rational filter (QNMRF) analysis applying a filter to determine the QNMs present in the post-inspiral signal (Section 7.3). Section 8 provides overall concluding remarks.

Table 1. Summary of testing GR pipelines and new results.

Test	Section	Quantities	Parameters	Main Result	Improvement
RT	4	p-value for residual signal	p-value	Consistent with uniform dist.	⋯
POL	5	Bayes factors between different polarization hypotheses	$\log_{10}\mathcal{B}^{X}_{\mathrm{T}}$	$-19.50^{+0.73}_{-0.73}$–$0.91^{+0.71}_{-0.71}$	−0.82–4.78
TIGER	6	PN deformation parameters	$\lvert\delta\hat{\varphi}_k\rvert$	$\lesssim 4.4\times10^{-4}$–0.8	1.2–2.2
FTI	6	PN deformation parameters	$\lvert\delta\hat{\varphi}_k\rvert$	$\lesssim 6.0\times10^{-4}$–0.9	1.7–2.6
RD	7.1	Amplitude of subdominant mode(s)	A_{221}	$4.68^{+7.84}_{-3.63}\times10^{-21}$	New
pSEOBNR	7.2	Fractional deviations in frequency and damping time	$\{\delta\hat{f}_{220},\delta\hat{\tau}_{220}\}$	$\{0.01^{+0.03}_{-0.03},0.09^{+0.13}_{-0.10}\}$	{1.95, 1.48}
QNMRF	7.3	Detection statistic for subdominant ringdown modes	$\mathcal{D}-\mathcal{D}_{1\%}$	221 : $\lesssim 2.46$	⋯

Note—This table summarizes the names of the tests performed, the corresponding sections, the parameters involved in the test, the new results, and the improvements with respect to the GWTC-4.0 analyses (Abac et al. 2026b,c,d). The analyses performed are: RT = residual test; POL = polarization tests; TIGER and FTI = tests of GW generation; RD, pSEOBNR and QNMRF = ringdown tests. The improvement is defined as $X_{\mathrm{GWTC\text{–}4.0}}/X_{\mathrm{GWTC\text{–}5.0}}$, where X denotes the width of the 90% credible interval for the parameters of each test. For RT, the ⋯ marker is used for the improvement to indicate that the result of overall consistency is maintained. For POL, the improvement is defined as the difference of the $\log_{10}$ Bayes factors (GWTC-4.0 minus GWTC-5.0), where a positive (negative) improvement favors (disfavors) GR. The main results and improvements for TIGER, FTI and pSEOBNR are obtained using the combined results from hierarchical analyses. The main result of RD is the amplitude of the subdominant ($\ell = m = 2, n = 1$) mode at the earliest analysis start time for which the amplitude is measured self-consistently, $t_{>} = 2t_{M_{\mathrm{f}}}$, for GW240621_195059. The label "New" refers to the tests that were not performed in the GWTC-4.0 analyses. Finally, the QNMRF parameter is the detection statistic for the subdominant 221 mode minus the threshold corresponding to a 1% false alarm probability, computed over a period when including the 221 mode improves consistency with the IMR results, here for GW240621_195059. For QNMRF, the ⋯ marker indicates that no improvement has been observed relative to GWTC-4.0.

2. THE GRAVITATIONAL-WAVE EVENTS

The sample of signals analyzed in this work consists of CBCs reported in GWTC-5.0 that satisfy the selection criteria adopted for the tests of GR. Table 2 lists the events from O4b that pass these requirements, while Table 3 shows additional events from O4a that were not included in the GWTC-4.0 tests of GR but satisfy the same selection criteria when using the reanalyzed PyCBC and GstLAL search results with updated configurations (see Appendix A of Abac et al. 2026i). The event GW240925_005809 (Abac et al. 2026g) is included using the C01 recalibrated strain data, which corrects the calibration error present in the observed data. Comparisons with astrophysical calibration measurements indicate consistency with the expected detector response, and the remaining calibration uncertainty is incorporated through the standard calibration-marginalization procedure. Combined with events analyzed in earlier catalog studies, these observations provide the dataset used in the tests presented in this paper.

The tables summarize several basic properties of each event. For each signal we report the network matched-filter SNR and the interferometers (IFOs) that contributed to the analysis (H = LIGO Hanford, L = LIGO Livingston, V = Virgo). We also include several quantities inferred from the parameter es-

timation (PE) analyses performed for the GW catalogs (Abac et al. 2026h), namely the redshifted chirp mass $(1+z)\mathcal{M}$, the effective inspiral spin $\chi_{\rm eff}$, the redshifted final mass $(1+z)M_{\rm f}$, and the final spin $\chi_{\rm f}$ of the remnant object.

Not all analyses use the full set of events listed in the tables. Several of the tests impose additional requirements related to signal strength, binary parameters, or the portion of the waveform being analyzed. The final columns of the tables therefore indicate which events contribute to each analysis. The detailed selection criteria used by the individual tests are described in the corresponding sections of this paper.

Descriptions of the search pipelines, detector characterization, PE methods, and data-quality investigations can be found in the GWTC-5.0 catalog and methods papers (Abac et al. 2026i,j).

Table 2. Event selection table for the analyses in this paper, from the second part of the fourth observing run (O4b) (Abac et al. 2026i).

O4b			Binary properties				Performed tests						
Event Name	SNR	IFOs	$(1+z)\mathcal{M}\,[M_\odot]$	$\chi_{\rm eff}$	$(1+z)M_{\rm f}\,[M_\odot]$	$\chi_{\rm f}$	RT	POL	TIGER	FTI	RD	pSEOBNR	QNMRF
GW240413_022019	$17.3^{+0.2}_{-0.2}$	HLV	$6.42^{+0.02}_{-0.02}$	$0.07^{+0.14}_{-0.04}$	$14.31^{+1.83}_{-0.30}$	$0.70^{+0.03}_{-0.04}$	✓	✓	✓	…	…	…	…
GW240414_054515	$10.7^{+0.2}_{-0.4}$	HV	$35.0^{+4.5}_{-3.8}$	$0.03^{+0.22}_{-0.21}$	$79.9^{+10.1}_{-6.8}$	$0.69^{+0.11}_{-0.12}$	✓	✓	…	…	…	…	…
GW240426_031451	$9.1^{+0.3}_{-0.5}$	HL	$62.1^{+8.3}_{-11.0}$	$-0.05^{+0.24}_{-0.28}$	140^{+16}_{-16}	$0.67^{+0.11}_{-0.13}$	✓	✓	…	…	…	…	…
GW240428_225440	$15.6^{+0.2}_{-0.3}$	HV	$17.21^{+0.32}_{-0.29}$	$0.03^{+0.10}_{-0.08}$	$38.27^{+2.82}_{-0.93}$	$0.69^{+0.04}_{-0.04}$	✓	✓	…	…	…	…	…
GW240501_033534	$9.7^{+0.4}_{-0.5}$	HLV	$49.2^{+8.3}_{-8.3}$	$0.04^{+0.26}_{-0.28}$	110^{+17}_{-15}	$0.70^{+0.10}_{-0.12}$	✓	✓	…	…	…	…	…
GW240507_041632	$9.7^{+0.5}_{-0.7}$	HLV	$17.48^{+2.87}_{-0.94}$	$0.01^{+0.20}_{-0.23}$	$50.0^{+9.0}_{-7.8}$	$0.55^{+0.19}_{-0.13}$	✓	✓	…	…	…	…	…
GW240511_031507	$15.6^{+0.2}_{-0.3}$	HLV	$42.4^{+3.7}_{-3.3}$	$0.05^{+0.17}_{-0.16}$	$93.9^{+7.5}_{-6.2}$	$0.71^{+0.07}_{-0.07}$	✓	✓	…	…	…	✓	…
GW240512_024139	$10.8^{+0.3}_{-0.5}$	HLV	$10.58^{+0.08}_{-0.10}$	$0.14^{+0.13}_{-0.07}$	$23.88^{+3.73}_{-0.85}$	$0.72^{+0.06}_{-0.04}$	✓	✓	…	…	…	…	…
GW240513_183302	$13.6^{+0.2}_{-0.4}$	HLV	$24.4^{+1.1}_{-1.0}$	$0.14^{+0.13}_{-0.14}$	$54.8^{+4.0}_{-2.5}$	$0.74^{+0.06}_{-0.06}$	✓	✓	…	…	…	…	…
GW240514_121713	$16.2^{+0.2}_{-0.3}$	HLV	$51.6^{+5.1}_{-6.1}$	$0.01^{+0.17}_{-0.18}$	$114.6^{+10.1}_{-9.4}$	$0.69^{+0.08}_{-0.09}$	✓	✓	…	…	…	✓	…
GW240515_005301	$10.3^{+0.3}_{-0.4}$	HLV	$36.7^{+3.3}_{-3.3}$	$0.47^{+0.17}_{-0.19}$	$85.4^{+11.7}_{-8.3}$	$0.84^{+0.06}_{-0.09}$	✓	✓	…	…	…	…	…
GW240520_213616	$10.5^{+0.3}_{-0.5}$	HLV	$10.49^{+0.14}_{-0.11}$	$-0.00^{+0.12}_{-0.08}$	$23.46^{+2.40}_{-0.60}$	$0.67^{+0.04}_{-0.04}$	✓	✓	…	…	…	…	…
GW240527_183429	$9.4^{+0.3}_{-0.5}$	HLV	70^{+11}_{-19}	$0.31^{+0.27}_{-0.33}$	157^{+21}_{-25}	$0.79^{+0.09}_{-0.16}$	✓	✓	…	…	…	…	…
GW240527_230910	$10.7^{+0.3}_{-0.4}$	HLV	$17.33^{+0.62}_{-0.55}$	$-0.03^{+0.20}_{-0.24}$	$44.8^{+8.5}_{-6.3}$	$0.58^{+0.10}_{-0.08}$	✓	✓	…	…	…	…	…
GW240530_012417	$10.6^{+0.5}_{-0.6}$	HLV	$11.10^{+0.12}_{-0.11}$	$0.04^{+0.25}_{-0.11}$	$25.4^{+7.8}_{-1.2}$	$0.68^{+0.07}_{-0.05}$	✓	✓	…	…	…	…	…
GW240601_231004	$9.8^{+0.3}_{-0.5}$	HLV	$9.72^{+0.07}_{-0.07}$	$0.09^{+0.16}_{-0.07}$	$21.56^{+3.79}_{-0.41}$	$0.72^{+0.06}_{-0.05}$	✓	✓	…	…	…	…	…
GW240615_113620	$23.8^{+0.1}_{-0.2}$	HLV	$33.5^{+1.6}_{-1.8}$	$-0.07^{+0.11}_{-0.13}$	$74.6^{+3.2}_{-3.3}$	$0.66^{+0.05}_{-0.06}$	✓	✓	…	✓	✓	✓	✓
GW240615_160735	$9.9^{+0.3}_{-0.5}$	HLV	$33.6^{+3.7}_{-3.5}$	$0.28^{+0.20}_{-0.24}$	$75.2^{+7.9}_{-6.5}$	$0.78^{+0.08}_{-0.11}$	✓	✓	…	…	…	…	…
GW240621_195059	$27.6^{+0.1}_{-0.2}$	HLV	$35.0^{+1.6}_{-1.5}$	$-0.00^{+0.10}_{-0.10}$	$77.5^{+2.9}_{-2.7}$	$0.68^{+0.04}_{-0.04}$	✓	✓	…	✓	✓	✓	✓
GW240622_004008	$12.3^{+0.3}_{-0.4}$	HLV	$15.97^{+0.28}_{-0.25}$	$0.25^{+0.13}_{-0.10}$	$36.1^{+6.7}_{-1.6}$	$0.77^{+0.07}_{-0.05}$	✓	✓	…	…	…	…	…
GW240629_145256	$12.3^{+0.3}_{-0.4}$	HLV	$9.98^{+0.06}_{-0.06}$	$0.04^{+0.11}_{-0.06}$	$22.26^{+1.98}_{-0.48}$	$0.69^{+0.03}_{-0.04}$	✓	✓	…	…	…	…	…
GW240630_101703	$10.1^{+0.3}_{-0.5}$	HLV	$33.1^{+3.7}_{-3.9}$	$-0.04^{+0.23}_{-0.27}$	$73.9^{+7.6}_{-6.8}$	$0.69^{+0.10}_{-0.11}$	✓	✓	…	…	…	…	…
GW240703_191355	$10.1^{+0.3}_{-0.6}$	LV	$35.2^{+3.3}_{-3.1}$	$0.05^{+0.21}_{-0.20}$	$78.4^{+6.9}_{-5.9}$	$0.71^{+0.09}_{-0.08}$	✓	✓	…	…	…	…	…
GW240705_053215	$14.3^{+0.2}_{-0.3}$	HLV	$56.5^{+6.4}_{-6.3}$	$-0.03^{+0.20}_{-0.21}$	126^{+13}_{-11}	$0.69^{+0.10}_{-0.09}$	✓	✓	…	…	…	✓	…
GW240716_034900	$13.0^{+0.3}_{-0.4}$	LV	$37.9^{+4.6}_{-4.7}$	$-0.02^{+0.22}_{-0.25}$	$85.9^{+8.7}_{-7.7}$	$0.67^{+0.10}_{-0.13}$	✓	✓	…	…	…	✓	…
GW240825_055146	$9.7^{+0.4}_{-0.6}$	HLV	$10.91^{+0.13}_{-0.28}$	$0.07^{+0.19}_{-0.13}$	$24.9^{+4.9}_{-1.2}$	$0.69^{+0.07}_{-0.06}$	✓	✓	…	…	…	…	…
GW240830_211120	$10.7^{+0.4}_{-0.5}$	HLV	$10.19^{+0.10}_{-0.09}$	$0.07^{+0.13}_{-0.07}$	$22.90^{+2.46}_{-0.69}$	$0.69^{+0.04}_{-0.04}$	✓	✓	…	…	…	…	…
GW240902_143306	$9.6^{+0.4}_{-0.5}$	HLV	$27.1^{+4.6}_{-3.2}$	$-0.03^{+0.28}_{-0.30}$	$60.9^{+9.1}_{-6.7}$	$0.69^{+0.11}_{-0.12}$	✓	✓	…	…	…	…	…
GW240907_153833	$9.1^{+0.3}_{-0.5}$	HLV	$48.6^{+7.7}_{-6.4}$	$0.04^{+0.26}_{-0.25}$	109^{+16}_{-12}	$0.71^{+0.10}_{-0.10}$	✓	✓	…	…	…	…	…
GW240908_082628	$10.0^{+0.3}_{-0.6}$	HLV	$44.0^{+7.3}_{-7.6}$	$0.14^{+0.24}_{-0.25}$	99^{+15}_{-13}	$0.73^{+0.10}_{-0.13}$	✓	✓	…	…	…	…	…
GW240910_103535	$17.4^{+0.2}_{-0.2}$	HL	$8.23^{+0.03}_{-0.03}$	$0.12^{+0.12}_{-0.07}$	$18.88^{+2.29}_{-0.92}$	$0.70^{+0.04}_{-0.04}$	✓	✓	✓	…	…	…	…
GW240915_001357	$16.2^{+0.2}_{-0.3}$	HLV	$9.41^{+0.05}_{-0.04}$	$0.13^{+0.16}_{-0.05}$	$20.92^{+4.06}_{-0.43}$	$0.72^{+0.04}_{-0.03}$	✓	✓	…	…	…	…	…
GW240915_105151	$13.7^{+0.3}_{-0.4}$	HV	$9.11^{+0.11}_{-0.06}$	$0.07^{+0.22}_{-0.09}$	$20.55^{+5.80}_{-0.70}$	$0.70^{+0.06}_{-0.04}$	✓	✓	…	…	…	…	…
GW240919_061559	$16.2^{+0.2}_{-0.3}$	HLV	$38.9^{+3.0}_{-2.6}$	$-0.02^{+0.16}_{-0.17}$	$86.7^{+6.1}_{-5.3}$	$0.68^{+0.07}_{-0.06}$	✓	✓	…	…	…	✓	…
GW240920_073424	$13.3^{+0.3}_{-0.4}$	HLV	$19.87^{+0.93}_{-0.91}$	$-0.09^{+0.19}_{-0.16}$	$46.6^{+5.8}_{-2.9}$	$0.64^{+0.11}_{-0.09}$	✓	✓	…	…	…	…	…
GW240920_124024	$37.0^{+0.1}_{-0.1}$	HL	$36.1^{+1.3}_{-1.2}$	$-0.04^{+0.08}_{-0.09}$	$79.6^{+2.6}_{-2.4}$	$0.68^{+0.04}_{-0.03}$	✓	✓	✓	✓	✓	✓	✓
GW240921_201835	$11.1^{+0.4}_{-0.7}$	LV	$18.46^{+0.64}_{-0.50}$	$0.00^{+0.17}_{-0.22}$	$51.9^{+8.5}_{-7.6}$	$0.52^{+0.08}_{-0.06}$	✓	✓	…	…	…	…	…
GW240922_142106	$12.4^{+0.3}_{-0.4}$	HLV	$10.06^{+0.06}_{-0.06}$	$0.07^{+0.12}_{-0.06}$	$22.52^{+2.69}_{-0.59}$	$0.70^{+0.04}_{-0.03}$	✓	✓	…	…	…	…	…
GW240923_204006	$11.7^{+0.2}_{-0.4}$	HLV	$56.9^{+6.8}_{-8.1}$	$0.01^{+0.20}_{-0.21}$	126^{+14}_{-13}	$0.70^{+0.08}_{-0.10}$	✓	✓	…	…	…	…	…
GW240924_000316	$9.9^{+0.4}_{-0.5}$	HLV	$55.8^{+7.6}_{-7.5}$	$0.10^{+0.23}_{-0.24}$	124^{+16}_{-13}	$0.73^{+0.10}_{-0.11}$	✓	✓	…	…	…	…	…

Table 2 *continued*

Table 2 *(continued)*

O4b			Binary properties				Performed tests						
Event Name	SNR	IFOs	$(1+z)\mathcal{M}\,[M_\odot]$	$\chi_{\rm eff}$	$(1+z)M_{\rm f}\,[M_\odot]$	$\chi_{\rm f}$	RT	POL	TIGER	FTI	RD	pSEOBNR	QNMRF
GW240925_005809	$31.9^{+0.1}_{-0.1}$	HLV	$7.37^{+0.01}_{-0.01}$	$0.02^{+0.07}_{-0.02}$	$16.25^{+0.87}_{-0.14}$	$0.69^{+0.02}_{-0.02}$	✓	✓	✓	✓	···	···	···
GW240930_035959	$15.2^{+0.2}_{-0.3}$	HLV	$17.23^{+0.48}_{-0.29}$	$-0.05^{+0.14}_{-0.13}$	$42.3^{+4.2}_{-3.0}$	$0.58^{+0.05}_{-0.04}$	✓	✓	···	···	···	···	···
GW241002_030559	$11.4^{+0.3}_{-0.4}$	LV	$36.9^{+3.5}_{-3.5}$	$-0.10^{+0.20}_{-0.23}$	$82.3^{+7.4}_{-7.1}$	$0.66^{+0.08}_{-0.09}$	✓	✓	···	···	···	···	···
GW241006_015333	$16.4^{+0.2}_{-0.3}$	HLV	$25.9^{+1.3}_{-1.0}$	$0.07^{+0.13}_{-0.13}$	$57.2^{+2.7}_{-2.2}$	$0.71^{+0.05}_{-0.04}$	✓	✓	···	···	···	✓	···
GW241009_084816	$11.8^{+0.3}_{-0.4}$	HLV	$10.72^{+0.10}_{-0.09}$	$-0.05^{+0.14}_{-0.08}$	$24.28^{+2.99}_{-0.83}$	$0.67^{+0.06}_{-0.04}$	✓	✓	···	···	···	···	···
GW241011_233834	$35.8^{+0.1}_{-0.1}$	HV	$9.51^{+0.03}_{-0.03}$	$0.50^{+0.06}_{-0.04}$	$25.5^{+2.3}_{-1.6}$	$0.84^{+0.02}_{-0.02}$	✓	✓	✓	✓	···	···	···
GW241101_220523	$10.6^{+0.3}_{-0.5}$	LV	$32.8^{+8.7}_{-4.7}$	$0.15^{+0.33}_{-0.29}$	80^{+17}_{-11}	$0.70^{+0.16}_{-0.17}$	✓	✓	···	···	···	···	···
GW241102_124058	$21.6^{+0.1}_{-0.2}$	HLV	$8.79^{+0.03}_{-0.02}$	$0.06^{+0.11}_{-0.04}$	$19.50^{+1.80}_{-0.31}$	$0.70^{+0.03}_{-0.04}$	✓	✓	✓	✓	···	···	···
GW241102_144729	$11.3^{+0.3}_{-0.5}$	HLV	$50.4^{+6.2}_{-8.6}$	$-0.09^{+0.21}_{-0.27}$	114^{+12}_{-12}	$0.65^{+0.09}_{-0.15}$	✓	✓	···	···	···	···	···
GW241109_115924	$11.6^{+0.2}_{-0.4}$	HV	$5.95^{+0.04}_{-0.02}$	$0.01^{+0.16}_{-0.05}$	$13.27^{+1.77}_{-0.27}$	$0.68^{+0.04}_{-0.04}$	✓	✓	···	···	···	···	···
GW241110_124123	$9.9^{+0.3}_{-0.5}$	HLV	$11.33^{+0.22}_{-0.23}$	$-0.31^{+0.23}_{-0.18}$	$27.1^{+4.1}_{-2.0}$	$0.52^{+0.07}_{-0.06}$	✓	✓	···	···	···	···	···
GW241111_111552	$15.1^{+0.2}_{-0.3}$	HL	$25.2^{+1.4}_{-1.3}$	$0.10^{+0.15}_{-0.14}$	$55.6^{+3.0}_{-2.5}$	$0.72^{+0.06}_{-0.06}$	✓	✓	···	···	···	···	···
GW241113_163507	$12.0^{+0.2}_{-0.3}$	LV	$18.12^{+0.58}_{-0.57}$	$0.50^{+0.11}_{-0.11}$	$39.5^{+3.3}_{-1.3}$	$0.85^{+0.04}_{-0.04}$	✓	✓	···	···	···	···	···
GW241114_024711	$11.1^{+0.4}_{-0.5}$	HLV	$42.9^{+6.3}_{-7.0}$	$0.12^{+0.21}_{-0.24}$	100^{+12}_{-10}	$0.74^{+0.09}_{-0.14}$	✓	✓	···	···	···	···	···
GW241114_235258	$12.2^{+0.2}_{-0.4}$	LV	$9.33^{+0.10}_{-0.10}$	$0.06^{+0.18}_{-0.08}$	$20.88^{+3.82}_{-0.58}$	$0.69^{+0.05}_{-0.04}$	✓	✓	···	···	···	···	···
GW241125_010116	$9.8^{+0.4}_{-0.5}$	HLV	80^{+10}_{-10}	$0.09^{+0.27}_{-0.27}$	176^{+20}_{-17}	$0.73^{+0.10}_{-0.11}$	✓	✓	···	···	···	···	···
GW241127_061008	$31.2^{+0.1}_{-0.2}$	HLV	$37.3^{+5.3}_{-2.9}$	$-0.16^{+0.13}_{-0.14}$	$100.9^{+7.1}_{-7.1}$	$0.56^{+0.09}_{-0.10}$	✓	✓	···	✓	✓	✓	✓
GW241129_021832	$14.9^{+0.2}_{-0.3}$	HLV	$32.9^{+2.6}_{-2.5}$	$0.17^{+0.16}_{-0.16}$	$73.1^{+5.7}_{-4.9}$	$0.75^{+0.08}_{-0.08}$	✓	✓	···	···	···	✓	···
GW241130_034908	$13.3^{+0.3}_{-0.4}$	HLV	$31.7^{+1.9}_{-1.7}$	$0.05^{+0.15}_{-0.15}$	$70.1^{+3.9}_{-3.5}$	$0.71^{+0.06}_{-0.05}$	✓	✓	···	···	···	···	···
GW241130_110422	$9.2^{+0.3}_{-0.5}$	HLV	$9.15^{+0.09}_{-0.10}$	$0.04^{+0.20}_{-0.10}$	$20.97^{+4.47}_{-0.98}$	$0.67^{+0.05}_{-0.05}$	✓	✓	···	···	···	···	···
GW241210_060606	$11.7^{+0.3}_{-0.4}$	HLV	$31.0^{+4.5}_{-2.7}$	$0.04^{+0.22}_{-0.19}$	$69.3^{+9.8}_{-5.4}$	$0.70^{+0.10}_{-0.09}$	✓	✓	···	···	···	···	···
GW241225_042553	$17.0^{+0.2}_{-0.3}$	LV	$9.83^{+0.04}_{-0.04}$	$0.12^{+0.09}_{-0.05}$	$22.11^{+1.95}_{-0.67}$	$0.72^{+0.06}_{-0.04}$	✓	✓	···	···	···	···	···
GW241225_082815	$17.7^{+0.3}_{-0.3}$	HL	$56.5^{+5.4}_{-6.1}$	$-0.17^{+0.21}_{-0.22}$	126^{+12}_{-11}	$0.67^{+0.08}_{-0.11}$	✓	✓	···	···	✓	✓	···
GW241230_084504	$11.8^{+0.2}_{-0.3}$	LV	$48.8^{+5.7}_{-5.7}$	$-0.14^{+0.21}_{-0.23}$	109^{+12}_{-11}	$0.65^{+0.08}_{-0.10}$	✓	✓	···	···	···	···	···
GW241231_054133	$16.9^{+0.2}_{-0.3}$	HLV	$9.55^{+0.05}_{-0.06}$	$0.08^{+0.12}_{-0.08}$	$22.0^{+2.5}_{-1.1}$	$0.68^{+0.04}_{-0.05}$	✓	✓	✓	···	···	···	···
GW250104_015122	$11.0^{+0.3}_{-0.4}$	HLV	$60.1^{+8.0}_{-8.7}$	$0.11^{+0.22}_{-0.23}$	133^{+16}_{-15}	$0.73^{+0.09}_{-0.10}$	✓	✓	···	···	···	···	···
GW250108_152221	$10.3^{+0.3}_{-0.4}$	HLV	62^{+10}_{-11}	$0.05^{+0.28}_{-0.28}$	139^{+20}_{-17}	$0.70^{+0.12}_{-0.15}$	✓	✓	···	···	···	···	···
GW250109_010541	$11.2^{+0.3}_{-0.4}$	HLV	$43.3^{+5.1}_{-6.0}$	$-0.11^{+0.27}_{-0.29}$	$97.4^{+12.3}_{-9.6}$	$0.66^{+0.12}_{-0.12}$	✓	✓	···	···	···	···	···
GW250114_082203	$76.9^{+0.0}_{-0.1}$	HL	$31.20^{+0.41}_{-0.49}$	$-0.03^{+0.04}_{-0.05}$	$68.31^{+0.83}_{-0.99}$	$0.68^{+0.01}_{-0.01}$	✓	✓	✓	✓	✓	✓	···
GW250118_170523	$13.1^{+0.2}_{-0.4}$	HL	$39.1^{+3.5}_{-3.2}$	$0.12^{+0.18}_{-0.19}$	$86.8^{+7.3}_{-6.3}$	$0.73^{+0.07}_{-0.07}$	✓	✓	···	···	···	···	···
GW250119_025138	$13.7^{+0.2}_{-0.3}$	HLV	$39.2^{+3.9}_{-3.8}$	$0.06^{+0.18}_{-0.18}$	$86.8^{+7.8}_{-6.7}$	$0.72^{+0.08}_{-0.08}$	✓	✓	···	···	···	✓	···
GW250119_190238	$20.4^{+0.2}_{-0.3}$	HLV	$10.01^{+0.04}_{-0.05}$	$0.09^{+0.05}_{-0.03}$	$21.91^{+0.74}_{-0.14}$	$0.71^{+0.02}_{-0.02}$	✓	✓	✓	✓	···	···	···

NOTE—A ✓ indicates an event meeting the selection criteria for an analysis, and thus included in the results. A ··· indicates that an event does not meet an analysis's selection criteria. More details about the events appear in Abac et al. (2026i).

Table 3. Event selection table for O4a events not analyzed in Abac et al. (2026b,c,d).

O4a			Properties				Tests performed						
Event Name	SNR	IFOs	$(1+z)\mathcal{M}\,[M_\odot]$	$\chi_{\rm eff}$	$(1+z)M_{\rm f}\,[M_\odot]$	$\chi_{\rm f}$	RT	POL	TIGER	FTI	RD	pSEOBNR	QNMRF
GW230707_124047	$10.6^{+0.2}_{-0.4}$	HL	$60.5^{+6.9}_{-7.6}$	$-0.04^{+0.26}_{-0.28}$	134^{+14}_{-13}	$0.69^{+0.10}_{-0.11}$	✓	✓	···	···	···	···	···
GW230709_122727	$8.5^{+0.3}_{-0.5}$	HL	56^{+10}_{-15}	$0.08^{+0.31}_{-0.31}$	126^{+20}_{-23}	$0.71^{+0.12}_{-0.16}$	✓	✓	···	···	···	···	···
GW230723_101834	$9.7^{+0.4}_{-0.5}$	HL	$14.89^{+0.46}_{-0.51}$	$-0.17^{+0.19}_{-0.16}$	$33.9^{+4.1}_{-1.7}$	$0.61^{+0.11}_{-0.08}$	✓	✓	···	···	···	···	···
GW230805_034249	$9.0^{+0.4}_{-0.5}$	HL	$36.2^{+8.8}_{-6.0}$	$0.08^{+0.26}_{-0.28}$	83^{+18}_{-12}	$0.73^{+0.10}_{-0.13}$	✓	✓	···	···	···	···	···
GW231118_071402	$8.5^{+0.4}_{-0.5}$	HL	$52.1^{+9.9}_{-9.7}$	$0.13^{+0.29}_{-0.28}$	117^{+21}_{-16}	$0.73^{+0.11}_{-0.14}$	✓	✓	···	···	···	···	···

NOTE—Notation as in Table 2. More details about the events appear in Abac et al. (2026i).

3. METHODS FOR TESTING GR

3.1. *Bayesian Inference*

For each event, we analyze a short data segment $d(t)$ around the coalescence time and compare it with waveform predic-

tions using a likelihood function $p(d|\boldsymbol{\theta}, \mathbb{M})$, where $\mathbb{M}$ denotes the assumed signal model and $\boldsymbol{\theta}$ represents the associated parameters. Most of the analyses presented in this work rely on the Bilby inference framework (Ashton et al. 2019; Romero-Shaw et al. 2020), often supplemented by the BilbyTGR extension designed for tests of GR (Ashton et al. 2025b). A detailed description of the general Bayesian PE framework used in the GWTC analyses is provided in the companion methods paper (Abac et al. 2026j). In the analyses considered here, the detector data are modeled as the sum of Gaussian, stationary noise and a GW signal (Abbott et al. 2020a). Under this assumption, subtracting the best-fit waveform from the data should leave residuals consistent with Gaussian noise. For a network of detectors, the likelihood is constructed as the product of the individual detector likelihoods, assuming statistical independence.

We adopt weakly informative priors $p(\boldsymbol{\theta}|\mathbb{M})$ that are sufficiently broad to encompass the region where the posterior distribution has significant support. Uniform priors are typically used for most parameters, with ranges chosen to avoid posterior probability accumulating near prior boundaries. The priors used for the parameters describing compact binaries in GR follow those adopted in the catalog analyses described in Abac et al. (2026j). Note that we use the ΛCDM cosmological model given in Abac et al. (2026h), where CDM refers to cold dark matter. Priors for parameters describing deviations from GR are specified independently for each individual analysis. Uncertainties in detector calibration are incorporated through frequency-dependent amplitude and phase corrections modeled as spline functions with Gaussian priors on their coefficients. These calibration parameters are marginalized over during sampling following the procedure described in Abac et al. (2026j).

The posterior distribution of the parameters is obtained from Bayes' theorem,

$$p(\boldsymbol{\theta}|d, \mathbb{M}) = \frac{p(d|\boldsymbol{\theta}, \mathbb{M})\, p(\boldsymbol{\theta}|\mathbb{M})}{\mathcal{Z}(d|\mathbb{M})}, \tag{1}$$

where $\mathcal{Z}(d|\mathbb{M})$ denotes the Bayesian evidence for the model $\mathbb{M}$. Posterior sampling is typically performed using nested-sampling algorithms (Skilling 2006). Most analyses employ the Dynesty sampler (Speagle 2020) through its implementation in Bilby.

In several tests, the evidence is used to compare different signal models. For example, when evaluating a model that extends GR by introducing additional parameters $\boldsymbol{\theta}_{\rm nGR}$, the relative support for the two hypotheses can be quantified through the Bayes factor

$$\mathcal{B}^{\rm nGR}_{\rm GR} = \frac{\mathcal{Z}(d|\boldsymbol{\theta}_{\rm GR}, \boldsymbol{\theta}_{\rm nGR})}{\mathcal{Z}(d|\boldsymbol{\theta}_{\rm GR})}. \tag{2}$$

In other analyses, deviations from GR are characterized through posterior distributions of phenomenological parameters. Consistency with GR is then quantified by evaluating the posterior quantile of the GR prediction. When multiple deviation parameters are considered simultaneously, we compute a multidimensional quantile defined as the fraction of posterior volume with probability density greater than or equal to that of the GR value.

3.2. *Hierarchical Inference*

While the analyses described above infer parameters independently for each event, additional information can be obtained by combining results across multiple detections. As the number of observed GW events grows, combining information across multiple detections becomes increasingly important for placing tight constraints on possible deviations from GR. This can be done either by directly combining posterior distributions for parameters assumed to be common across events, or more generally through hierarchical inference, in which the distribution of parameters across the population is modeled in terms of hyperparameters (Isi et al. 2019a; Mandel et al. 2019; Zimmerman et al. 2019).

For parameters describing effects that are expected to be identical for all sources, the combined constraint is obtained by jointly analyzing the measurements from all events. In contrast, for parameters that may vary between individual observations, such as phenomenological deviations in waveform coefficients, we infer the parameters by describing the ensemble of measurements.

In practice, we model the distribution of deviation parameters across events using a Gaussian population characterized by a mean μ and standard deviation σ (Isi et al. 2019a). When multiple deviation parameters are combined simultaneously, as in the pSEOBNR analysis, the full covariance structure of the population is inferred, including correlation coefficients ρ between parameters (Zhong et al. 2024). The FTI, TIGER and pSEOBNR tests employ hierarchical inference for combining results. The set of GWTC-4.0 events included in the hierarchical analyses, in addition to the new events listed in Tables 2 and 3, is reported in the corresponding tables of Abac et al. (2026c,d).

4. RESIDUAL TEST

The residual test checks for any excess coherent power that remains in the detector network after the best-fit GR template is subtracted out from the data. If GR correctly describes the GW signal, the GR-based template would capture all astrophysical features in the signal. In this case, the resulting residual would be consistent with instrumental noise. We use the same method as in previous analyses for this test (Abbott et al. 2019b, 2021b, 2025; Abac et al. 2026b).

The noise in ground-based GW detectors arises from various sources (Aasi et al. 2015; Abbott et al. 2016d). The detector noise is typically assumed to be stationary and Gaussian, so the data time series $\boldsymbol{d}(t)$ is modeled as the sum of the Gaussian noise $\boldsymbol{n}(t)$ and the GW model waveform $\boldsymbol{h}(t)$:

$$\boldsymbol{d}(t) = \boldsymbol{h}(t) + \boldsymbol{n}(t), \tag{3}$$

where the boldface notation indicates that each quantity has a component corresponding to each detector. The best-fit signal model is defined here to be the waveform with the set

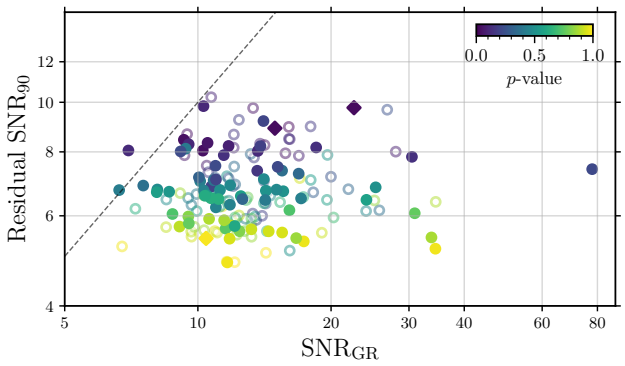


Figure 1. Results of the residuals analysis. Scatter plot of the SNR of the maximum-likelihood template ($\mathrm{SNR_{GR}}$) and the 90% credible upper bound on the residual network SNR ($\mathrm{SNR_{90}}$) for each event. The color bar denotes the p-values of individual events. Solid (empty) circles represent the O4b (pre-O4b) events, where the pre-O4b events are from Abbott et al. (2019b, 2021b, 2025); Abac et al. (2026b). The events with the highest and lowest p-values are shown by diamonds. The dashed line represents $\mathrm{SNR_{GR}} = \mathrm{SNR_{90}}$. We see that $\mathrm{SNR_{GR}} \geq \mathrm{SNR_{90}}$ for all events considered, except for GW240507_041632 and GW240601_231004.

of parameters that maximize the likelihood of the recorded data under the assumption that a signal is present. The resulting maximum-likelihood parameters minimize the difference between the data $\boldsymbol{d}(t)$ and the template $\boldsymbol{h}(t)$.

Once the best-fit (maximum-likelihood) waveform $\boldsymbol{h}_{\mathrm{maxL}}(t)$ is obtained using a GR template—here IMRPhenomXPHM_SpinTaylor (Pratten et al. 2021; Colleoni et al. 2025) for the O4b events—the residual is calculated as $\boldsymbol{r}(t) = \boldsymbol{d}(t) - \boldsymbol{h}_{\mathrm{maxL}}(t)$. The residual $\boldsymbol{r}(t)$ should be consistent with noise if $\boldsymbol{h}_{\mathrm{maxL}}$ provides an accurate GR-based estimate of the signal. We assess this by analyzing the residual with BayesWave algorithm (Cornish & Littenberg 2015; Cornish et al. 2021, 2026). BayesWave employs a template-independent model based on Morlet–Gabor wavelets to characterize any coherent features in the detector network. It produces a discretized probability distribution over the wavelet parameter space, with each point corresponding to a residual reconstruction with a well-defined SNR. Any loud, multi-detector coherent features in the data $\boldsymbol{d}(t)$ that are not captured by the GR-based model $\boldsymbol{h}(t)$ are reconstructed. If the true underlying signal is well reconstructed by the maximum-likelihood waveform $\boldsymbol{h}_{\mathrm{maxL}}$, the BayesWave reconstruction of the residual SNR posterior will be consistent with noise (Johnson-McDaniel et al. 2022).

The residuals are quantified by calculating the 90th percentile of the network SNR distribution produced by BayesWave, denoted $\mathrm{SNR_{90}}$. This means there is a 90% probability that, after subtracting $\boldsymbol{h}_{\mathrm{maxL}}$, the residual signal has an optimal network SNR $\leq \mathrm{SNR_{90}}$. For the case of Gaussian noise in the LIGO Hanford, LIGO Livingston, and Virgo network, this value is typically $\lesssim 5$ (Johnson-McDaniel et al. 2022).

We perform a more systematic assessment by defining a p-value under the null hypothesis that the residual is consistent with noise. This is done by running an identical BayesWave analysis on 200 randomly selected data segments around the event that contain neither a known GW signal nor transient non-Gaussian features, and then calculating the probability that instrumental noise alone could produce an SNR^n_{90} such that $\mathrm{SNR}^n_{90} \geq \mathrm{SNR_{90}}$. This probability defines the p-value, $P(\mathrm{SNR}^n_{90} \geq \mathrm{SNR_{90}})$. A higher probability that the residual power originates from instrumental noise corresponds to a higher p-value, whereas a smaller p-value indicates that the residual power is less likely to arise from noise alone.

Finally, we quantify how well the GR-based template fits the signal in the data. Since our GR templates are not perfect, and we have a probabilistic measurement of the signal in the data, we can quantify this fit using the 90% lower bound on the fitting factor between the model and the signal:

$$\mathrm{FF_{90}} = \frac{\mathrm{SNR_{GR}}}{\sqrt{\mathrm{SNR^2_{GR}} + \mathrm{SNR^2_{90}}}}, \tag{4}$$

where $\mathrm{SNR_{GR}}$ is the optimal network SNR for $\boldsymbol{h}_{\mathrm{maxL}}$. A value of $\mathrm{FF_{90}} = 1$ indicates perfect agreement between the GR template and the signal in the data.

All events that pass this paper's selection criteria and are analyzed with BBH waveforms are considered for this test. We summarize the results for $\mathrm{SNR_{GR}}$, the residual $\mathrm{SNR_{90}}$, $\mathrm{FF_{90}}$, and the p-values for each event in Table 4. Figure 1 shows $\mathrm{SNR_{90}}$ versus $\mathrm{SNR_{GR}}$ for both O4b and pre-O4b events. The residual $\mathrm{SNR_{90}}$ is smaller than the original $\mathrm{SNR_{GR}}$ for all events except GW240507_041632 and GW240601_231004. This is due to the low values of $\mathrm{SNR_{GR}}$ for these events, where the use of point estimates for both $\mathrm{SNR_{90}}$ and $\mathrm{SNR_{GR}}$ leads to $\mathrm{SNR_{90}} \geq \mathrm{SNR_{GR}}$. The residual $\mathrm{SNR_{90}}$ is consistent with noise, with p-values of 0.12 and 0.42 for GW240507_041632 and GW240601_231004, respectively.

Among the 72 O4b events, GW240428_225440 and GW241102_124058 have the lowest p-values of 0.03. GW240428_225440 has a residual $\mathrm{SNR_{90}} = 8.89$, and GW241102_124058 has a residual $\mathrm{SNR_{90}} = 9.76$. GW240622_004008 has the highest p-value of 0.99 with a residual $\mathrm{SNR_{90}} = 5.42$. GW240507_041632 has the lowest $\mathrm{FF_{90}}$ of 0.72, and a p-value of 0.12. The mean $\mathrm{FF_{90}}$ for O4b events is 0.87, which indicates agreement between the GR templates and the signals in the data.

If our best-fit template has successfully recovered the GW signals, the contribution of the residual $\mathrm{SNR_{90}}$ should arise solely from detector noise. In this case, the p-values are expected to be uniformly distributed between $[0, 1]$. To verify this, we show a probability–probability (PP) plot in Figure 2. The measurement of p-values is subject to uncertainty due to the finite number of background runs. If the p-value $\hat{p} = n/N$ is estimated by running on N background segments, of which n produce $\mathrm{SNR_{90}}$ greater than that of the event, the likelihood of the estimated p-value is given by the binomial distribution function:

$$\mathcal{L}(\hat{p}) = \binom{N}{n} p^n (1-p)^{N-n}. \tag{5}$$

Assuming a uniform prior distribution on the p-value, the posterior distribution is given by a beta distribution,

$$P(p|N,n) = \mathrm{Beta}(n+1, N-n+1). \tag{6}$$

In Figure 2, the shaded region around the PP curve represents the 90% credible interval of the p-value posterior.

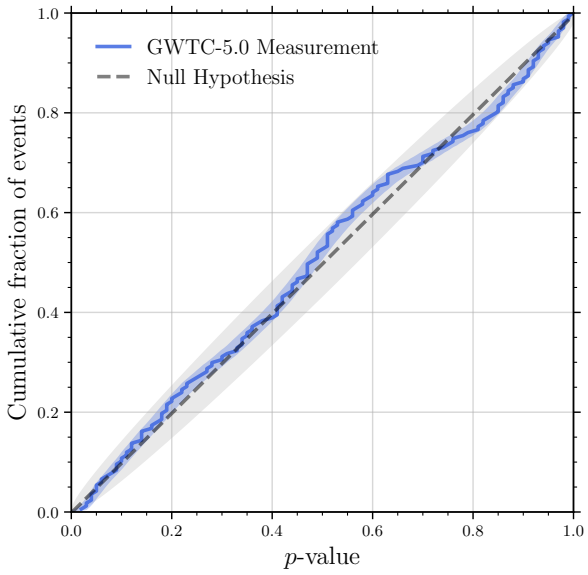


Figure 2. Results of the residuals analysis. The blue line shows the fraction of events with p-values of the residual SNR less than or equal to the value of the abscissa. The shaded blue region represents the 90% credible interval arising from the finite number of noise-only realizations. The null hypothesis that the p-value is uniformly distributed over [0, 1] is shown by the diagonal dashed line, with the shaded light grey area denoting the 90% credible interval around the null hypothesis due to the finite number of events.

5. TESTS OF POLARIZATION

Polarization tests (Wong et al. 2021) involve constructing a linear combination of strain data, called the null stream, to eliminate excess power and compare the residual data to noise, allowing one to constrain possible non-tensorial polarizations of GWs. The null stream is determined geometrically, relying solely on the beam pattern function (Forward 1978; Gürsel & Tinto 1989; Nishizawa et al. 2009; Błaut 2012; Isi et al. 2015; Isi & Weinstein 2017), which describes detector responses to GWs, independent of the waveform morphology.

While the residuals analysis is in principle sensitive to any deviations from GR to which the other tests considered are sensitive, in practice it is less sensitive to a given GR deviation than a more specific test, particularly when the deviation is not well localized in time, as is the case for low-mass compact binary signals (Johnson-McDaniel et al. 2022). Nevertheless, in summary, the residuals for all analyzed events are consistent with detector noise, with no evidence for deviations from the GR-based waveform model.

In the frequency domain, the observation model is expressed as

$$\tilde{\boldsymbol{d}}(f;\Delta\boldsymbol{t}) = \boldsymbol{F}(\alpha,\delta,\psi,t_{\rm event})\tilde{\boldsymbol{h}}(f) + \tilde{\boldsymbol{n}}(f;\Delta\boldsymbol{t})\,, \tag{7}$$

where $\tilde{\boldsymbol{d}}(f;\Delta\boldsymbol{t})$ is the time-shifted strain data at the geocenter, $\boldsymbol{F}(\alpha,\delta,\psi,t_{\rm event})$ is the beam pattern matrix, $\tilde{\boldsymbol{h}}(f)$ represents polarization modes at the geocentric time, and $\tilde{\boldsymbol{n}}(f;\Delta\boldsymbol{t})$ is the detector noise, also time-shifted to the geocenter. Here the elements of $\tilde{\boldsymbol{d}}$ and $\tilde{\boldsymbol{n}}$ represent the strain data and detector noise for each individual detector, respectively (time-shifted to the geocenter), so for an N-detector network, these are N-dimensional vectors. $\tilde{\boldsymbol{h}}(f)$ is a vector of the different polarization modes, so an M-dimensional vector for M polarization modes. α and δ denote the source's right ascension and declination, ψ is the polarization angle, and $\Delta\boldsymbol{t} = \Delta\boldsymbol{t}(\alpha,\delta,t_{\rm event})$ represents the time delays from the geocenter to the detectors. Since the beam pattern function varies negligibly over the duration of the GW transients considered, it is approximated as constant throughout.

The null stream construction can be formulated as a projection using the projector constructed from the beam pattern matrix $\boldsymbol{F}$ (Wong et al. 2021). The projector removes data within the hyperplane spanned by the column vectors of $\boldsymbol{F}$, irrespective of the waveform morphology of $\boldsymbol{h}(f)$. Here $\boldsymbol{F}$ defines the polarization model; for instance, in a scalar-tensor model:

$$\boldsymbol{F} = \left[\boldsymbol{f}_+ \;\; \boldsymbol{f}_\times \;\; \boldsymbol{f}_{\rm b}\right], \tag{8}$$

where the column vectors $\boldsymbol{f}_+$, $\boldsymbol{f}_\times$, and $\boldsymbol{f}_{\rm b}$ represent the plus, cross, and scalar breathing polarization modes, respectively. The i-th entry of the column vectors, $\boldsymbol{f}_{+/\times/{\rm b}}$, corresponds to the beam pattern function of the i-th detector for the +/×/scalar breathing polarization mode. The scalar longitudinal mode, being degenerate with the scalar breathing mode, is excluded from the matrix.

The number of detectors must exceed the number of columns in the beam pattern matrix. However, data from only two detectors are available for part of the events analyzed, constraining the analysis to a single effective polarization mode. We assume the polarization modes are similar to chosen basis modes indicated in Table 5, differing at most by a complex scaling factor (Wong et al. 2021), in a manner similar to the elliptical parameterization in Lee et al. (2025). This assumption defines the effective beam pattern matrix used in the analysis and imposes a simplifying structure on the polarization subspace (e.g., in the single basis mode case for tensorial signals,

Table 4. Results from the residuals analysis.

Events	SNR_{GR}	SNR_{90}	FF_{90}	p-value
GW230707_124047*	11.71	5.48	0.91	0.95
GW230709_122727*	7.53	6.87	0.76	0.37
GW230723_101834*	9.09	5.72	0.85	0.9
GW230805_034249*	9.54	5.8	0.85	0.76
GW231118_071402*	8.76	6.05	0.82	0.72
GW240413_022019	15.6	6.7	0.92	0.51
GW240414_054515	10.72	6.85	0.84	0.2
GW240426_031451	9.3	8.45	0.76	0.04
GW240428_225440	14.93	8.9	0.86	0.03
GW240501_033534	8.61	6.69	0.79	0.53
GW240507_041632	6.97	8.05	0.72	0.12
GW240511_031507	15.06	6.75	0.91	0.47
GW240512_024139	10.12	7.01	0.82	0.42
GW240513_183302	14.24	6.72	0.9	0.51
GW240514_121713	16.1	6.15	0.93	0.72
GW240515_005301	10.05	7.13	0.82	0.31
GW240520_213616	10.93	6.49	0.86	0.61
GW240527_183429	9.51	5.98	0.85	0.82
GW240527_230910	10.3	6.73	0.84	0.44
GW240530_012417	10.6	5.92	0.87	0.88
GW240601_231004	6.64	6.74	0.75	0.42
GW240615_113620	24.04	6.45	0.97	0.57
GW240615_160735	10.48	6.58	0.85	0.52
GW240621_195059	25.23	6.82	0.97	0.5
GW240622_004008	10.43	5.42	0.89	0.99
GW240629_145256	13.21	5.65	0.92	0.92
GW240630_101703	11.92	5.74	0.9	0.91
GW240703_191355	11.55	5.66	0.9	0.74
GW240705_053215	14.07	9.19	0.84	0.21
GW240716_034900	13.62	7.35	0.88	0.16
GW240825_055146	10.26	8.04	0.79	0.04
GW240830_211120	10.86	7.04	0.84	0.38
GW240902_143306	10.52	8.33	0.78	0.06
GW240907_153833	8.1	6.75	0.78	0.48
GW240908_082628	11.22	6.74	0.86	0.49
GW240910_103535	17.35	5.35	0.96	0.97
GW240915_001357	15.5	7.35	0.9	0.36
GW240915_105151	14.2	6.44	0.91	0.41
GW240919_061559	16.9	7.25	0.92	0.34
GW240920_073424	15.15	7.48	0.9	0.22
GW240920_124024	34.43	5.17	0.99	0.98
GW240921_201835	10.98	6.23	0.87	0.45
GW240922_142106	11.13	6.47	0.86	0.63
GW240923_204006	10.3	9.83	0.74	0.11
GW240924_000316	9.53	8.28	0.76	0.11
GW240925_005809	30.47	7.82	0.97	0.14
GW240930_035959	13.66	8.03	0.86	0.12
GW241002_030559	12.42	5.6	0.91	0.81
GW241006_015333	15.5	5.57	0.94	0.94
GW241009_084816	12.5	6.56	0.89	0.56
GW241011_233834	33.72	5.44	0.99	0.87
GW241101_220523	11.66	4.86	0.92	0.98
GW241102_124058	22.53	9.76	0.92	0.03
GW241102_144729	11.73	8.22	0.82	0.09
GW241109_115924	10.99	7.07	0.84	0.2
GW241110_124123	10.7	6.49	0.85	0.63
GW241111_111552	14.45	5.6	0.93	0.93
GW241113_163507	11.82	6.31	0.88	0.44
GW241114_024711	10.37	6.57	0.84	0.61
GW241114_235258	12.11	5.73	0.9	0.6
GW241125_010116	9.38	8.12	0.76	0.41
GW241127_061008	30.91	6.07	0.98	0.76
GW241129_021832	14.09	7.08	0.89	0.47
GW241130_034908	13.85	8.21	0.86	0.19
GW241130_110422	8.07	6.67	0.8	0.56
GW241210_060606	9.14	8.02	0.76	0.23
GW241225_042553	16.68	5.43	0.95	0.85
GW241225_082815	17.11	6.49	0.94	0.43
GW241230_084504	12.61	6.47	0.89	0.33
GW241231_054133	15.59	5.94	0.93	0.79
GW250104_015122	11.83	6.88	0.86	0.47
GW250108_152221	10.96	7.51	0.82	0.23
GW250109_010541	11.46	5.88	0.89	0.87
GW250114_082203	77.85	7.4	1.0	0.22
GW250118_170523	11.79	5.41	0.91	0.94
GW250119_025138	11.44	7.88	0.82	0.14
GW250119_190238	18.51	8.16	0.92	0.14

Note— For each event we list the SNR of the best-fit waveform (SNR_{GR}), the 90% credible upper limit on the residual coherent network SNR (SNR_{90}), the 90% credible lower limit on the fitting factor (FF_{90}), and the p-value calculated from the background analysis. Additional O4a events analyzed are marked with an asterisk (∗).

it is equivalent to testing consistency with $h_\times = Ch_+$ for a complex constant C). For sources with higher-order modes, however, a complex constant may not fully capture the relation between the polarization modes. Consequently, for such events, the complex constant scaling may be inaccurate—we will discuss this in the context of specific events below. Nevertheless, the null-stream construction is geometrical and therefore more tolerant of inaccuracies in the polarization model than fully modeled (template-based) searches, which require precise agreement in both polarization content and waveform evolution.

A concrete illustration for non-tensorial signals is provided by scalar-tensor models, where one can choose the plus mode as the basis and assume similarity with the cross and scalar breathing modes, leading to the effective beam pattern matrix

$$\boldsymbol{F}_{\text{scalar-tensor,eff}} = \left[\boldsymbol{f}_+ + C_\times \boldsymbol{f}_\times + C_\text{b}\boldsymbol{f}_\text{b}\right], \tag{9}$$

where $C_\times$ and C_b are complex scaling factors. This method, independent of the waveform of the basis mode $\tilde{h}_+$, is sensitive only to the geometric projection of the polarization modes in the detector responses. Assuming similarity between polarization modes addresses the limitation posed by the number of detectors. Any violation of this assumption would indicate a signal component orthogonal to the signal space of the GR waveform model, which should appear in the residual test described in Section 4 – however, dedicated analyses (such as a template-based approach) are more sensitive to given GR deviations than the residuals test is.

We use a Bayesian framework to compute the evidence for each polarization hypothesis by sampling parameters $\boldsymbol{\theta} = \{\alpha, \delta, \psi, t_\text{event}, \boldsymbol{C}\}$ using the Dynesty sampler. Here, $\boldsymbol{C}$ are the complex scaling factors of the polarization modes relative to the basis mode(s). The number of scaling factors depends on the assumed number of polarization and basis modes. For each polarization hypothesis, we choose a particular polarization mode as the basis, but its waveform morphology is not assumed or involved in the analysis. Instead, only the relative amplitudes $\boldsymbol{C}$ are used to construct the effective beam pattern matrix for null stream projection. After constructing the null stream time-shifted to the geocenter, we then perform the time–frequency transform (Necula et al. 2012) to obtain the time–frequency representation of the null stream.

The Bayes factor quantifies the relative plausibility of the hypotheses. We adopt the same uniform priors for sky position, polarization angle, and event time as in previous analyses. For the complex scaling factors $\boldsymbol{C}$, the amplitude prior is uniform from 0 to 1, and the phase prior is uniform from 0 to 2π. We report the base-10 logarithm of the Bayes factor $\log_{10}\mathcal{B}^X_\text{T}$ comparing the non-tensor hypothesis X to the tensor hypothesis T.

The likelihood is calculated by summing the contributions of a predefined set of time–frequency pixels $\mathcal{I}$, known as the time–frequency cluster. The time–frequency cluster is constructed by performing the time–frequency transform on the maximum-likelihood GR waveform. This approach focuses the analysis on the time–frequency region where the signal is expected to be strongest, enhancing sensitivity to potential non-tensorial polarizations while reducing the influence of instrumental noise in other regions. However, as this method relies on the results of the GR template-based analysis, it may reduce sensitivity when the non-tensorial polarization modes have a significantly different time–frequency structure from the tensorial polarization modes. For example, this can occur for dominant dipolar scalar radiation at the orbital frequency during the inspiral in scalar-tensor theories (Bernard et al. 2022), when the non-tensorial radiation frequency significantly deviates from that of the tensorial radiation by more than the analysis's frequency resolution of $\Delta f = 16\,\text{Hz}$ in the time–frequency representation. This frequency resolution was selected as a practical choice that still sufficiently resolves the merger phase containing the dominant excess power in CBC signals, providing a corresponding time resolution of $\Delta t \simeq 1/(2\Delta f) \simeq 31\,\text{ms}$, which is commensurate with the characteristic duration of the merger phase in stellar-mass BBH coalescences, which lasts only a few tens of milliseconds for GW150914-like systems (Abbott et al. 2016e).

We analyze all the events satisfying the selection criteria of this paper using the same implementation used in Abac et al. (2026b). Each event is analyzed independently, yielding a per-event log Bayes factor $\log_{10}\mathcal{B}^X_\text{T}$. We then combine the events under the assumption that they are statistically independent, so that the joint Bayes factor for a hypothesis is the product of the individual-event Bayes factors. The combined log Bayes factor is therefore simply the sum of the per-event log Bayes factors,

$$\log_{10}\mathcal{B}^X_\text{T}\big|_\text{combined} = \sum_i \log_{10}\mathcal{B}^X_{\text{T},i}, \tag{10}$$

where the index i runs over the analyzed events. Figure 3 compares the results with and without the new events included in this paper, showing the improvements in constraints on non-tensorial polarizations. Table 6 lists the combined log Bayes factors for the same events and specifically shows the log Bayes factors for only the new events included in this paper. The scalar, vector, and vector–scalar hypotheses are strongly disfavored compared to the tensor hypothesis, indicating a clear preference for the tensorial polarization modes. After including the new events, these hypotheses are even more disfavored, with improved constraints on their viability due to the additional data. In contrast, the new events yield positive log Bayes factors for the tensor–scalar, tensor–vector, and tensor–vector–scalar hypotheses, shifting the combined values upward slightly. Since the tensor hypothesis is a subset of these combined hypotheses, they are not expected to be strongly disfavored when the tensor hypothesis represents the ground truth. In particular, the GWTC-5.0 combined log Bayes factor for the tensor–vector hypothesis is (slightly) positive. Particularly noteworthy is that the 90% credible interval excludes zero. Because this combined value is a sum over events, we checked whether it is dominated by a single outlying event. The largest individual contribution comes from GW241102_124058, followed by GW241011_233834, but we found no reason to exclude either event from the combined analysis. Even with GW241102_124058 excluded, the com-

Table 5. Polarization modes and their corresponding basis choices for the polarization analysis.

Hypothesis	Polarization modes	Pol. basis modes
Scalar (S)	b	b
Vector (V)	x, y	x
Vector–scalar (VS)	x, y, b	x
Tensor (T)	+, ×	+
Tensor–scalar (TS)	+, ×, b	+
Tensor–vector (TV)	+, ×, x, y	+
Tensor–vector–scalar (TVS)	+, ×, x, y, b	+

NOTE— The tensor plus and cross modes are denoted as "+" and "×," respectively, while the vector x and y modes are labeled "x" and "y." The scalar breathing mode is represented by "b." Only the breathing mode is considered for scalar polarizations, as the longitudinal mode is degenerate with it.

bined value remains positive, although zero then falls within its 90% credible interval, indicating that the positive result is not attributable to any single event. The effective-polarization approximation discussed above could contribute to the positive value for GW241011_233834, which shows evidence for an unequal mass ratio and spin precession (Abac et al. 2026i), but it does not obviously account for the largest contribution from GW241102_124058. In any case, these credible intervals reflect only the uncertainty in the evidence estimate arising from the nested sampling runs, and do not account for systematic effects such as non-Gaussian noise. Non-Gaussian or non-stationary detector noise, including glitches and PSD drift, can produce excess power not captured by the stationary Gaussian likelihood and can bias model comparison toward more flexible hypotheses (Zackay et al. 2021; Macas & Lundgren 2023). These caveats also apply to the combined log Bayes factors. Verifying these interpretations would require dedicated background studies. Given the modest magnitude of the observed positive values, we do not pursue such studies, and we regard these results as inconclusive.

6. TESTS OF GW GENERATION WITH TIGER AND FTI

During the adiabatic inspiral, when the orbital separation greatly exceeds the compact objects' characteristic size, we can model the compact objects as point particles and approximate both their orbital dynamics and their emitted GW signal within GR using the PN formalism (Blanchet 2024). In this approximation, relativistic corrections to the Newtonian solution are incorporated systematically in powers of the dimensionless parameter v/c, where v denotes the characteristic relative orbital speed and c the speed of light in vacuum. Contributions of relative order $O([v/c]^n)$ beyond the leading term are labeled as $(n/2)$PN and for fixed intrinsic binary parameters, GR uniquely determines the phase coefficients at each PN order. The frequency domain phase of the GW signal during early inspiral has the form

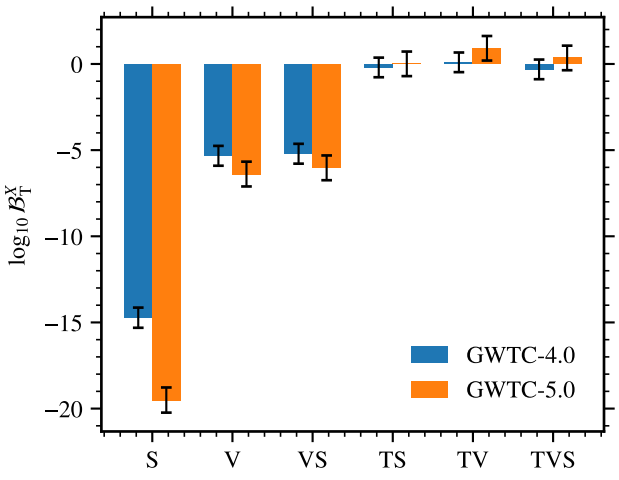


Figure 3. Comparison of $\log_{10}$ Bayes factors $\mathcal{B}^{X}_{\rm T}$ for various polarization hypotheses (S, V, VS, TS, TV, TVS) against the tensor hypothesis, for GWTC-4.0 and GWTC-5.0. Polarization states are projected onto one basis mode as detailed in Table 5. Positive (negative) values indicate that the hypothesis in the superscript is favored (disfavored) relative to the tensor hypothesis. Error bars represent 90% credible intervals.

$$\Psi_{\ell m}(f) = 2\pi f t_c - \phi_c - \frac{\pi}{4} + \frac{3c^5}{128\eta v^5}\frac{|m|}{2}\sum_{n=0}^{7}\left(\psi_n + \psi_{nl}\log\frac{v}{c}\right)\frac{v^n}{c^n} \quad (11)$$

for quasi-circular, spin-aligned binaries. Here $v = [2\pi G(1+z)Mf/|m|]^{1/3}$ with f the GW frequency, M the total mass, η the symmetric mass ratio, z the redshift, and t_c, ϕ_c the time and phase at coalescence (Abac et al. 2026h). The coefficients ψ_n and ψ_{nl} (where the l denotes the coefficients of the logarithmic terms) are the $(n/2)$PN coefficients and depend on the masses and spins of the binary. The subscript ℓm denotes the (ℓ, m) multipole in the spherical harmonic decomposition of the waveform.

This structure provides a natural framework to test GR (Blanchet & Sathyaprakash 1995, 1994; Arun et al. 2006a,b; Yunes & Pretorius 2009; Mishra et al. 2010; Li et al. 2012a,b). If GR correctly describes the dynamics, the measured phase evolution must agree with these uniquely predicted coefficients. Generic deviations from GR modify them, and we can capture such effects by introducing controlled fractional deviations in the PN phase coefficients.

Parameterized GW-based inspiral tests of GR, such as FTI (Mehta et al. 2023) and TIGER (Agathos et al. 2014; Meidam et al. 2018; Roy et al. 2026), implement this strategy and probe the consistency of the adiabatic inspiral with the PN prediction. More explicitly, these tests introduce controlled deviations of the form

$$\psi_i \to (1 + \delta\hat{\varphi}_i)\psi_i^{\rm GR,NS} + \psi_i^{\rm GR,S}, \quad (12)$$

Table 6. Results from the polarization analysis.

Events	$\log_{10}\mathcal{B}^{\mathrm{S}}_{\mathrm{T}}$	$\log_{10}\mathcal{B}^{\mathrm{V}}_{\mathrm{T}}$	$\log_{10}\mathcal{B}^{\mathrm{VS}}_{\mathrm{T}}$	$\log_{10}\mathcal{B}^{\mathrm{TS}}_{\mathrm{T}}$	$\log_{10}\mathcal{B}^{\mathrm{TV}}_{\mathrm{T}}$	$\log_{10}\mathcal{B}^{\mathrm{TVS}}_{\mathrm{T}}$
GWTC-4.0	$-14.72^{+0.59}_{-0.59}$	$-5.33^{+0.58}_{-0.58}$	$-5.21^{+0.58}_{-0.58}$	$-0.20^{+0.57}_{-0.57}$	$0.10^{+0.57}_{-0.57}$	$-0.31^{+0.57}_{-0.57}$
O4b + additional events from O4a	$-4.78^{+0.44}_{-0.44}$	$-1.06^{+0.43}_{-0.43}$	$-0.82^{+0.43}_{-0.43}$	$0.21^{+0.43}_{-0.43}$	$0.82^{+0.43}_{-0.43}$	$0.66^{+0.43}_{-0.43}$
GWTC-5.0	$-19.50^{+0.73}_{-0.73}$	$-6.39^{+0.72}_{-0.72}$	$-6.03^{+0.72}_{-0.72}$	$0.01^{+0.71}_{-0.71}$	$0.91^{+0.71}_{-0.71}$	$0.35^{+0.71}_{-0.71}$

Note— Combined $\log_{10}$ Bayes factors $\mathcal{B}$ for various polarization hypotheses against the tensor hypothesis, using both 2-detector and 3-detector events. Polarization states have been projected onto one basis mode as detailed in Table 5. The combined values are obtained by summing the individual $\log_{10}\mathcal{B}^{\mathrm{X}}_{\mathrm{T}}$ under the assumption that the events are independent. Positive (negative) values indicate that the hypothesis indicated in the superscript is favored (disfavored) with respect to the tensor hypothesis. Error bars refer to 90% credible intervals.

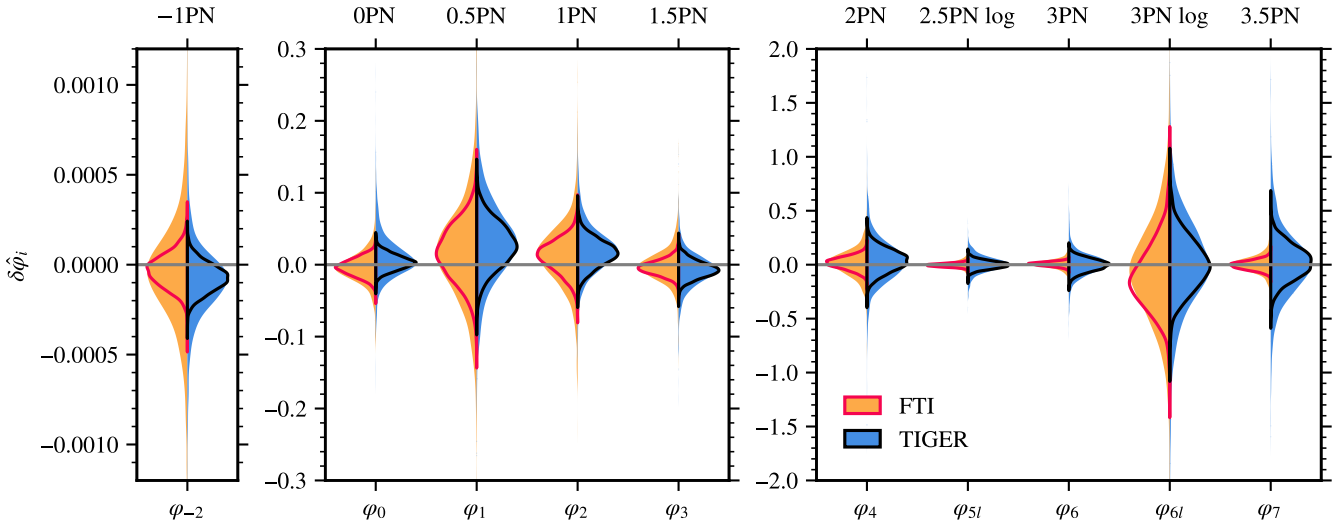


Figure 4. The constraints on the GR deviation parameters in the inspiral, marginalized over all analyzed events, in orange on the left for the FTI pipeline using the SEOBNRv5HM_ROM waveform model and in blue on the right for the TIGER pipeline using the IMRPhenomXPHM_SpinTaylor waveform model. The horizontal line shows the GR value. Assuming independent deviation parameters across events, using a hierarchical analysis yields the filled probability densities. Instead, assuming that the deviation parameters have the same values across different events leads to the distributions shown with solid lines.

to the non-spinning PN phase coefficients $\psi_i^{\mathrm{GR,NS}}$ in GR. Here $\delta\hat{\varphi}_i$ is the deviation parameter and $\psi_i^{\mathrm{GR,S}}$ is the spin-dependent part of the PN coefficient in GR. This is the same parameterization used previously (Abbott et al. 2016b, 2019a,b, 2021b, 2025; Sänger et al. 2026; Abac et al. 2026c), and it avoids possible issues when the spin-dependent terms cancel the non-spinning terms. Since the spin terms are part of the model, excluding them from the normalization of the deviations does not introduce pathologies, even for highly spinning systems.

Here, consistent with the GWTC-4.0 analysis (Abac et al. 2026c), we constrain the following inspiral parameters

$$\{\delta\hat{\varphi}_{-2}, \delta\hat{\varphi}_0, \delta\hat{\varphi}_1, \delta\hat{\varphi}_2, \delta\hat{\varphi}_3, \delta\hat{\varphi}_4, \delta\hat{\varphi}_{5l}, \delta\hat{\varphi}_6, \delta\hat{\varphi}_{6l}, \delta\hat{\varphi}_7\}\,, \quad (13)$$

for FTI and TIGER with l denoting the logarithmic terms. The non-logarithmic 2.5PN term is degenerate with the reference phase and cannot be constrained. Also, in GR, the −1PN and 0.5PN terms vanish, so $\delta\hat{\varphi}_{-2}$ and $\delta\hat{\varphi}_1$ are treated as absolute deviations normalized to the 0PN coefficient. All remaining parameters represent fractional deviations relative to their non-spinning GR values, and both pipelines apply these deviations to the signal's inspiral portion and turn them off as the signal transitions to the post-inspiral regime.

Across pipelines, the implementation primarily differs in the choice of baseline GR approximant and the transition frequency. For the analysis presented here, FTI uses the non-precessing non-eccentric frequency domain multipolar waveform model SEOBNRv5HM_ROM (Pompili et al. 2023) and adopts a tapering frequency which is equal to the peak frequency of the dominant $(2,2)$ multipole, allowing it to probe deeper into the late inspiral, thereby placing relatively tighter constraints on higher-order PN coefficients. On the other hand, TIGER adopts a lower cutoff and uses the non-eccentric multipolar phenomenological waveform model IMRPhenomXPHM_SpinTaylor as its GR baseline (Pratten et al. 2021; Colleoni et al. 2025).

Here, we apply the FTI test to events observed during O4b that have at least 5 inspiral GW cycles and an inspiral SNR ≥ 20. This minimum SNR is stricter than what was used in GWTC-4.0 (Abac et al. 2026c) because the increased number of observed events as well as limited computing re-

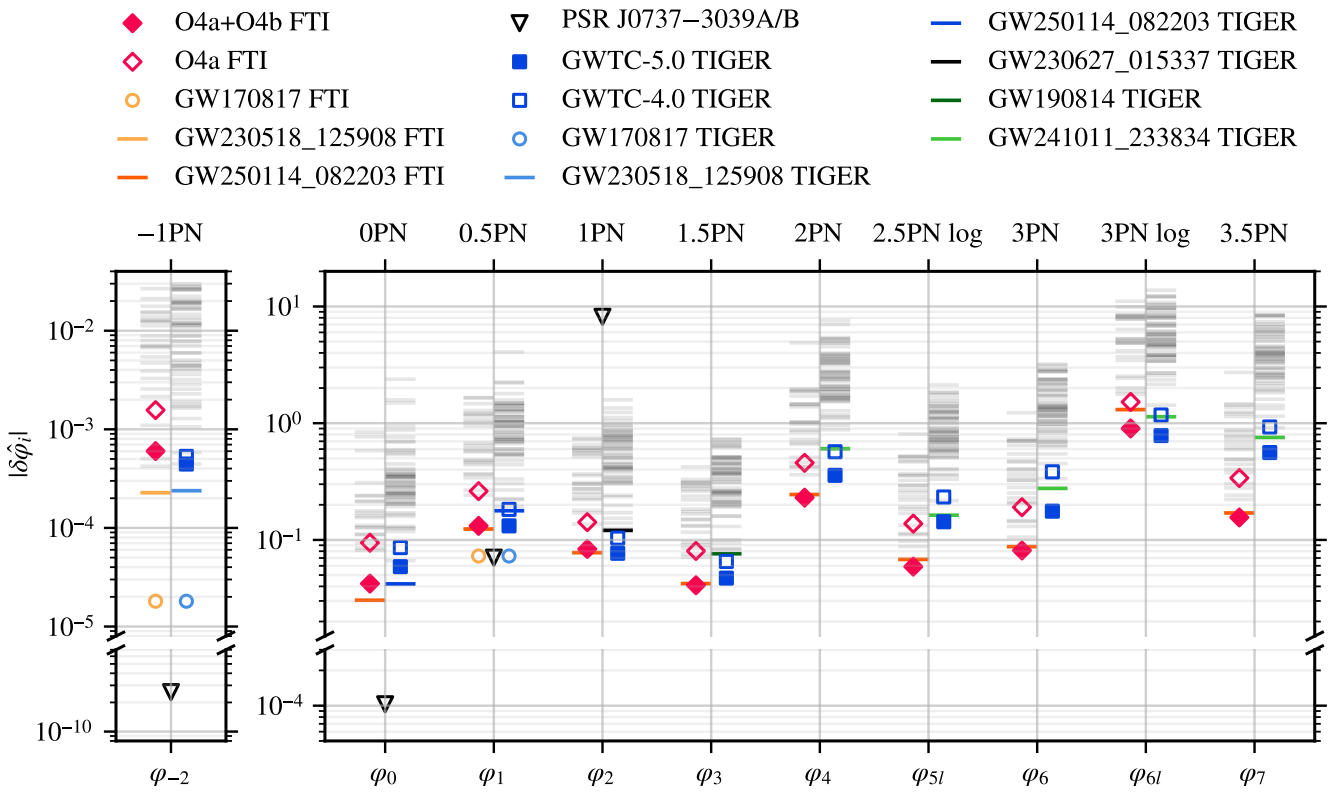


Figure 5. Upper limits on the absolute values of the inspiral deviation coefficients $|\delta\hat{\varphi}_i|$ from −1PN to 3.5PN. Shown are 90% upper bounds obtained from selected individual GW events and from combined analyses. The horizontal gray markers indicate bounds from individual O4a and O4b events analyzed with FTI using the SEOBNRv5HM_ROM waveform model on the left and from individual GWTC-5.0 events analyzed with TIGER using the IMRPhenomXPHM_SpinTaylor waveform model on the right. The filled red diamonds show the hierarchically combined FTI constraints from the combined O4a+O4b analysis, while the unfilled red diamonds correspond to the combined FTI constraints from only the O4a events (Abac et al. 2026c). For comparison, the filled (unfilled) blue squares denote the hierarchically combined bounds from GWTC-5.0 (GWTC-4.0) obtained with TIGER. The colored bars highlight specific events when they provide the most stringent individual-event constraint at a given PN order for FTI or TIGER. The bounds for the NSBH event GW230518_125908 were obtained using the SEOBNRv4_ROM_NRTidalv2_NSBH waveform model for FTI and the IMRPhenomXPHM_SpinTaylor waveform model for TIGER (Abac et al. 2026c). The orange circles show constraints obtained from the BNS event GW170817 using the SEOBNRv4_ROM_NRTidalv2 waveform model when competitive, which are not included in the combined bounds shown (Abbott et al. 2019a). The black inverted triangles indicate constraints from the double pulsar PSR J0737−3039A/B (Kramer et al. 2021). For PN orders ≥ 1.5PN, the pulsar bounds are significantly weaker than those from GW observations and are therefore not shown.

sources meant that we had to be more selective in choosing the events to analyze. We checked that these stricter selection criteria still give us events whose masses roughly span the range of masses of the full set of O4b events meeting the less strict selection criterion used in GWTC-4.0. Unless otherwise specified, we apply corrections to frequencies from 20 Hz up to a tapering frequency equal to the peak frequency of the dominant $(2, 2)$ multipole. Since the bounds from the FTI test depend on the tapering frequency, results obtained with this setup are directly comparable to those from O4a (Abac et al. 2026c), GW230529_181500 (Sänger et al. 2026), GW230814_230901 (Abac et al. 2026e), GW250114 (Abac et al. 2026f), GW240925_005809 and GW250207_115645 (Abac et al. 2026g), but not to earlier analyses that employed different tapering prescriptions (Abbott et al. 2016b, 2019a,b, 2021b, 2025). The first two events, being observed only by one detector, do not satisfy our event selection criteria, which require that the signal be coherent across at least two detectors. The last event was observed during the third part of the fourth observing run (O4c) and is therefore not part of GWTC-5.0 so it is not included in combined results in this work. GW250207_115645 provided some of the best constraints on the higher-order PN coefficients to date due to its high SNR combined with favorable source properties.

The SNR used for FTI event selection is computed from the inspiral portion of the signal using the best-fit parameters from the GR analysis performed with the IMRPhenomXPHM_SpinTaylor waveform model, following the GWTC-4.0 methodology. Consistent with GWTC-4.0 analyses, we adopt

flat, broad priors of $\delta\hat{\varphi}_{-2} \in [-1, 1]$ and $\delta\hat{\varphi}_i \in [-20, 20]$ for all other parameters. When presenting combined or hierarchical FTI results, we report constraints obtained jointly from events analyzed in O4a and O4b. This is motivated by the use of a consistent FTI setup and waveform modeling across both observing runs, which allows the results to be meaningfully combined.

In the TIGER analysis, we impose an SNR threshold of greater than 15 in the inspiral part of the signal to select an event for testing inspiral phasing coefficients. This threshold is higher than what was used in GWTC-4.0 (Abac et al. 2026c) due to limited computing resources combined with an increased number of observed events. We checked that these stricter selection criteria still give us events whose masses broadly cover the mass range of the full set of O4b events meeting the less strict selection criterion used in GWTC-4.0, but with more concentration at lower masses. Following the prescription of the IMRPhenomXAS model, we use the $(2, 2)$ multipole frequency at the minimum energy circular orbit (MECO) to divide the waveform between the inspiral and post-inspiral regimes. Small corrections are incorporated in IMRPhenomXAS to obtain the exact end-of-inspiral frequency, but their impact is negligible (Abac et al. 2026c). The median SNR values are computed using posterior samples obtained from standard GR analyses with the IMRPhenomXPHM_SpinTaylor model. As in the GWTC-4.0 analyses, we adopt flat, broad prior ranges: $\delta\hat{\varphi}_{-2} \in [-0.1, 0.1]$, $\delta\hat{\varphi}_{0,1,2,3} \in [-5, 5]$, $\delta\hat{\varphi}_{4,5\ell,6} \in [-20, 20]$, and $\delta\hat{\varphi}_{6\ell,7} \in [-30, 30]$. When presenting combined or hierarchical TIGER results, we report constraints obtained jointly from all events analyzed in GWTC-5.0.

The results for the inspiral deviation parameters $\delta\hat{\varphi}_i$ are consistent with GR for all events analyzed for both FTI and TIGER. When reviewing results, we investigated more closely all results where there are strong multimodalities or GR is outside the 90% credible interval. For FTI, we also investigated the impact of unmodelled spin-precession on the results for events with precessing spins. When we do not trust the results for an event to be accurate, we exclude the event from any combined results. For two events, namely GW240621_195059 and GW241011_233834, we were unable to obtain well converged results for FTI.

For GW240621_195059 the posteriors are multimodal and show signs of not being well converged for all deviation parameters. While there were no obvious data quality issues observed for GW240621_195059, we suspect that the bimodality for this event is caused by minor non-Gaussian transients. Since this event has a relatively short duration, its analyses are more susceptible to data-quality issues causing multimodalities or biased results. This event is excluded when combining results from multiple events because of this. This event does not pass the selection criteria for TIGER due to its short inspiral so it was not analyzed by TIGER.

GW241011_233834 has precessing spins with $\chi_{\rm p} = 0.38^{+0.13}_{-0.12}$ and a relatively high SNR of $35.8^{+0.1}_{-0.1}$ (Abac et al. 2025b), which causes the FTI results to be biased since it does not model spin-precession. The posteriors are also multimodal and showing signs of not being well converged for some of the deviation parameters. We verified that spin-precession is indeed causing the biases by simulating a GR signal with source parameters consistent with those observed for this event, for which FTI observes similar biases. This event is excluded from the FTI results when combining multiple events because of the observed biases and convergence issues. Further studies are needed to fully understand possible biases from including or excluding such spin-precessing events when combining FTI results from multiple events.

The TIGER analysis uses a waveform model that includes spin precession, and therefore is not affected by the same spin-precession related bias seen in the FTI analysis. For GW241011_233834, TIGER finds all inspiral deviation parameters to be consistent with GR. This event provides the strongest bounds in higher PN coefficients among the TIGER analyses in GWTC-5.0. The event has a positive effective inspiral spin, $\chi_{\rm eff} = 0.49^{+0.06}_{-0.07}$, which allows the binary to remain in the inspiral regime up to higher orbital velocities compared to a non-spinning binary. This follows from the spin contributions to the orbital frequency evolution, including spin–orbit and spin–spin couplings, which modify the time spent in band and the accumulated phase (Kidder et al. 1993). In particular, a positive $\chi_{\rm eff}$ increases the signal duration relative to a non-spinning binary from the same reference frequency. As a result, higher-PN-order terms make a larger contribution to the accumulated waveform phase. Spin effects also contribute to the phasing, with the leading spin–orbit terms entering at 1.5PN order and the leading spin-spin terms entering at 2PN order. The combination of high SNR, positive effective spin, and measurable higher-order multipole moment content helps reduce degeneracies between the binary parameters and the inspiral deviation parameters, leading to stronger constraints, particularly at higher PN orders. Evidence for spin precession in this event can also contribute to this degeneracy breaking, although we do not separately quantify its impact here.

Overlapping the time of GW241127_061008, the LIGO Livingston detector was affected by a glitch which particularly affects the strain data below 30 Hz for this event. While it does not significantly affect the baseline GR analysis, parameterized tests such as FTI, which introduce additional degrees of freedom, can potentially absorb such features, leading to apparent deviations or multimodal posteriors (Kwok et al. 2022). We therefore analyzed this event with an increased minimum frequency of 50 Hz for the LIGO Livingston detector. In addition, GW241127_061008 includes significant imprints of precession dynamics with $\chi_{\rm p} = 0.57^{+0.14}_{-0.21}$ (Abac et al. 2026i), which could bias the FTI analysis. We verified that the precession is not causing biases in this case by simulating a GR signal with source parameters consistent with those observed for this event, for which unbiased posteriors were obtained when analyzing with FTI. Since no issues were observed with the FTI results for this event we do include it in combined results. TIGER did not analyze this event since it does not pass the event selection criteria due to its high masses.

The combined posteriors for the inspiral deviation parameters are shown in Figure 4. We use two ways of combining the

events. The first uses a restricted approach assuming all events share a common value of the deviation parameter, shown with unfilled violins. The second approach uses a hierarchical analysis leading to the combined posteriors shown with filled violins. We find that the hyperparameters of the hierarchical analyses are all consistent with GR. Each is consistent with GR at the 90% credible level across all parameters. The combined posteriors are generally dominated by the events giving the tightest individual constraints.

In Figure 5, we show the 90% upper bounds on the magnitude of the PN deviation parameters in the inspiral. The bounds for individual events obtained with FTI are shown as grey stripes. We highlight the events GW250114 and GW250207_115645, which are the BBH signals that give the best constraints due to their relatively high SNRs. We also show the bounds for the BNS GW170817 (Abbott et al. 2019a) when they are competitive. We also show the hierarchically combined bounds for both FTI and TIGER. For comparison, we also include the combined FTI and TIGER results from O4a and GWTC-4.0, respectively (Abac et al. 2026c). The list of events included in the combined bounds is different between FTI and TIGER, making it impossible to directly compare the combined bounds between the two. The differences between the bounds for individual events from FTI and TIGER should be seen as an estimate for the systematic uncertainties that can be expected from inspiral tests, with FTI pushing the transition from inspiral to post-inspiral to the highest possible frequency and TIGER providing more conservative bounds. The best bound from GWs at −1PN is still from GW170817, and is an order of magnitude better than the combined constraints.

It is also possible to put constraints on the PN deviation parameters using binary pulsars (Yunes & Hughes 2010; Nair & Yunes 2020). The best overall bound on dipole radiation comes from the double pulsar PSR J0737−3039A/B (Kramer et al. 2021), which also puts the most stringent constraint on deviations from GR at Newtonian order. At 0.5PN, the double pulsar bound is on par with the bound from GW170817. From 1PN onwards the bounds from the double pulsar are less constraining than the bounds from GWs. The bounds from PSR J0737−3039A/B are shown as black upside-down triangles in Figure 5.

A useful framework for interpreting agnostic PN deformation tests is the parameterized post-Einsteinian formalism, which provides a systematic description of leading deviations from general relativity in the gravitational waveform phase and amplitude (Yunes & Pretorius 2009; Cornish et al. 2011). In this approach, deviations are introduced as power-law corrections at specific PN orders, allowing one to connect agnostic constraints on PN coefficients to a broad class of modified gravity theories.

To provide intuition for the meaning of the agnostic constraints on deviations in the PN coefficients we have obtained, we thus follow the ppE formalism and perform event-by-event illustrative mappings of the inferred PN deformation posteriors onto parameters appearing in selected modified gravity theories, with the results given in Table 7. The example theories considered are: scalar–tensor theories with an evolving scalar field (Jacobson 1999; Horbatsch & Burgess 2012), Einstein–dilaton–Gauss–Bonnet (EdGB; Gross & Sloan 1987; Kanti et al. 1996), dynamical Chern–Simons (dCS; Alexander & Yunes 2009), and Einstein–Maxwell with BH charge.

These mappings should not be interpreted as theory-specific statistical constraints, but rather as illustrative translations of agnostic deviations into the language of particular models, obtained under simplifying assumptions (e.g., leading-order modifications and no deviations in the merger–ringdown phase).

Each mapping is performed independently for each GW event using the corresponding posterior samples of both intrinsic binary parameters and the agnostic PN deformation parameter, with consistent reweighting to account for prior differences.

In order to map a PN parameter constraint to a parameter in a given theory, we map use the posteriors of the PN coefficient constraint along with the intrinsic parameters for each analyzed O4b event and the PN expressions derived to leading order in Yagi et al. (2012a) for EdGB, Yagi et al. (2012a,b) for dCS gravity, Khalil et al. (2018) for Einstein–Maxwell with BH charge, and Horbatsch & Burgess (2012); Yunes et al. (2016) for scalar–tensor theories with an evolving scalar field.

The mapping on theories we quote in Table 7 are the 90% upper bound for the GW event from O4b that gives the tightest translation on a given parameter. For comparison, we also provide existing bounds on each theory. Current constraints on scalar–tensor theories come from observations of the quasar OJ 287 (Horbatsch & Burgess 2012). For EdGB gravity and Einstein–Maxwell, the strongest bound arises from the low-mass, long-inspiral signal GW230529_181500 (Sänger et al. 2026). For dCS gravity, the current bound of 8.5 km comes from combining NICER observations of a pulsar with the NS tidal deformability inferred from the binary NS event GW170817 (Silva et al. 2021).

The expression from Horbatsch & Burgess (2012) used to obtain the translation to the evolution of the scalar field in the scalar–tensor case requires $\epsilon_{\rm ST} = 2Gm_1|\dot{\phi}|/c^3 \ll 1$. While $|\dot{\phi}|$ in our translation is large in units of s^{-1}, the 90% upper bound on $\epsilon_{\rm ST}$ is 0.3 and 0.88 for TIGER and FTI respectively, which are a bit larger than the value of 0.13 for the primary of OJ 287 used to obtain the constraints in Horbatsch & Burgess (2012). The much better constraints on $|\dot{\phi}|$ from OJ 287 are due to its much larger total mass (its primary's mass is $\sim 10^{10} M_\odot$). Additionally, the assumptions made in obtaining the frequency-domain dephasing used in the translation assume that the dipole radiation is a small perturbation to the quadrupole radiation (as discussed in, e.g., Yunes & Pretorius 2009), but this is also the case here: The 90% upper bounds on the ratio of the leading dipole to quadrupole energy fluxes are 0.23 and 0.41 for TIGER and FTI at the low-frequency cutoff of 20 Hz and decrease at higher frequencies.

Similarly, the PN expressions used for dCS and EdGB gravity are derived under the perturbative assumption that the dimensionless coupling $\zeta_{\rm dCS,EdGB} =$

Table 7. Event-by-event illustrative mappings of agnostic constraints on PN coefficients onto parameters of selected modified gravity theories. (Section 6).

Theory	φ_i	Parameter	FTI mapping	FTI event	TIGER mapping	TIGER event	Existing constraint
Scalar–tensor	φ_{-2}	$\lvert\dot{\phi}\rvert$ [s^{-1}]	1.5×10^{3}	GW241127_061008	1.6×10^{3}	GW241011_233834	10^{-6} (Horbatsch & Burgess 2012)
Einstein–dilaton–Gauss–Bonnet	φ_{-2}	$\sqrt{\alpha_{\rm EdGB}}$ [km]	12.1	GW241127_061008	1.32	GW241011_233834	0.28 (Sänger et al. 2026)
Einstein–Maxwell	φ_{-2}	$\zeta=\left\lvert\frac{q_1}{m_1}-\frac{q_2}{m_2}\right\rvert$	0.080	GW240925_005809	0.086	GW240925_005809	0.024 (Gao et al. 2024)

Note— The values shown are event-by-event mappings of posteriors on agnostic PN deformation parameters (from FTI and TIGER analyses) into 90% credibility results for parameters in modified gravity theories. These mappings are performed independently for each event using posterior samples for the intrinsic binary parameters along with the agnostic PN deformation parameter, and includes consistent prior reweighting. The results therefore reflect how agnostic constraints project onto theory parameters under simplifying assumptions, rather than a joint or theory-specific inference. Specifically the binary parameters used are those of the events indicated in the table. These values should not be interpreted as statistical constraints on the underlying theories, nor as exclusions. In particular, real modified gravity theories generically induce correlated deviations across multiple PN coefficients and modify the merger–ringdown signal, effects not included in this mapping. The table is therefore intended only to provide intuition about the agnostic PN bounds we have obtained by showing how they would translate to the parameters of some commonly considered alternative theories, together with constraints on those parameters from the literature.

$16\pi\alpha^2_{\rm dCS/EdGB}/\left(Gm_2/c^2\right)^4 \ll 1$ throughout the inspiral (Yagi et al. 2012a), where we have evaluated this quantity using m_2, since that sets the smallest length scale in the binary. Evaluating this condition for EdGB gravity first, we find that although GW241102_124058 gives the tightest value for the coupling of theory, it does not satisfy the small coupling approximation. Therefore we quote the value obtained by mapping GW241127_061008 which does satisfy the small-coupling requirement within the 90% credible region with a value for $\zeta_{\rm EdGB}$ of 0.25. The dCS translation however does not satisfy the condition for any GW event.

Indeed, the dCS mapping gives a value ~ 3 times larger than the best current bounds on the theory (Silva et al. 2021), but the 90% upper bound on small-coupling parameter is $O(10^3)$. As a result, the perturbative waveform is not self-consistent for the systems considered, and any inferred PN-to-theory translation for dCS gravity would lie outside the regime of validity of the effective field theory expansion. We therefore do not report a dCS entry in Table 7. In order to obtain constraints for dCS that are within small-coupling approximation from GW observations, one would need a low-mass binary with at least one rapidly spinning object.

The values given in Table 7 are for the purposes of illustration, and there are several caveats to them being directly taken as constraints on the specific theories. To begin with, a given modified gravity theory will give deviations in multiple PN parameters, while the analyses consider each PN parameter deviation in isolation. In addition, these analyses make use of the merger–ringdown signal, but without taking into account how a specific theory would modify that portion of the GW signal from GR. For instance, Johnson-McDaniel et al. (2022) analyzes phenomenological non-GR signals with FTI and TIGER, finding that these methods can yield constraints on a PN deviation parameter that are much smaller than the true deviation. Thus, these mappings will not be equivalent to those obtained with an analysis that assumes a specific theory.

7. RINGDOWN TESTS

The ringdown-based pipelines presenting results in this section are RINGDOWN (Section 7.1), PSEOBNRV5PHM (Section 7.2), and QNMRF (Section 7.3). The various analyses probe the nature of the spacetimes of remnant BHs born from BBH mergers. The perturbed remnant radiates away its excess energy in the form of GWs during the ringdown (post-inspiral–merger) phase to reach a stable Kerr solution (Kerr 1963; Press & Teukolsky 1973; Teukolsky 1973; Whiting 1989; Buonanno et al. 2007). In the late stages of ringdown, the dynamics of the remnant's spacetime are governed by linear perturbations, such that the emitted gravitational radiation takes the form of superposed exponentially damped sinusoids called QNMs when projected onto a basis of spin-weighted spheroidal harmonics (Vishveshwara 1970; Teukolsky 1973; Cunningham et al. 1978; Chandrasekhar 1983; Berti et al. 2026). The RINGDOWN and QNMRF analyses adhere to this perturbation theory-based QNM template for post-merger signals, while PSEOBNRV5PHM models the entire signal with the transitions between pre- and post-merger portions of the template informed by numerical relativity (NR). RINGDOWN and QNMRF employ time-domain likelihoods (though the latter first applies frequency-domain filters to the data), while PSEOBNRV5PHM operates in the frequency domain.

The complex spectrum of QNMs is determined solely by the mass and spin of the remnant BH (Penrose 1969; Carter 1971; Hawking 1972; Robinson 1975; Penrose 2002); we do not expect astrophysical BHs to be charged (Wald 1974; Gibbons 1975; Blandford & Znajek 1977), and charges have a demonstrably negligible effect in the ringdown for present detector sensitivities (Carullo et al. 2022; Gu et al. 2024). Consequently, resolving a single mode in the ringdown signals of detected events with QNM models like those of RINGDOWN and QNMRF permits independent measurements of remnant properties that can corroborate parameter estimates from IMR analyses. Furthermore, through BH spectroscopy, multiple resolved modes in a ringdown signal provide a direct probe of the Kerr prediction for the nature of remnant BHs (Detweiler 1980; Dreyer et al. 2004; Berti et al. 2006, 2026). A complementary test is that of PSEOBNRV5PHM, which is itself a coherent IMR analysis that constrains parameterized deformations of the Kerr QNM spectrum within an NR-calibrated mapping between remnant and binary properties.

The reliability of GR tests with BH spectroscopy hinges on the assumption that the ringdown is consistent with lin-

ear perturbations on a Kerr background, and that a sum of QNMs with constant amplitudes is an appropriate description of the signal. This asymptotic regime is preceded by merger-dependent initial conditions for QNM excitations (Berti & Cardoso 2006; Albanesi et al. 2023; Lagos & Hui 2023; Chavda et al. 2025; De Amicis et al. 2026), nonlinear dynamics (Gleiser et al. 1996; London et al. 2014; Bhagwat et al. 2018; Sberna et al. 2022; Baibhav et al. 2023; Bucciotti et al. 2023; Cheung et al. 2023; Lagos & Hui 2023; Mitman et al. 2023; Bucciotti et al. 2024; Ma & Yang 2024; Perrone et al. 2024; Redondo-Yuste et al. 2024a; Bourg et al. 2025), and remnant relaxation (Sberna et al. 2022; Capuano et al. 2024; May et al. 2024; Redondo-Yuste et al. 2024b; Zhu et al. 2024); QNM templates do not include these effects and thus could bias analysis results if applied too early in the post-merger phase of the signal. For this reason, ringdown-only analyses like RINGDOWN and QNMRF perform scans over analysis start times to determine the onset of measured linear QNM amplitudes decaying self-consistently. Because pSEOBNRv5PHM models the entire signal, the analysis bypasses the matter of start time validity up to the uncertainty of the merger-ringdown matching time, which has a negligible impact on measured parameters for even the loudest signals observed to date (Maggio et al. 2023; Pompili et al. 2025a; Grimaldi et al. 2026).

We label the spectrum of linear regime ringdown QNMs by a set of indices (ℓ, m, n). The (ℓ, m) indices of the harmonics capture the angular structure of the emission, while the additional index n orders the discrete solutions to the radial perturbation equations by their damping rates, with $n = 0$ denoting *fundamental* modes and $n > 0$ the corresponding *overtones*. So, we express the GWs emitted during this phase as the following (ignoring counter-rotating modes),

$$h(t) = \sum_{\ell=2}^{\infty} \sum_{m=-\ell}^{\ell} \sum_{n=0}^{\infty} \mathcal{A}_{\ell mn}\, {}_{-2}S_{\ell mn}(\iota, \varphi, \chi_{\rm f}) \times \exp\left\{-\frac{t}{(1+z)\tau_{\ell mn}}\right\} \exp\left\{\frac{2\pi i f_{\ell mn} t}{(1+z)}\right\}, \quad (14)$$

where $f_{\ell mn}$ and $1/\tau_{\ell mn}$ are the real and imaginary parts of the complex QNM frequency, $\mathcal{A}_{\ell mn}$ is the QNM amplitude, ${}_{-2}S_{\ell mn}$ are the spin-weighted spheroidal harmonic basis functions (Teukolsky 1973) evaluated at the QNM frequency, and ι and φ are the inclination and azimuthal angles, respectively, relative to the final spin of the remnant $\chi_{\rm f}$. IMR templates typically model the full waveform on a spin-weighted spherical harmonic basis (Gelfand et al. 1958; Newman & Penrose 1966; Creighton & Anderson 2011), resulting in mode mixing in the ringdown due to the basis mismatch (Kelly & Baker 2013; Berti & Klein 2014). This effect is handled in pSEOBNRv5PHM with explicitly computed mixing coefficients (Berti & Klein 2014; Pompili et al. 2025b), and it is not a factor in analyses like RINGDOWN and QNMRF whose templates only leverage complex QNM frequencies without modeling the angular structure of the signal.

RINGDOWN and pSEOBNRv5PHM are able to constrain the Kerr nature of remnants by applying deviations to the QNM spectrum. Both RINGDOWN and QNMRF find evidence for a sub-dominant mode present in the post-merger signal of one event, GW240621_195059. The pSEOBNRv5PHM constraints on Kerr deviations for this event are also among the tightest of the GWTC-5.0 events, with even more stringent constraints reported for the loudest events in O4b.

7.1. *The* RINGDOWN *analysis*

The RINGDOWN (Isi et al. 2019b; Isi & Farr 2021; Siegel et al. 2025) pipeline carries out Bayesian, time-domain analyses of the postmerger portions of GW signals detected from coalescences of BBHs to infer remnant BH properties through measurements of QNMs and their properties. This analysis permits spectroscopic tests of the Kerr nature of the remnant if more than one ringdown mode is resolvable in a given signal.

We model the postmerger signal as a sum of elliptically polarized damped sinusoidal modes that are each comprised of contributions from a pair of circularly polarized $\pm m$ QNMs (Isi 2023). We express the polarizations of each such ringdown mode as the following:

$$h^{+}_{\ell|m|n}(t) = A_{\ell|m|n}\Big[\cos\theta_{\ell|m|n}\cos(2\pi f_{\ell|m|n}t - \phi_{\ell|m|n}) - \epsilon_{\ell|m|n}\sin\theta_{\ell|m|n}\sin(2\pi f_{\ell|m|n}t - \phi_{\ell|m|n})\Big]e^{-\gamma_{\ell|m|n}t} \quad (15a)$$

$$h^{\times}_{\ell|m|n}(t) = A_{\ell|m|n}\Big[\sin\theta_{\ell|m|n}\cos(2\pi f_{\ell|m|n}t - \phi_{\ell|m|n}) + \epsilon_{\ell|m|n}\cos\theta_{\ell|m|n}\sin(2\pi f_{\ell|m|n}t - \phi_{\ell|m|n})\Big]e^{-\gamma_{\ell|m|n}t}\,. \quad (15b)$$

RINGDOWN also offers a QNM template that restores the angular harmonic factors and implements equatorial reflection symmetry constraints, which is applicable for non-precessing systems and particularly useful for inferring their signals' polarization structure (Khusid et al. 2026). However, to avoid assumptions of source configurations, we proceed with the template given in Equation (15). We do not directly sample in the amplitude $A_{\ell|m|n}$, polarization ellipticity $\epsilon_{\ell|m|n}$, polarization angle $\theta_{\ell|m|n}$, and phase $\phi_{\ell|m|n}$ of each ringdown mode; rather, we sample the linear quadrature coefficients on the sinusoidal terms in the above expressions from Gaussian priors and transform them back into the overall ringdown mode properties (Isi & Farr 2021; Isi 2023). The effective priors are flat in $A_{\ell|m|n}$, isotropic in phase and polarization angle, and disfavoring $\epsilon_{\ell|m|n} = \pm 1$. We also incorporate Kerr assumptions of the QNM spectrum by fixing the mode frequencies $f_{\ell|m|n}$ and damping rates $\gamma_{\ell|m|n} = 1/\tau_{\ell|m|n}$ to be functions of the detector-frame final mass $(1+z)M_{\rm f}$ and final spin $\chi_{\rm f}$ of the remnant, on which we place flat priors. In the event that multiple ringdown modes are detected in a given postmerger signal, we can test the Kerr nature of the remnant by extending this model to include deviations on the sub-dominant mode properties,

$$f_{\ell|m|n} = f^{\rm Kerr}_{\ell|m|n} e^{\delta f_{\ell|m|n}}, \quad \gamma_{\ell|m|n} = \gamma^{\rm Kerr}_{\ell|m|n} e^{\delta\gamma_{\ell|m|n}}\,. \quad (16)$$

While the dominant (2,2,0) mode fixes the final mass and spin of the remnant, the sub-dominant mode is free to depart

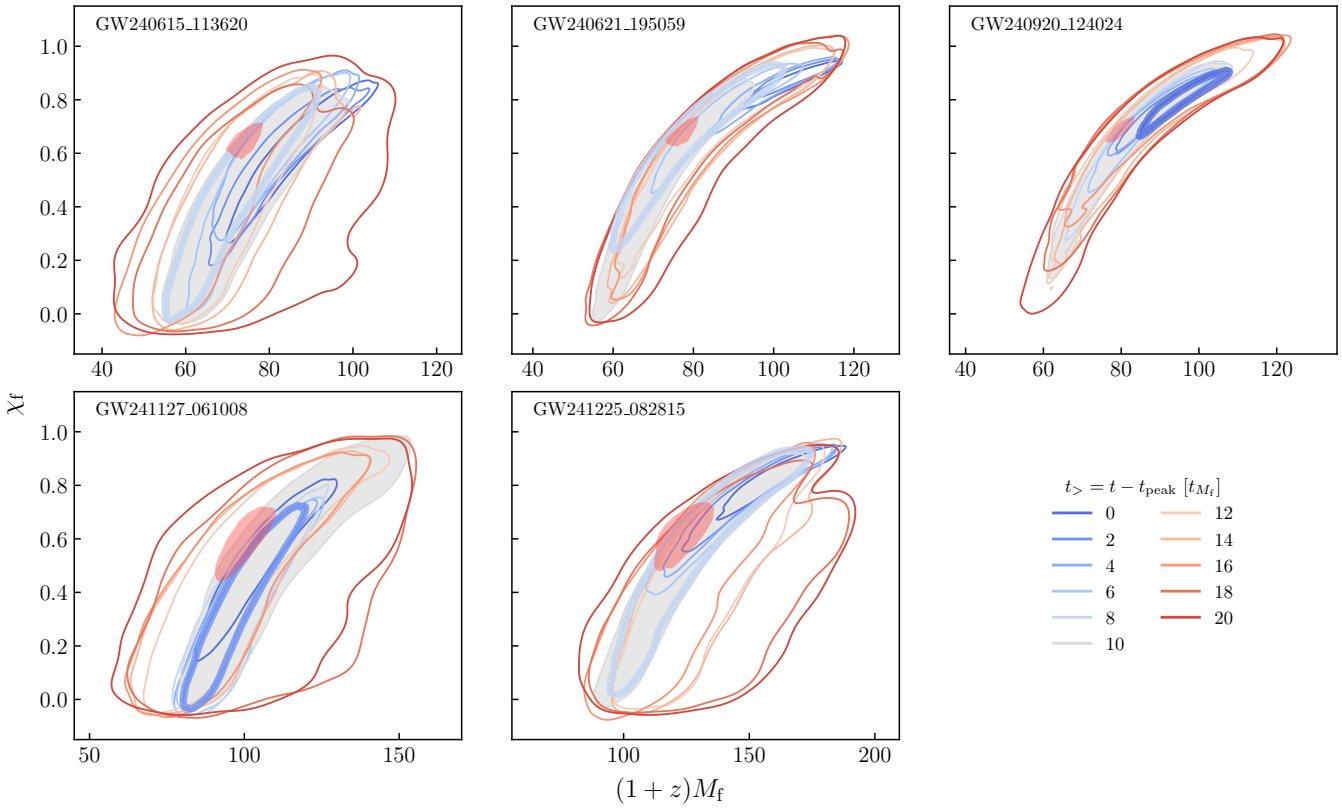


Figure 6. Joint posteriors on detector-frame remnant final mass $(1+z)M_f$ and remnant final spin χ_f from analyses by the RINGDOWN pipeline of the postmerger signal of selected events using the Kerr (2,2,0) QNM model. All curves shown are 90% credible contours. The outlined contours correspond to various analysis start times with respect to the peak of each signal, with the gray posterior filled to indicate the $t_> = 10t_{M_f}$ analysis result. Additionally, one of the contours per event is thickened to indicate $t_>^{\rm ref}$, the earliest time of self-consistent posteriors according to expected amplitude decay. The filled red contour is the posterior on remnant final mass and spin from the full-signal analysis using NRSUR7DQ4. For late analysis start times, the remnant properties inferred using a single damped sinusoid QNM template are consistent with those of the IMR analysis.

from the Kerr prediction. We place flat priors on the deviation parameters, $\mathcal{U}(-0.8, 0.8)$, by default and extend the range as necessary in cases of railing posteriors.

The analysis takes as input a handful of parameters drawn from the posterior distribution of a full-signal analysis. These include the geocenter time corresponding to the time of coalescence as defined by NRSUR7DQ4 (Varma et al. 2019) to determine the reference start time of the analysis; $(1+z)M_f$ to map time in seconds to the natural timescale of the perturbed remnant's oscillations ($t_{M_f} = G(1+z)M_f/c^3$); and sky location parameters right ascension (α), declination (δ), and polarization angle (ψ, which we fix because it is degenerate with $\theta_{\ell|m|n}$) to compute the time delay of the analysis segment in each interferometer's data stream and determine the antenna patterns. We omit information prior to the analysis start time by truncating the strain data and excluding those earlier samples from likelihood evaluation.

Our selection criterion for analysis intends for a given signal to have a great enough post-peak optimal SNR, $\rho_{\rm pp}$, to resolve at least one long-lived mode and an additional sub-dominant mode in the signal. To determine which events meet this threshold, we compute the optimal SNR of the truncated NRSUR7DQ4 waveform constructed from 500 posterior sample draws, and we choose the events for which the median $\rho_{\rm pp}$ is 10 or greater. This particular threshold is motivated by ringdown analyses of GW150914 (Abbott et al. 2016a,e), which find varying degrees of support for an identifiable overtone in the postmerger signal (Giesler et al. 2019; Isi et al. 2019b; Cotesta et al. 2022; Finch & Moore 2022; Carullo et al. 2023; Isi & Farr 2023; Ma et al. 2023a; Correia et al. 2024); for a summary of these works, see Berti et al. (2026). For all selected events, we analyze their postmerger signals with the (2,2,0) and (2,2,0)+(2,2,1) Kerr and beyond-Kerr models from the peak until late times. Systems that are consistent with particular configurations based on IMR parameter estimation qualify for analyses with additional physically motivated mode combinations; precession can amplify other $\ell = 2$ fundamental QNMs relative to the typically dominant (2,2,0) (Zhu et al. 2025), and the resolvability of higher-order modes in the ringdown is more optimistic for unequal mass ratio systems (Berti et al. 2007; Jiménez Forteza et al. 2020). So, for systems with at least one component spin inferred to be misaligned from the system's orbital angular momentum at the 95% credible level (CL),

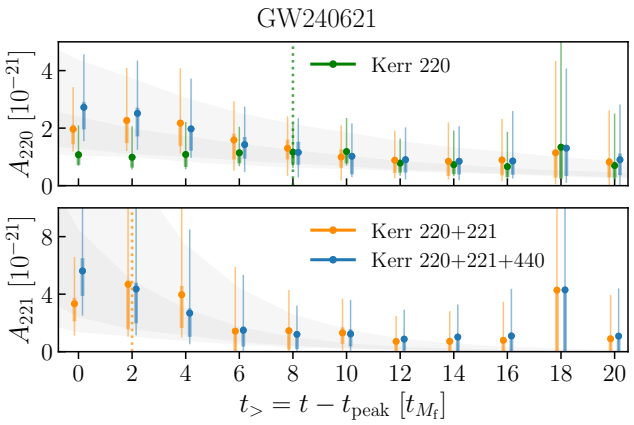


Figure 7. QNM amplitude posteriors from analyses of the postmerger signal of GW240621_195059 using both default Kerr (2,2,0) and (2,2,0)+(2,2,1) models shown in green and orange, respectively, as well as an additional Kerr three-mode model including the (4,4,0) mode shown in blue. In both panels, the darker and lighter gray bands correspond to the 50% and 90% credible regions, respectively, of the inferred QNM amplitudes projected forward and back from the reference $t_>$ indicated by the vertical dotted lines. Each marker is the median of the posterior distribution, the thicker vertical bars are 50% credible intervals, and the thinner vertical bars are the 90% credible intervals for each analysis start time. All credible regions are highest posterior density intervals. *Top:* The posteriors on the amplitude of the (2,2,0) mode. *Bottom:* The posteriors on the amplitude of the (2,2,1) mode from analyses with the multi-mode models.

we perform additional analyses with the (2,2,0)+(2,1,0) Kerr and beyond-Kerr models. Finally, for systems inferred to have unequal masses at the 95% CL, we perform additional analyses with the (2,2,0)+(3,3,0) models. Analysis start times are expressed with respect to the peak, $t_> = t - t_{\rm peak}$, and all inference is performed over a range $t_> \in [0, 20]t_{M_{\rm f}}$ in increments of $2t_{M_{\rm f}}$. The events selected for analysis in addition to the special event GW250114 (Abac et al. 2025c, 2026f) are GW240615_113620, GW240621_195059, GW240920_124024, GW241127_061008, and GW241225_082815. In the following results, we discuss measurements of $(1+z)M_{\rm f}$ and $\chi_{\rm f}$, $A_{\ell|m|n}$, $\delta f_{\ell|m|n}$, and $\delta\gamma_{\ell|m|n}$.

We expect most postmerger signals to be consistent at late times with a single damped sinusoid, typically the quadrupolar fundamental (2,2,0) mode. Analyzing the selected signals over a grid of start times with a Kerr model consisting of this single ringdown mode, we identify such a late-time behavior by determining the earliest analysis start time, $t_>^{\rm ref}$, for which we subsequently make self-consistent measurements of the (2,2,0) amplitude. Here, self-consistency means the amplitude posteriors from inferences beginning at $t_> > t_>^{\rm ref}$ match the projected exponential decay $A_{\ell|m|n}(t_>) = A_{\ell|m|n}(t_>^{\rm ref})\exp\{-\gamma_{\ell|m|n}(t_> - t_>^{\rm ref})\}$ at the 90% CL. For these same ranges of times, we compare the $(1+z)M_{\rm f}$ and $\chi_{\rm f}$ posteriors independently inferred from the QNM analyses to the remnant BH properties measured by the IMR analyses.

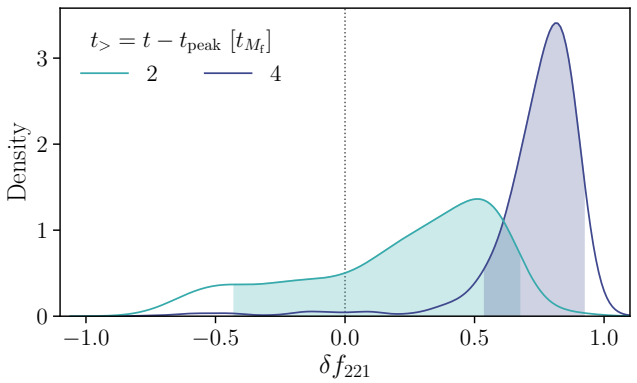


Figure 8. Posteriors on the frequency deviation to the (2,2,1) mode from the beyond-Kerr analysis of GW240621_195059, at start times for which the self-consistent Kerr two-mode model posteriors show evidence of a sub-dominant mode detection. The filled regions under the posteriors correspond to 90% credible intervals, which at $t_> = 4t_{M_{\rm f}}$ excludes $\delta f_{221} = 0$.

Figure 6 summarizes the inference results for each of the selected events' postmerger signals using the Kerr (2,2,0) QNM model. We show the joint posteriors on $(1+z)M_{\rm f}$ and $\chi_{\rm f}$ from the QNM analyses as outlined contours. We fill the 90% credible contour for the analyses beginning at $t_> = 10t_{M_{\rm f}}$, which all agree with the IMR measurements at the ~90% CL. For reference, the postmerger signal of GW250114 is shown to be consistent with a single long-lived damped sinusoid at $t_> = 10.5t_{M_{\rm f}}$ (Abac et al. 2025c, 2026f), and the events selected here have smaller $\rho_{\rm pp}$. We also plot a thicker contour to indicate the $t_>^{\rm ref}$ for which inferences within the scan over start time produce self-consistent posteriors according to the expected decay of the (2,2,0) amplitude.

At early $t_>$ near the peak estimates (varying blue contours), we see biased measurements of the remnant BH properties as compared to those of the IMR analysis. This effect may be due to physical content unaccounted for by our fixed-amplitude QNM template—additional sub-dominant ringdown modes excluded from the model, non-QNM contributions arising in linear BH pertubration theory (Arnaudo et al. 2026; Küchler et al. 2026; Ma 2026; Su et al. 2026), and other transient features discussed in Section 7. For all events, $t_>^{\rm ref} < 10t_{M_{\rm f}}$, and these times also mark the onset of self-consistent remnant mass and spin measurements after which the posteriors broaden approximately concentrically rather than continuing to systematically shift. The varying red QNM model contours at later $t_>$ overlap with the IMR posteriors at the 90% CL. While the posteriors broaden as the signal decays and the SNR decreases, the damped sinusoid in the QNM template is still confidently identified at these late times. Both self-consistent and IMR-consistent measurements of remnant and mode properties are achieved with a single, long-lived (2,2,0) for each event.

Beyond the single-mode QNM model, we also analyze all events using the Kerr (2,2,0)+(2,2,1) template. For all events but one, the posterior on A_{221} is consistent with zero,

meaning the data do not require the inclusion of an additional damped sinusoid with frequency and damping rate consistent with those of the Kerr (2,2,1) in the model. The exception is GW240621_195059 whose inferred A_{221} peaks confidently away from zero at analysis start times $t_> < t_>^{\rm ref}$ for which the Kerr (2,2,0) model produces biased measurements of the remnant final mass and spin.

In Figure 7, we show the $A_{\ell|m|n}$ posteriors from analyses of GW240621_195059 over a range of start times with both default Kerr QNM models, with green (orange) markers for the single- (two-) mode inference. The measured A_{220} from $t_> \geq 8t_{M_{\rm f}}$ are consistent between the two models and with the extrapolated decay, and A_{221} is consistent with zero for the same range of analysis start times, suggesting that the signal is well-described by a single damped sinusoid over that regime. For earlier times when the (2,2,0) amplitude departs from the $t_>^{\rm ref} = 8t_{M_{\rm f}}$ extrapolation, including the (2,2,1) in the QNM template reconciles the inferred A_{220} posteriors with the backwards-projected trend. Furthermore, for $t_> < 6t_{M_{\rm f}}$ we measure A_{221} to be non-zero at the ≥90% CL. The (2,2,1) turns over from an exponential decay at the peak, but later QNM analyses measure its amplitude in agreement with the extrapolation from $A_{221}(t_> = 2t_{M_{\rm f}}) = 4.68^{+7.84}_{-3.63} \times 10^{-21}$ until it becomes indistinguishable from noise by $t_> = 6t_{M_{\rm f}}$. So, we find support for additional signal content at early $t_>$ that resembles a free-ringing damped sinusoid if we label this mode as the Kerr (2,2,1).

With evidence for appreciable non-(2,2,0) mode content early in the postmerger signal of GW240621_195059 that behaves like the Kerr (2,2,1) mode, we analyze the data at $t_> \in [2, 4]t_{M_{\rm f}}$ with introduced deviations on the (2,2,1) mode properties given by Equation (16) to measure deviations from the Kerr spectrum. Figure 8 shows the resulting δf_{221} marginal distributions ($\delta\gamma_{221}$ is unconstrained for all $t_>$ and thus not shown). While GR is supported within the 90% CL at $t_> = 2t_{M_{\rm f}}$, the $t_> = 4t_{M_{\rm f}}$ posterior exhibits railing against the default upper prior bound, favoring $\ell = 4$ frequencies.

We compare these posteriors against those inferred from the reference (maximum a posteriori) NRSur7dq4 reconstruction injected into zero noise, which yield A_{221} measurements consistent with zero and thus broad and poorly constrained δf_{221} posteriors consistent with GR. We also inject the same signal into ten different realizations of Gaussian noise, of which two give rise to (2,2,1) amplitude measurements that mimic those of the real data analysis. For one of these realizations, the measured deviation posteriors peak around $\delta f_{221} = 0$, while the other gives rise to a GR violation at the 90% CL with posterior support concentrated at negative values of δf_{221}. While statistical fluctuations due to Gaussian noise do capture deviations from Kerr, we cannot reproduce the particular tightly constrained high-frequency feature of the real data inference.

Motivated by the support for higher-frequency mode content in the beyond-Kerr analysis, we analyze the GW240621_195059 postmerger data with yet another Kerr model including an additional sub-dominant mode, the (4,4,0). The resulting three-mode model posteriors on remnant properties and all mode amplitudes (blue markers in Figure 7) are self-consistent for $t_> \geq 0t_{M_{\rm f}}$. Moreover, all three damped sinusoids are confidently identified, inferred to have non-zero amplitudes at the ≥90% CL for $t_> < 6t_{M_{\rm f}}$, with $A_{440}(t_> = 0t_{M_{\rm f}}) = 1.37^{+6.97}_{-1.21} \times 10^{-21}$. The same three-mode model applied to the zero-noise NRSur7dq4 injection does not reproduce the same statistical support for the presence of three modes, and in fact the overtone amplitude is inferred to be consistent with zero at the 90% CL. For the two Gaussian noise injections whose (2,2,1) content is identified with confidence comparable to the measurements from real data, the (4,4,0) is never resolved. The inconsistencies between the real data and injection analyses suggest that the (4,4,0)-like content in the data may be a non-Gaussian noise artifact, but further investigation with additional mode combinations and injections remains open for future work.

While we do identify multiple damped sinusoids in the postmerger signal of GW240621_195059 at early $t_>$, the inability of injection studies to replicate the behavior of the real data results cautions us against using this event to constrain deviations from the Kerr spectrum. Analyses of GW240621_195059 and the other qualified events with additional (2,1,0) and (3,3,0) modes included in the QNM template do not produce any meaningful measurements of the sub-dominant modes; in other words, the data do not require their presence in the model to sufficiently describe the signal.

7.2. *The* pSEOBNR *analysis*

The pSEOBNRv5PHM analysis (Brito et al. 2018; Ghosh et al. 2021; Maggio et al. 2023; Toubiana et al. 2024; Pompili et al. 2025b) introduces fractional deviations ($\delta\hat{f}_{\ell m0}$, $\delta\hat{\tau}_{\ell m0}$) to the frequency and decay time of the fundamental QNMs in the ringdown description of the underlying SEOBNRv5PHM waveform model (Khalil et al. 2023; Pompili et al. 2023; Ramos-Buades et al. 2023; van de Meent et al. 2023) as:

$$f_{\ell m0} = f^{\rm GR}_{\ell m0}\,(1 + \delta\hat{f}_{\ell m0})\,, \tag{17}$$

$$\tau_{\ell m0} = \tau^{\rm GR}_{\ell m0}\,(1 + \delta\hat{\tau}_{\ell m0})\,. \tag{18}$$

The final mass and spin of the remnant BH, computed from the masses and spins of the components using NR fits (Hofmann et al. 2016; Jiménez-Forteza et al. 2017), are used to predict the GR values of the frequency and damping time of the $(\ell, m, 0)$ QNM (Berti et al. 2009).

The SEOBNRv5PHM waveform model describes BBHs with spin precession and includes the subdominant modes $(\ell, |m|) = (2, 1), (3, 3), (3, 2), (4, 4)$, and $(4, 3)$, in addition to the dominant $(2, 2)$ mode, in the coprecessing frame. While the $(5, 5)$ mode is also modeled, it is not included by default for computational reasons and is not used in the analyses presented here, as its contribution is typically negligible. We denote the spherical-harmonic modes in the coprecessing frame as $\tilde{h}_{\ell m}$. Negative-m modes are derived from the positive-m ones using the reflection symmetry $\tilde{h}_{\ell m} = (-1)^\ell \tilde{h}^*_{\ell -m}$, which is exact for aligned-spin binaries but not for precessing-spin binaries (Boyle et al. 2014), even in the coprecessing frame. Mode asymmetries in the co-precessing frame are a subdominant effect and have only recently been included in SEOBNRv5PHM (Estellés et al. 2026). They are not included in

Table 8. Results of the pSEOBNR analysis

Event	$\delta\hat{f}_{220}$	$\delta\hat{\tau}_{220}$	f_{220} [Hz]	τ_{220} [ms]	$(1+z)M_{\rm f}/M_\odot$	$\chi_{\rm f}$
GW240511_031507	$0.04^{+0.14}_{-0.12}$	$-0.02^{+0.42}_{-0.32}$	$188.2^{+20.5}_{-17.6}$ ($184.5^{+8.2}_{-9.2}$)	$5.75^{+2.30}_{-1.86}$ ($5.74^{+0.70}_{-0.49}$)	$93.8^{+18.5}_{-21.9}$ ($93.9^{+7.5}_{-6.2}$)	$0.73^{+0.15}_{-0.41}$ ($0.71^{+0.07}_{-0.07}$)
GW240514_121713	$0.02^{+0.16}_{-0.15}$	$-0.05^{+0.39}_{-0.29}$	$149.5^{+14.9}_{-12.8}$ ($149.5^{+9.5}_{-9.0}$)	$6.76^{+3.01}_{-2.20}$ ($6.96^{+0.89}_{-0.72}$)	$113.8^{+24.7}_{-26.3}$ ($114.6^{+10.1}_{-9.4}$)	$0.68^{+0.18}_{-0.45}$ ($0.69^{+0.08}_{-0.09}$)
GW240615_113620	$-0.05^{+0.10}_{-0.09}$	$0.13^{+0.28}_{-0.24}$	$215.3^{+15.4}_{-16.8}$ ($223.9^{+6.8}_{-9.1}$)	$4.85^{+1.08}_{-0.95}$ ($4.47^{+0.27}_{-0.26}$)	$80.3^{+10.1}_{-11.8}$ ($74.6^{+3.2}_{-3.3}$)	$0.70^{+0.12}_{-0.25}$ ($0.66^{+0.05}_{-0.06}$)
GW240621_195059	$-0.01^{+0.12}_{-0.06}$	$0.18^{+0.29}_{-0.26}$	$215.8^{+9.0}_{-8.5}$ ($219.4^{+4.9}_{-7.3}$)	$5.44^{+1.30}_{-1.18}$ ($4.68^{+0.27}_{-0.21}$)	$85.5^{+9.7}_{-12.2}$ ($77.5^{+2.9}_{-2.7}$)	$0.78^{+0.09}_{-0.19}$ ($0.68^{+0.04}_{-0.04}$)
GW240705_053215	$0.13^{+0.21}_{-0.16}$	$-0.04^{+0.39}_{-0.29}$	$139.5^{+17.9}_{-11.3}$ ($135.3^{+9.3}_{-8.3}$)	$8.41^{+3.23}_{-2.55}$ ($7.64^{+1.19}_{-0.84}$)	$131.6^{+25.0}_{-28.1}$ ($126.0^{+13.0}_{-11.1}$)	$0.78^{+0.11}_{-0.32}$ ($0.69^{+0.10}_{-0.09}$)
GW240716_034900	$0.00^{+0.13}_{-0.18}$	$0.18^{+0.56}_{-0.40}$	$191.8^{+21.5}_{-28.4}$ ($195.5^{+11.3}_{-13.6}$)	$6.00^{+2.89}_{-2.02}$ ($5.17^{+0.81}_{-0.60}$)	$95.6^{+20.9}_{-22.3}$ ($85.9^{+8.7}_{-7.7}$)	$0.76^{+0.15}_{-0.41}$ ($0.67^{+0.10}_{-0.13}$)
GW240919_061559	$0.01^{+0.13}_{-0.10}$	$0.14^{+0.45}_{-0.35}$	$195.9^{+19.6}_{-15.7}$ ($196.5^{+7.3}_{-9.3}$)	$5.98^{+2.21}_{-1.76}$ ($5.25^{+0.55}_{-0.41}$)	$93.8^{+17.3}_{-19.5}$ ($86.7^{+6.1}_{-5.3}$)	$0.78^{+0.11}_{-0.30}$ ($0.68^{+0.07}_{-0.06}$)
GW240920_124024	$0.03^{+0.05}_{-0.05}$	$0.03^{+0.25}_{-0.20}$	$217.6^{+8.0}_{-6.8}$ ($213.4^{+4.2}_{-4.0}$)	$5.06^{+1.29}_{-1.03}$ ($4.81^{+0.24}_{-0.20}$)	$81.9^{+10.6}_{-11.6}$ ($79.6^{+2.6}_{-2.4}$)	$0.74^{+0.11}_{-0.20}$ ($0.68^{+0.04}_{-0.03}$)
GW241006_015333	$0.08^{+0.30}_{-0.12}$	$-0.14^{+0.55}_{-0.33}$	$324.9^{+94.5}_{-33.3}$ ($305.2^{+8.1}_{-12.3}$)	$3.14^{+1.99}_{-1.16}$ ($3.51^{+0.26}_{-0.19}$)	$52.0^{+17.4}_{-14.2}$ ($57.2^{+2.7}_{-2.2}$)	$0.72^{+0.18}_{-0.46}$ ($0.71^{+0.05}_{-0.04}$)
GW241127_061008	$0.00^{+0.15}_{-0.11}$	$0.03^{+0.45}_{-0.33}$	$157.8^{+9.7}_{-8.0}$ ($153.9^{+8.8}_{-7.8}$)	$6.43^{+2.65}_{-2.25}$ ($5.88^{+0.50}_{-0.49}$)	$108.4^{+22.4}_{-26.2}$ ($100.9^{+7.1}_{-7.1}$)	$0.69^{+0.17}_{-0.44}$ ($0.56^{+0.09}_{-0.10}$)
GW241129_021832	$-0.03^{+0.16}_{-0.13}$	$0.10^{+0.45}_{-0.35}$	$237.7^{+35.5}_{-27.2}$ ($246.9^{+12.4}_{-15.1}$)	$4.86^{+1.87}_{-1.46}$ ($4.58^{+0.64}_{-0.44}$)	$76.4^{+15.2}_{-16.2}$ ($73.1^{+5.7}_{-4.9}$)	$0.77^{+0.13}_{-0.34}$ ($0.75^{+0.08}_{-0.08}$)
GW241225_082815	$0.08^{+0.13}_{-0.19}$	$0.38^{+0.52}_{-0.40}$	$129.3^{+8.7}_{-5.9}$ ($133.5^{+7.6}_{-6.4}$)	$10.70^{+4.23}_{-3.37}$ ($7.60^{+1.02}_{-0.89}$)	$154.3^{+24.7}_{-30.5}$ ($126.0^{+11.8}_{-11.3}$)	$0.85^{+0.08}_{-0.22}$ ($0.67^{+0.08}_{-0.11}$)
GW250114_082203	$0.02^{+0.02}_{-0.02}$	$-0.01^{+0.10}_{-0.09}$	$251.5^{+5.2}_{-5.1}$ ($247.6^{+1.6}_{-1.3}$)	$4.06^{+0.41}_{-0.38}$ ($4.13^{+0.07}_{-0.08}$)	$67.7^{+4.1}_{-4.3}$ ($68.3^{+0.8}_{-1.0}$)	$0.68^{+0.07}_{-0.09}$ ($0.68^{+0.01}_{-0.01}$)
GW250119_025138	$0.09^{+0.20}_{-0.13}$	$0.23^{+0.62}_{-0.47}$	$207.8^{+16.4}_{-13.3}$ ($201.7^{+11.9}_{-11.4}$)	$7.52^{+3.11}_{-2.83}$ ($5.34^{+0.75}_{-0.54}$)	$101.2^{+14.3}_{-21.3}$ ($86.8^{+7.8}_{-6.7}$)	$0.89^{+0.06}_{-0.24}$ ($0.72^{+0.08}_{-0.08}$)

NOTE—Median values and symmetric 90% credible intervals for the one-dimensional marginalised posteriors of the fractional deviations in the frequency and damping time of the $(2,2,0)$ QNM, $(\delta\hat{f}_{220}, \delta\hat{\tau}_{220})$; the reconstructed frequency and damping time of the $(2,2,0)$ QNM; and the redshifted mass and spin of the remnant object, from different events that are analyzed using the pSEOBNRv5PHM method. For the last four quantities, estimates from IMR analyses assuming GR are shown in parentheses for comparison.

the present analyses to maintain consistency with previous observing runs, and are expected to have a marginal impact on our results. In the following, we restrict the discussion to (ℓ, m) modes with $m > 0$. The current implementation of the pSEOBNR test has a pTEOBResumS analogue for eccentric and precessing binary black holes (Chiaramello et al. 2025; Chandra et al. 2025).

The SEOBNRv5PHM waveform is constructed by attaching the merger–ringdown waveform, $\tilde{h}^{\rm merger\text{-}RD}_{\ell m}(t)$, to the inspiral–plunge waveform, $\tilde{h}^{\rm insp-plunge}_{\ell m}(t)$, in the coprecessing frame at a matching time, $t_{\rm match}$, corresponding to the peak amplitude of the $(2,2)$ harmonic. The merger–ringdown for each mode can be written as

$$\tilde{h}^{\rm merger-RD}_{\ell m} = \eta \tilde{A}_{\ell m}(t) \exp\left[i\tilde{\phi}_{\ell m}(t)\right] \exp\left[-i\tilde{\sigma}_{\ell m 0}\,(t - t_{\rm match})\right], \tag{19}$$

where η is the symmetric mass ratio of the binary and $\tilde{\sigma}_{\ell m 0}$ are the complex frequencies of the least-damped QNM of the remnant BH, in the coprecessing frame. The functions $\tilde{A}_{\ell m}(t)$ and $\tilde{\phi}_{\ell m}(t)$ are constrained by the requirement that the amplitude and phase of $\tilde{h}_{\ell m}(t)$ are continuously differentiable at the matching time and are calibrated to a large set of NR simulations of BBHs with aligned spins (Pompili et al. 2023). These time-dependent functions implicitly absorb the contribution of overtones, so the analysis enables constraints on the fundamental QNMs but not on overtones, which are not parameterized explicitly.

In BH perturbation theory, the mode decomposition used to define QNMs assumes a frame where the z-axis is aligned with the final angular momentum of the system ($\boldsymbol{J}_{\rm f}$-frame). The QNM frequencies in this frame, $\sigma_{\ell m 0}$, can be mapped to effective QNM frequencies in the coprecessing frame, $\tilde{\sigma}_{\ell m 0}$ (Hamilton et al. 2023). The complex frequencies are related to the QNM oscillation frequency and damping time as:

$$f_{\ell m 0} = \frac{1}{2\pi}\,{\rm Re}\,(\sigma_{\ell m 0})\,, \quad \tau_{\ell m 0} = -\frac{1}{{\rm Im}\,(\sigma_{\ell m 0})}. \tag{20}$$

Because the pSEOBNR analysis directly modifies parameters in an IMR waveform model, it takes full advantage of the signal length and its SNR, and removes the dependency on an externally chosen ringdown start time. This analysis requires SNR ≥ 8 in both the inspiral and post-inspiral parts of the signal, since a reasonable inspiral SNR is needed to constrain the remnant properties expected in the GR prediction of the QNMs, and break the degeneracy between the fundamental ringdown frequency deviation parameter and the remnant mass (Ghosh et al. 2021). More specifically, for events starting from O4a that have been analyzed with the SEOBNRv5PHM model in Abac et al. (2026i), we use the median SNR from the GR analyses performed with this waveform. For events up to O3b, where results with SEOBNRv5PHM are not available, we follow Abbott et al. (2025) and use the maximum a posteriori estimate from the IMR analyses performed with IMRPhenomXPHM_MSA (Pratten et al. 2021).

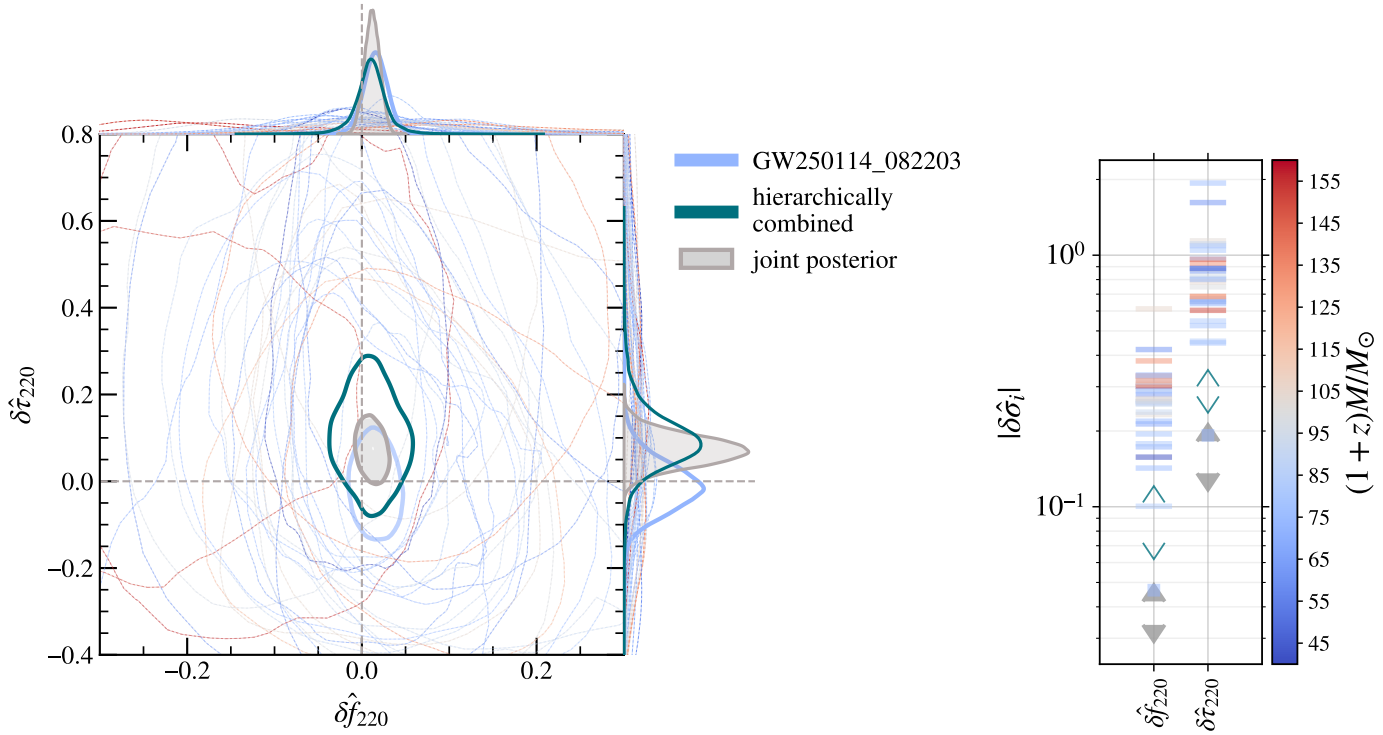


Figure 9. *Left panel*: The 90% credible regions of the posterior probability distribution of the fractional deviations in the frequency and damping time of the $(2, 2, 0)$ QNM, $(\delta\hat{f}_{220}, \delta\hat{\tau}_{220})$, and their corresponding one-dimensional marginalized posterior distributions, for events passing a SNR threshold of 8 in both the inspiral and post-inspiral signal. We highlight the posteriors for GW250114_082203. The joint constraints on $(\delta\hat{f}_{220}, \delta\hat{\tau}_{220})$ obtained multiplying the posteriors (given a flat prior) from individual events are given by the filled grey contours, while the hierarchical method of combination yields the teal curves. *Right panel*: Width of the 90% credible interval for the one-dimensional marginalised posteriors of $(\delta\hat{f}_{220}, \delta\hat{\tau}_{220})$, colored by the median redshifted total mass $(1 + z)M/M_\odot$, inferred assuming GR. Filled gray (unfilled teal) downward triangles mark the constraints obtained when all the events are combined by multiplying posteriors (hierarchically). For comparison, we mark bounds from GWTC-4.0 results (Abac et al. 2026d) with filled/unfilled upward triangles. The bounds from GW250114_082203 (square) are indicated with separate markers.

The current version of the analysis focuses exclusively on constraining the fractional deviations of the dominant (least-damped) QNM by sampling over $\delta\hat{f}_{220}$ and $\delta\hat{\tau}_{220}$ in addition to the GR parameters of the waveform model. The prior range for the fractional deviations is uniform in the ranges $\delta\hat{f}_{220} \in [-0.8, 2]$ and $\delta\hat{\tau}_{220} \in [-0.8, 2]$. In cases exhibiting railing, we extend the prior range to $[-0.8, 4]$. The internal consistency of the model requires $\delta\hat{\tau}_{220}$ to be larger than -1.

Table 8 presents the median values and symmetric 90% credible intervals on the one-dimensional marginalised posteriors of the fractional deviations in the frequency and damping time of the $(2, 2, 0)$ QNM, $(\delta\hat{f}_{220}, \delta\hat{\tau}_{220})$ for the events that are new with respect to GWTC-4.0. Additionally, the table reports the reconstructed frequency and damping time of the $(2, 2, 0)$ QNM, derived using Equations (17) and (18), and the mass and spin of the remnant BH, estimated from the complex QNM frequencies by inverting the fitting formulas from Berti et al. (2006). Since the fitting formulas map the GR QNM frequencies to remnant mass and spin, samples with large fractional deviations, which can arise when the posterior is poorly informative, may map to unphysical remnant parameters. The $(1 + z)$ M_f and χ_f entries in the table are therefore restricted to the physical range of remnant mass and spin. For all events analyzed, the two-dimensional posteriors for the reconstructed frequency and damping time of the $(2, 2, 0)$ QNM, as well as the inferred remnant mass and spin, are consistent at the 90% CL with the estimates from IMR analyses (Abac et al. 2026i).

The results of the analysis are summarised in Figure 9. The left panel of the figure shows the two-dimensional posteriors (along with the corresponding marginalized one-dimensional posteriors) for the fractional deviations in the frequency and damping time of the $(2, 2, 0)$ QNM, for all events analyzed. The contours are colored according to the median detector-frame total mass $(1 + z)M$ of the corresponding binary. We specifically highlight the posteriors from one event, GW250114 (Abac et al. 2026f), which is the loudest observed so far and provides the strongest single-event bounds on the frequency and damping time deviations. Combined constraints are shown both by multiplying individual posteriors, depicted as the filled grey contours, and by hierarchically combining events, represented by the teal curves. For the hierarchical analysis, we model the population-level fractional deviations $\delta\hat{f}_{220}$ and $\delta\hat{\tau}_{220}$ as a two-dimensional Gaussian distribution, with means $(\mu_{\delta\hat{f}_{220}}, \mu_{\delta\hat{\tau}_{220}})$, standard deviations

$(\sigma_{\delta\hat{f}_{220}}, \sigma_{\delta\hat{\tau}_{220}})$, and a correlation coefficient $\rho_{\delta\hat{f}_{220}\delta\hat{\tau}_{220}}$ (Zhong et al. 2024). The right panel of Figure 9 summarizes the 90% credible intervals on the one-dimensional marginalized posteriors, color-coded by the median detector-frame mass of each binary.

When using the same analysis settings adopted for GR analyses in GWTC-5.0, the event GW241127_061008 shows a bimodal posterior in total mass and $\delta\hat{f}_{220}$, with one mode near the GR prediction and another at negative $\delta\hat{f}_{220}$ (around -0.1) and lower total mass (around $90M_\odot$), with the two parameters strongly correlated with each other. This is a loud event with median SNR of 31.2, occurring in a region of parameter space (with support for unequal masses, high total mass, and evidence for strong spin precession) where waveform systematics are expected to be significant (Dhani et al. 2025). However, analyses of signal simulated generated using NRSur7dq4 in zero noise do not show such features, indicating that waveform systematics are unlikely to be the origin of the observed bimodality.

Instead, the features are more plausibly attributed to noise systematics in the LIGO Livingston detector. Data-quality reports indicate the presence of a glitch in the Livingston data near the time of the event, and default analyses adopt a low-frequency cutoff of 30 Hz to mitigate its impact. We find, however, that the results are sensitive to the choice of this cutoff: increasing the low-frequency cutoff enhances the bimodality of the posterior. Conversely, removing Livingston data entirely yields a unimodal posterior consistent with GR. This behavior suggests the presence of a residual in the data, which the waveform model accommodates by shifting parameters in a correlated manner. Further evidence for non-stationary noise comes from analyses performed using a PSD estimated with BilbyPipe via median averaging (Allen et al. 2012; Ashton et al. 2019; Romero-Shaw et al. 2020), computed from 256 s of data prior to the event and an 8 s segment length, with all other settings left at their defaults. In this case, the posterior distributions for both the total mass and $\delta\hat{f}_{220}$ differ significantly from those obtained with the default BayesWave PSD based on on-source data, losing their bimodal structure and instead peaking around the non-GR mode. Taken together, these results point to non-stationary noise in the Livingston detector as the most likely explanation for the observed features. In light of these findings, our final results for this event are obtained excluding the Livingston detector data.

Motivated by the findings of other ringdown pipelines, we extend the pSEOBNR analysis of the GW240621_195059 event to probe also the sub-dominant $(4,4,0)$ QNM. We sample over $\delta\hat{f}_{440}$ and $\delta\hat{\tau}_{440}$ in addition to $\delta\hat{f}_{220}$, $\delta\hat{\tau}_{220}$ and the GR parameters of the waveform model. We choose a prior range for the $(4,4,0)$ QNM fractional deviations uniform in the ranges $\delta\hat{f}_{440} \in [-0.8, 4]$ and $\delta\hat{\tau}_{440} \in [-0.8, 4]$. We find that the posteriors on the two additional parameters are uninformative. Therefore, the pSEOBNR analysis does not find support for the presence of the $(4,4,0)$ QNM in GW240621_195059.

When considering the joint posterior results, a shift toward positive values in the damping time deviation, previously noted in both the GWTC-3.0 and GWTC-4.0 analyses (Abbott et al. 2025; Abac et al. 2026d), is now recovered, with the GR prediction lying slightly outside the 90% credible region. The distribution for the frequency deviation is more sharply peaked around zero compared to GWTC-4.0, while the damping time deviation shows a shift towards positive values. A similar trend is observed in the hierarchical posterior for $\delta\hat{\tau}_{220}$, which also peaks at positive values but features larger statistical uncertainties compared to the corresponding joint posterior.

The joint bounds read

$$\delta\hat{f}_{220} = 0.01^{+0.01}_{-0.02}; \quad \delta\hat{\tau}_{220} = 0.07^{+0.06}_{-0.05} \tag{21}$$

by multiplying the posteriors and

$$\begin{aligned} \delta\hat{f}_{220} &= 0.01^{+0.03}_{-0.03} \quad \left[\mu_{\delta\hat{f}_{220}} = 0.01^{+0.02}_{-0.02}; \sigma_{\delta\hat{f}_{220}} < 0.03\right]; \\ \delta\hat{\tau}_{220} &= 0.09^{+0.13}_{-0.10} \quad \left[\mu_{\delta\hat{\tau}_{220}} = 0.09^{+0.07}_{-0.06}; \sigma_{\delta\hat{\tau}_{220}} < 0.10\right]; \\ \rho_{\delta\hat{f}_{220}\delta\hat{\tau}_{220}} &= -0.07^{+0.76}_{-0.69} \end{aligned} \tag{22}$$

by combining hierarchically. The values in square brackets correspond to the inferred hyperparameters.

Given a posterior distribution, the consistency with the null hypothesis can be quantified by the GR quantile, which corresponds to the fraction of the posterior enclosed by the iso-probability contour that passes through the GR value (Ghosh et al. 2018), and is defined such that 0% (100%) indicates full consistency (full inconsistency) with the null hypothesis. We denote by $Q^{\rm 2D}_{\rm GR}$ the quantile computed from the joint posterior in two dimensions, where the GR hypothesis corresponds to $\delta\hat{f}_{220} = \delta\hat{\tau}_{220} = 0$. For the hierarchical analysis, we instead refer to $Q^{\rm 4D}_{\rm GR}$, where the null hypothesis is given by $\mu_{\delta\hat{f}_{220}} = \mu_{\delta\hat{\tau}_{220}} = \sigma_{\delta\hat{f}_{220}} = \sigma_{\delta\hat{\tau}_{220}} = 0$. We summarize the GR quantiles in Table 9.

The analysis of GWTC-4.0 events using the pSEOBNRv5PHM model results in $Q^{\rm 2D}_{\rm GR} = 98.6\%$ when multiplying the posteriors, while the hierarchical combination yields $Q^{\rm 1D,\mu_{\delta\hat{\tau}_{220}}}_{\rm GR} = 99.3\%$ when just considering the mean of the deviation in the damping time which exhibits the largest deviation and $Q^{\rm 4D}_{\rm GR} = 85.1\%$ when considering all four parameters (two means and two standard deviations). Adding the O4b events results in a shift toward slightly smaller GR quantiles, indicating marginally increased consistency with GR. We find $Q^{\rm 2D}_{\rm GR} = 98.1\%$ when multiplying the posteriors, while the hierarchical combination yields $Q^{\rm 1D,\mu_{\delta\hat{\tau}_{220}}}_{\rm GR} = 98.7\%$ and $Q^{\rm 4D}_{\rm GR} = 92.5\%$. To quantify the variance due to the finite number of events in the catalog, we estimate uncertainties by bootstrapping over the event set (Pacilio et al. 2024). Specifically, we build 1000 synthetic catalogs by resampling events with replacement and recompute the combined quantiles for each realization. The central 90% interval of the resulting bootstrap distribution is reported as the uncertainty on $Q^{\rm 2D}_{\rm GR}$, $Q^{\rm 1D,\mu_{\delta\hat{\tau}_{220}}}_{\rm GR}$, and $Q^{\rm 4D}_{\rm GR}$ in Table 9.

This discrepancy with GR could arise from a variety of factors, including non-Gaussian or non-stationary noise (Abbott et al. 2021b; Ghosh et al. 2021), parameter correlations (Abbott et al. 2025), systematic errors from waveform modeling, intrinsic variance due to the limited number of events in the

Table 9. GR quantiles for combined pSEOBNR results

Catalog	Joint $Q_{\rm GR}^{\rm 2D}$	Hierarchical $Q_{\rm GR}^{\rm 1D,\mu_{\delta\hat{\tau}_{220}}}$	Hierarchical $Q_{\rm GR}^{\rm 4D}$
GWTC-4.0	$98.6^{+1.4}_{-9.4}\%$	$99.3^{+0.7}_{-4.5}\%$	$85.1^{+14.9}_{-19.7}\%$
GWTC-5.0	$98.1^{+1.9}_{-6.9}\%$	$98.7^{+1.3}_{-13.1}\%$	$92.5^{+7.5}_{-20.8}\%$

Note—Comparison of GR quantiles for joint ($Q_{\rm GR}^{\rm 2D}$) and hierarchical ($Q_{\rm GR}^{\rm 1D,\mu_{\delta\hat{\tau}_{220}}}$ and $Q_{\rm GR}^{\rm 4D}$) results of the pSEOBNRv5PHM analysis, where the GWTC-4.0 results are taken from Abac et al. (2026d). Smaller GR quantiles indicate better consistency with GR. Uncertainties quantify variance from the finite catalog size and are estimated via bootstrapping over the event set.

catalog (Pacilio et al. 2024), or selection effects. Incorporating additional events from future observing runs, as well as injection studies in the meantime, could help clarify this behavior.

As previously noted, several events exhibit non-trivial correlations between the deviations $\delta\hat{f}_{220}$ and $\delta\hat{\tau}_{220}$ and the binary's mass and spin parameters, which are themselves influenced by prior choices that may not reflect the true source population (Payne et al. 2023). To assess the impact of these correlations, we reweight the individual-event posteriors according to the GW astrophysical population inferred from GWTC-4.0 (Abac et al. 2025d). We find that the joint posterior distribution on $\delta\hat{f}_{220}$ shifts slightly towards negative values, becoming more consistent with GR, while the posterior distribution on $\delta\hat{\tau}_{220}$ narrows while peaking at approximately the same value, becoming slightly less consistent with GR ($Q_{\rm GR}^{\rm 2D} = 99.7\%$). When combining the reweighted posteriors hierarchically, we find that both the posterior distributions on $\delta\hat{f}_{220}$ and $\delta\hat{\tau}_{220}$ shift slightly towards negative values, becoming more consistent with GR ($Q_{\rm GR}^{\rm 1D,\mu_{\delta\hat{\tau}_{220}}} = 97.0\%$, $Q_{\rm GR}^{\rm 4D} = 67.8\%$). This behaviour is expected: the PE prior tends to assign relatively more weight to higher-mass events compared to the astrophysical population, and since $\delta\hat{f}_{220}$ is correlated with the total mass, the reweighting naturally shifts the posterior in this direction. By contrast, $\delta\hat{\tau}_{220}$ is less strongly correlated with the binary parameters, consistent with smaller changes observed at the level of individual events.

Additionally, selection effects are not accounted for in the current hierarchical inference. For example, the pSEOBNR selection criteria may favor events with positive damping time deviations, as these tend to have a larger ringdown SNR, potentially contributing to the observed deviation. We therefore expect that properly accounting for selection effects would shift $\delta\hat{\tau}_{220}$ toward lower values and reduce the apparent level of inconsistency with GR. These limitations should be addressed in future analyses to avoid potential biases.

When restricting the combination to the five highest-SNR events in GWTC-5.0 (namely, GW250114_082203, GW240920_124024, GW231226_101520, GW240621_195059, GW200129_065458, and excluding the loud special event GW250207_115645 observed in O4c; Abac et al. 2026g), the deviation on $\delta\hat{\tau}_{220}$ is significantly reduced. The 90% credible interval of the joint posterior is $\delta\hat{\tau}_{220} = 0.02^{+0.08}_{-0.08}$ with a joint GR quantile of $Q_{\rm GR}^{\rm 2D} = 69.6\%$. Including lower-SNR events shifts the posterior distribution towards higher $\delta\hat{\tau}_{220}$ values while also tightening it, eventually rendering it only marginally consistent with GR. However, removing GW250114 from the five highest SNR events in GWTC-5.0results in a posterior on $\delta\hat{\tau}_{220}$ shifted towards positive values ($\delta\hat{\tau}_{220} = 0.08^{+0.14}_{-0.12}$ at 90% credible interval). Combining all events in the GWTC-5.0 without GW250114 yields $\delta\hat{\tau}_{220} = 0.11^{+0.09}_{-0.07}$ at 90% credible interval, with a joint GR quantile $Q_{\rm GR}^{\rm 2D} = 99.0\%$. This highlights the significant influence of GW250114 on the combined result.

To give a rough sense of the constraining power of our results, we map the pSEOBNRv5PHM ringdown results to approximate bounds on two beyond-GR theories: Einstein–dilaton–Gauss–Bonnet gravity (Gross & Sloan 1987; Kanti et al. 1996) and dynamical Chern–Simons gravity (Alexander & Yunes 2009). Both theories modify the QNM spectrum of Kerr BHs, with corrections controlled by a coupling length scale $\sqrt{\alpha}$ (Chung & Yunes 2024; Chung et al. 2025). Unlike in GR, where the QNM spectrum is isospectral—perturbations of opposite parity (axial and polar) share the same frequencies—these theories generically break this degeneracy, so that axial and polar perturbations receive distinct corrections. For simplicity, we consider each parity sector independently and map the posterior on $\delta\hat{f}_{220}$ to the coupling length scale $\sqrt{\alpha}$ by inverting the leading-order relation between QNM frequency shifts and the coupling (Chung & Yunes 2024; Chung et al. 2025), reweighting to a prior uniform in $\sqrt{\alpha}$. More rigorous constraints would require a Bayesian analysis that directly employs waveform predictions in these theories (Silva et al. 2023; Chung & Yunes 2025). Across all 31 events, GW230927_153832 yields the tightest indicative constraint for both theories and both parities; the resulting 90% upper limits on $\sqrt{\alpha_{\rm dCS}}$ and $\sqrt{\alpha_{\rm EdGB}}$ for each parity sector are summarized in Table 10. We adopt the same conventions for $\sqrt{\alpha_{\rm EdGB}}$ and $\sqrt{\alpha_{\rm dCS}}$ as in Sänger et al. (2026) and Silva et al. (2021), respectively, which provide the current best bounds quoted both here and in Table 7. Although this event ranks only fourth in $\delta\hat{f}_{220}$ precision (90% credible interval half-width of 0.077, compared to 0.022 for GW250114), its source-frame remnant mass of $36.6^{+3.6}_{-2.3}\,M_\odot$ is the lowest in our sample, nearly half that of GW250114 ($62.9^{+1.2}_{-1.3}\,M_\odot$). Because the mapped constraint scales linearly with $M_{\rm f}$ but only as the fourth root of $\delta\hat{f}_{220}$, the low remnant mass dominates and makes GW230927_153832 the most constraining event despite its moderate SNR of ~ 20.

The leading-order QNM corrections are derived under the perturbative small-coupling assumption $\zeta^{\rm QNM} \ll 1$, where the dimensionless coupling is $\zeta^{\rm QNM}_{\rm EdGB} =$

Table 10. Illustrative mappings of agnostic constraints on the $(2,2,0)$ QNM frequency deviation $\delta\hat{f}_{220}$ onto parameters of selected modified gravity theories for the event that gives the best results, with axial and polar parity sectors mapped independently.

	Axial		Polar		
Theory	$\sqrt{\alpha}$ [km]	$\zeta^{\rm QNM}$	$\sqrt{\alpha}$ [km]	$\zeta^{\rm QNM}$	Existing $\sqrt{\alpha}$ [km]
Einstein–dilaton–Gauss–Bonnet	14.3	0.238	9.1	0.038	0.28 (Sänger et al. 2026)
dynamical Chern–Simons	17.0	0.030	21.0	0.070	8.5 (Silva et al. 2021)

Note— The values shown are mappings of posteriors on the $(2,2,0)$ QNM frequency deviation $\delta\hat{f}_{220}$ from the pSEOBNRv5PHM analysis into 90% credibility results for parameters in modified gravity theories for the event that gives the best result (GW230927_153832 in both theories). The mapping is obtained by inverting the leading-order relation between the $(2,2,0)$ QNM frequency shift and the coupling length scale $\sqrt{\alpha}$ (Chung & Yunes 2024; Chung et al. 2025), after reweighting the $\delta\hat{f}_{220}$ posterior to a prior uniform in $\sqrt{\alpha}$. For each parity sector we also report the 90% upper limits of the dimensionless coupling $\zeta^{\rm QNM}$ (defined in the text), which quantifies the validity of the perturbative small-coupling expansion. The values of $\sqrt{\alpha}$ should not be interpreted as statistical constraints or exclusions. Real modified gravity theories generically modify the full ringdown spectrum and the inspiral signal, while our mapping uses only the $\delta\hat{f}_{220}$ posterior and neglects additional field-induced modes that enter at the same order in the coupling (Lestingi et al. 2025); moreover, the axial and polar parity sectors, which are degenerate in GR but receive distinct corrections in these theories, are mapped independently, with each bound obtained assuming that the observed deviation is dominated by a single parity sector. Similar to the case of Table 7, the table is intended only to provide intuition about the agnostic $\delta\hat{f}_{220}$ bound by showing how it would translate to the parameters of some commonly considered alternative theories, together with constraints on those parameters from the literature.

$16\pi\,\alpha^2_{\rm EdGB}/\left(GM_{\rm f}/c^2\right)^4$ for Einstein–dilaton–Gauss–Bonnet and $\zeta^{\rm QNM}_{\rm dCS} = \pi\,\alpha^2_{\rm dCS}/\left(GM_{\rm f}/c^2\right)^4$ for dynamical Chern–Simons. The factor of 16 difference between the two theories arises from converting from the convention of Chung & Yunes (2024); Chung et al. (2025) to that typically adopted in the ppE literature, e.g., Sänger et al. (2026); Silva et al. (2021). For GW230927_153832, the mapped bounds correspond to $\zeta^{\rm QNM}_{\rm EdGB} = 0.238$ (0.038) and $\zeta^{\rm QNM}_{\rm dCS} = 0.030$ (0.070) for axial (polar) perturbations, all well below unity.

7.3. *The QNM Rational Filter analysis*

The QNMRF (Ma et al. 2022, 2023a,b; Lu et al. 2025) analyzes postmerger signals from BBH systems to identify the QNMs present and, if more than one mode is present, perform BH spectroscopy. A test of GR via BH spectroscopy aims to determine whether the observation of multiple QNMs in the ringdown signal is consistent with the spectrum predicted for a Kerr BH. In this context, the identification of a subdominant mode in addition to the dominant 220 QNM, along with detector-frame remnant mass $(1+z)M_{\rm f}$ and spin $\chi_{\rm f}$ estimates that agree with those from the full IMR analysis, constitutes a self-consistency test of GR. In the remainder of this section we denote the mass and spin pair as $\vartheta_{\rm f} = \{(1+z)M_{\rm f}, \chi_{\rm f}\}$. To test for this self-consistency, the QNMRF applies filters in the frequency domain to eliminate specific complex-valued QNMs from the ringdown signal, and then compares the residual with colored Gaussian noise. For a ringdown model with a chosen set of QNMs, a rational filter is constructed for a given pair $\vartheta_{\rm f}$ of remnant BH mass and spin. By applying filters to the frequency-domain strain data and transforming the filtered data back to the time domain, the method removes the QNMs associated with each $\vartheta_{\rm f}$ pair. Since the filters remove QNMs without requiring prior knowledge of their amplitudes or phases, the likelihood function $\mathcal{L}(\vartheta_{\rm f})$ remains two-dimensional regardless of the number of modes considered.

To determine the set of QNMs present in a ringdown signal, we compare the Bayesian evidences for different hypotheses, each of which contains a different set of QNMs and has a different analysis start time. We compute the Bayesian evidence $\mathcal{Z}(d|\mathcal{H})$ for the hypothesis $\mathcal{H}$ by integrating the likelihood over the remnant mass and spin of the BH for the given set of QNMs and analysis start time. We then define a detection statistic $\mathcal{D}$ which quantifies how much the data support a hypothesis $\mathcal{H}$ over an alternative hypothesis $\mathcal{H}'$:

$$\mathcal{D}(\mathcal{H}:\mathcal{H}') = \log_{10}\frac{\mathcal{Z}(d|\mathcal{H})}{\mathcal{Z}(d|\mathcal{H}')}\,. \tag{23}$$

Here $\mathcal{D}$ is analogous to a log Bayes factor but formally differs from the Bayes factors computed by other time-domain ringdown analyses. Specifically, the QNMRF likelihood is closely connected to the usual time-domain likelihood when using the maximum-likelihood estimation for mode amplitudes under the assumption of white noise (Lu et al. 2025, Appendix A).

For each hypothesis, we also compute the joint posterior quantile of the remnant mass and spin inferred from the full IMR analysis:

$$p_{\rm IMR} = \frac{\sum_{\mathcal{L}(d|\vartheta_{\rm f})>\mathcal{L}(d|\vartheta_{\rm f}^{\rm IMR})}\mathcal{L}(d|\vartheta_{\rm f})}{\sum_{\vartheta_{\rm f}}\mathcal{L}(d|\vartheta_{\rm f})}\,, \tag{24}$$

where summation over the grid on which the likelihoods are computed is used instead of integration. A lower $p_{\rm IMR}$ value indicates a better match between the IMR analysis and a specific QNM hypothesis. We set $\vartheta_{\rm f}^{\rm IMR}$ and other IMR-inferred quantities to the values with the maximum a posteriori in the NRSur7dq4 approximant PE from Abac et al. (2026i).

To evaluate a detection statistic preferring a specific two-mode hypothesis to the single-mode hypothesis, we compute a threshold on $\mathcal{D}$ which corresponds to a statistical significance with a 1% false-alarm probability (FAP). We do this by injecting a single QNM with randomized properties into the detector noise around the event (i.e., analyzing a simulated observation of a QNM added to the detector noise). We then compare the detection statistic between an overfiltered two-mode hypothesis (for a specific secondary mode) and the correct single-mode hypothesis. By performing 300 injections with the detector noise around each event we find

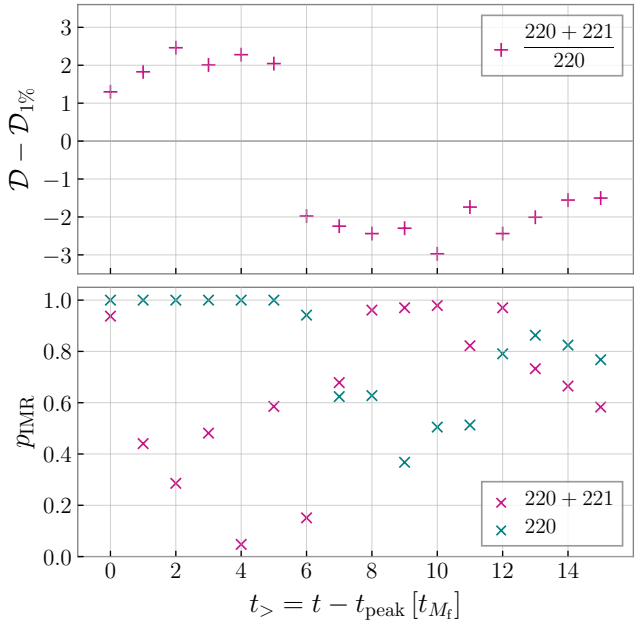


Figure 10. QNMRF results for GW240621_195059. The top panel shows $\mathcal{D} - \mathcal{D}_{1\%}$ when comparing different candidates of the secondary ringdown mode against the {220}-only hypothesis at different starting times. Values above zero are unlikely to be caused by the noise background and are denoted by filled markers, values below zero are denoted by unfilled markers. The bottom panel shows $p_{\rm IMR}$ for the different mode hypotheses. The {220, 221} combination yields a $\mathcal{D} - \mathcal{D}_{1\%} > 0$ for a range of times and improves $p_{\rm IMR}$ the most compared to all other modes.

a distribution of false-alarm detection statistic and then take the $\mathcal{D}$ value that corresponds to a 1% FAP as a threshold $\mathcal{D}_{1\%}$. Observing a $\mathcal{D} - \mathcal{D}_{1\%} > 0$ indicates that the preference for a secondary mode is unlikely to be caused by the noise background. However, it is common to find $\mathcal{D} - \mathcal{D}_{1\%} > 0$ for multiple two-mode hypotheses because the power of a specific subdominant QNM can be partially recovered when searching for other QNMs. In this case, the preferred secondary QNM is identified by selecting the hypothesis that reduces $p_{\rm IMR}$ the most (Lu et al. 2025). A subdominant mode is, therefore, statistically significant if *both* $\mathcal{D} - \mathcal{D}_{1\%} > 0$ *and* it leads to the largest decrease in $p_{\rm IMR}$.

Since the onset of the stationary QNM regime is uncertain, we evaluate at multiple reference times $\Delta t_0 > 0$, where $\Delta t_0 = 0$ marks the merger time as determined by the IMR analysis. The likelihood is computed with the data within $[\Delta t_0, \Delta t_0 + 0.2\,\mathrm{s}]$, with the sky position inferred from the IMR analysis. We select O4b events with the highest total SNRs above 20 and the largest redshifted final masses above $50M_\odot$. The four events selected for analysis are GW240615_113620, GW240621_195059, GW240920_124024, and GW241127_061008.

The main results for the analysis of GW240621_195059 are shown in Figure 10. For the 221 QNM, we find that for $\Delta t_0 < 5t_{M_{\rm f}}$ (where $t_{M_{\rm f}}$ is defined using $M_{\rm f}^{\rm IMR}$), the detection statistic exceeds the threshold and yields a FAP < 1% relative to the noise background, with $\mathcal{D} - \mathcal{D}_{1\%} \leq 2.46$. This suggests evidence for the presence of the 221 mode. While other subdominant modes also have $\mathcal{D} - \mathcal{D}_{1\%} > 0$, the 221 QNM improves $p_{\rm IMR}$ the most at times when $\mathcal{D} - \mathcal{D}_{1\%} > 0$. The caveats discussed in Section 7.1 still apply for early times right after $\Delta t_0 > 0$: (i) at such an early time, systematic uncertainties due to time-dependent effects, unaccounted for in QNM superpositions, may affect the results; (ii) there is a statistical uncertainty in the merger time itself ($t_{\rm geo}{}^{+1.7t_{M_{\rm f}}}_{-1.4t_{M_{\rm f}}}$). We also find qualitatively similar behaviour for the 440 mode in this event. The QNMRF statistic supports a 440-like component, in addition to the {220, 221} modes, in the early post-merger data, consistent with the RINGDOWN analysis. However, applying the same analysis procedure to the NRSur7dq4 waveform without additive noise does not yield a corresponding preference for the 440 mode. This is consistent with the interpretation that the apparent 440 content in the data may be due to a noise artifact; future studies of similar systems may help determine the origin of such features. No evidence for a subdominant QNM is found in the other three events.

8. CONCLUSIONS

The second part of the fourth observing run of the LVK network has substantially expanded the sample of GW signals from GWs. In this work we analyze 72 events detected during O4b and five events from O4a, together with previously observed events from earlier observing runs that satisfy the selection criteria adopted for tests of GR. This brings the total number of analyzed events for tests of GR in the cumulative GWTC to 168. These signals were subjected to a set of seven complementary analyses probing different aspects of GW generation, propagation, and the properties of the remnant BH.

Within the set of tests considered in this paper we find no significant evidence for deviations from the predictions of GR. The residual tests show that the reconstructed signals leave noise-like residuals in the detector data, while no significant evidence is found for additional polarization content beyond the one predicted by GR. Additionally, we improve the constraints on deviations from the GR PN coefficients by factors of 1.2–2.6. Similarly, the ringdown analyses improve the constraint on the dominant quadrupolar ($\ell = m = 2, n = 0$) mode by a factor of 1.95 and 1.48 for the fractional deviation in the frequency and damping time, respectively. When events are combined by multiplying likelihoods (hierarchically), the frequency-domain ringdown analysis places the GR prediction at the edge of the $98.1^{+1.9}_{-6.9}$% ($98.7^{+1.3}_{-13.1}$ %) credible region, where the quoted intervals are obtained via bootstrapping to account for the finite number of analyzed events. The agreement with GR is improved relative to GWTC-4.0, primarily due to the inclusion of GW250114. Furthermore, we infer a nonzero amplitude of the ($\ell = m = 2, n = 1$) overtone for GW240621_195059, though content beyond the two-mode template biases the inferred deviations from the Kerr spectrum in the overtone. Moreover, we do not find evidence for ad-

ditional subdominant ringdown modes in the events. Finally, Tables 7 and 10 provide illustrative mappings of inspiral and ringdown constraints onto parameters of selected modified gravity theories.

In a few individual cases, the posterior distributions of deviation parameters place the GR value outside the 90% credible interval. Given the number of events analyzed, such occurrences are expected and do not constitute evidence for a breakdown of GR. The growing number of high-significance detections also highlights the importance of carefully accounting for potential systematic effects in tests of strong-field gravity. In particular, instrumental artifacts, non-Gaussian noise features, and waveform-modeling uncertainties can influence PE and may mimic apparent deviations from GR. Continued improvements in detector characterization, waveform modeling, and analysis methodologies will therefore remain essential as the sensitivity of the detector network increases and louder signals become more common.

Several exceptional events from O4b have already motivated dedicated studies of strong-field gravity and BH dynamics. For example, the asymmetric high-spin binary GW241011_233834 (Abac et al. 2025b), the very loud event GW250114 (Abac et al. 2025c, 2026f), as well as GW240925_005809 and GW250207_11564 (Abac et al. 2026g) provide particularly stringent probes of relativistic dynamics and the properties of the remnant BH. Such events demonstrate the scientific potential of the current observing run and motivate the continued development of increasingly precise tests of gravitational theory.

Future analyses will incorporate detections from the remainder of the fourth observing run and subsequent observing runs as the global detector network continues to improve in sensitivity (Abbott et al. 2020b). [1] The increasing catalog of gravitational-wave events, together with advances in modeling and inference techniques, will enable progressively more stringent tests of GR in the strong-field, highly dynamical regime.

All strain data analyzed in this paper are publicly available through the Gravitational Wave Open Science Center (Abac et al. 2026k). The data products used to generate the figures and tables in this work are available through the LVK Document Control Center, while the configuration files and posterior samples for the analyses will be released through Zenodo.

ACKNOWLEDGEMENTS

This material is based upon work supported by NSF's LIGO Laboratory, which is a major facility fully funded by the National Science Foundation. The authors also gratefully acknowledge the support of the Science and Technology Facilities Council (STFC) of the United Kingdom, the Max-Planck-Society (MPS), and the State of Niedersachsen/Germany for support of the construction of Advanced LIGO and construction and operation of the GEO 600 detector. Additional support for Advanced LIGO was provided by the Australian Research Council. The authors gratefully acknowledge the Italian Istituto Nazionale di Fisica Nucleare (INFN), the French Centre National de la Recherche Scientifique (CNRS) and the Netherlands Organization for Scientific Research (NWO) for the construction and operation of the Virgo detector and the creation and support of the EGO consortium. The authors also gratefully acknowledge research support from these agencies as well as by the Council of Scientific and Industrial Research of India, the Department of Science and Technology, India, the Science & Engineering Research Board (SERB), India, the Ministry of Human Resource Development, India, the Spanish Agencia Estatal de Investigación (AEI), the Spanish Ministerio de Ciencia, Innovación y Universidades, the European Union NextGenerationEU/PRTR (PRTR-C17.I1), the ICSC - CentroNazionale di Ricerca in High Performance Computing, Big Data and Quantum Computing, funded by the European Union NextGenerationEU, the Comunitat Autonòma de les Illes Balears through the Conselleria d'Educació i Universitats, the Conselleria d'Innovació, Universitats, Ciència i Societat Digital de la Generalitat Valenciana and the CERCA Programme Generalitat de Catalunya, Spain, the Polish National Agency for Academic Exchange, the National Science Centre of Poland and the European Union - European Regional Development Fund; the Foundation for Polish Science (FNP), the Polish Ministry of Science and Higher Education, the Swiss National Science Foundation (SNSF), the Russian Science Foundation, the European Commission, the European Social Funds (ESF), the European Regional Development Funds (ERDF), the Royal Society, the Scottish Funding Council, the Scottish Universities Physics Alliance, the Hungarian Scientific Research Fund (OTKA), the French Lyon Institute of Origins (LIO), the Belgian Fonds de la Recherche Scientifique (FRS-FNRS), Actions de Recherche Concertées (ARC) and Fonds Wetenschappelijk Onderzoek - Vlaanderen (FWO), Belgium, the Paris Île-de-France Region, the National Research, Development and Innovation Office of Hungary (NKFIH), the National Research Foundation of Korea, the Natural Sciences and Engineering Research Council of Canada (NSERC), the Canadian Foundation for Innovation (CFI), the Brazilian Ministry of Science, Technology, and Innovations, the International Center for Theoretical Physics South American Institute for Fundamental Research (ICTP-SAIFR), the Research Grants Council of Hong Kong, the National Natural Science Foundation of China (NSFC), the Israel Science Foundation (ISF), the US-Israel Binational Science Fund (BSF), the Leverhulme Trust, the Research Corporation, the National Science and Technology Council (NSTC), Taiwan, the United States Department of Energy, and the Kavli Foundation. The authors gratefully acknowledge the support of the NSF, STFC, INFN and CNRS for provision of computational resources.

This work was supported by MEXT, the JSPS Leading-edge Research Infrastructure Program, JSPS Grant-in-Aid for Specially Promoted Research 26000005, JSPS Grant-in-Aid for Scientific Research on Innovative Areas 2402: 24103006, 24103005, and 2905: JP17H06358, JP17H06361 and JP17H06364, JSPS Core-to-Core Program A. Advanced

[1] LVK observing run plans https://observing.docs.ligo.org/plan

Research Networks, JSPS Grants-in-Aid for Scientific Research (S) 17H06133 and 20H05639, JSPS Grant-in-Aid for Transformative Research Areas (A) 26A204: JP26H00407, the joint research program of the Institute for Cosmic Ray Research, University of Tokyo, the National Research Foundation (NRF), the Computing Infrastructure Project of the Global Science experimental Data hub Center (GSDC) at KISTI, the Korea Astronomy and Space Science Institute (KASI), the Ministry of Science and ICT (MSIT) in Korea, Academia Sinica (AS), the AS Grid Center (ASGC) and the National Science and Technology Council (NSTC) in Taiwan under grants including the Science Vanguard Research Program, the Advanced Technology Center (ATC) of NAOJ, the Mechanical Engineering Center of KEK and Vietnam National Foundation for Science and Technology Development (NAFOSTED) 103.01-2025.147.

Additional acknowledgements for support of individual authors may be found in the following document: `https://dcc.ligo.org/LIGO-M2300033/public`. For the purpose of open access, the authors have applied a Creative Commons Attribution (CC BY) license to any Author Accepted Manuscript version arising. We request that citations to this article use 'A. G. Abac *et al.* (LIGO-Virgo-KAGRA Collaboration), ...' or similar phrasing, depending on journal convention.

The following open-source software has been used:

Calibration of the LIGO strain data was performed with a GstLAL-based calibration software pipeline (Viets et al. 2018). Data-quality products and event-validation results were computed using the BRiSTOL (Di Renzo et al. 2024), DMT (Zweizig 2006), DQRBuild (Davis et al. 2026), DQSEGDB (Fisher et al. 2021), Glitchfind (Vazsonyi & Davis 2023), GSpyNetTree (Alvarez-Lopez et al. 2024), gwdetchar (Macleod et al. 2021a), hveto (Smith et al. 2011), iDQ (Essick et al. 2020), LigoDV-web (Areeda et al. 2017), OmegaOverlap (Areeda et al. 2017), Omicron (Robinet et al. 2020), PEMcheck (Helmling-Cornell et al. 2024), PythonVirgoTools (Virgo Collaboration 2021) Stationarity (Mozzon et al. 2020), and Virgo DQR (Acernese et al. 2023) software packages and contributing software tools. Analyses in this catalog relied upon the LALSuite software library (LIGO Scientific, Virgo, and KAGRA Collaboration 2025; Wette 2020). The detection of the signals and subsequent significance evaluations in this catalog were performed with the GstLAL-based inspiral software pipeline (Messick et al. 2017; Sachdev et al. 2019; Hanna et al. 2020; Cannon et al. 2021), with the MBTA pipeline (Adams et al. 2016; Aubin et al. 2021), and with the PyCBC (Usman et al. 2016; Nitz et al. 2017; Davies et al. 2020) and the cWB (Klimenko et al. 2004, 2011, 2016) packages. Estimates of the noise spectra and glitch models were obtained using BayesWave (Cornish & Littenberg 2015; Littenberg et al. 2016; Cornish et al. 2021). Noise subtraction for one candidate was also performed with gwsubtract (Davis et al. 2022). Source-parameter estimation was performed with the Bilby library (Ashton et al. 2019), sometimes automated using the BilbyPipe package (Romero-Shaw et al. 2020), and the Dynesty nested sampling package (Speagle 2020). PESummary was used to postprocess and collate parameter-estimation results (Hoy & Raymond 2021). The various stages of the parameter-estimation analysis were managed with the Asimov library (Williams et al. 2023) together with CBCFlow (Ashton et al. 2025a). Plots were prepared with Matplotlib (Hunter 2007), seaborn (Waskom 2021) and GWpy (Macleod et al. 2021b). NumPy (Harris et al. 2020) and SciPy (Virtanen et al. 2020) were used in the preparation of the manuscript. SEOBNRv5PHM waveforms used in parameter estimation were generated using pySEOBNR (Mihaylov et al. 2025). FTI, TIGER, and pSEOBNR waveforms used for testing GR were generated using BilbyTGR (Ashton et al. 2025b). Other analyses used IGWN Wave Compare (Sullivan et al. 2025), nullpol (Ng et al. 2026), and ringdown (Isi & Farr 2021, 2024). Quasinormal mode frequencies were computed using QNM (Stein 2019). The QNMRF analysis used Ma et al. (2026). The multi-dimensional hierarchical analysis results were produced using hierfit (Zhong et al. 2026).

REFERENCES

Aasi, J., Abbott, B. P., Abbott, R., et al. 2015, CQGra, 32, 074001, doi: 10.1088/0264-9381/32/7/074001

Abac, A. G., Abbott, R., Abouelfettouh, I., et al. 2024, ApJL, 970, L34, doi: 10.3847/2041-8213/ad5beb

Abac, A. G., Abouelfettouh, I., Acernese, F., et al. 2025a, ApJL, 995, L18, doi: 10.3847/2041-8213/ae0c06

—. 2025b, ApJL, 993, L21, doi: 10.3847/2041-8213/ae0d54

—. 2025c, PhRvL, 135, 111403, doi: 10.1103/kw5g-d732

—. 2025d. https://arxiv.org/abs/2508.18083

—. 2026a, ApJL, 1004, L22, doi: 10.3847/2041-8213/ae2c74

—. 2026b. https://arxiv.org/abs/2603.19019

—. 2026c. https://arxiv.org/abs/2603.19020

—. 2026d. https://arxiv.org/abs/2603.19021

—. 2026e, ApJL, 1004, L23, doi: 10.3847/2041-8213/ae2ad3

—. 2026f, PhRvL, 136, 041403, doi: 10.1103/6c61-fm1n

—. 2026g, PhRvL, doi: 10.1103/gzrj-mwv3

—. 2026h. https://arxiv.org/abs/2605.27223

—. 2026i. https://arxiv.org/abs/2605.27225

—. 2026j. https://arxiv.org/abs/2605.27224

—. 2026k. https://arxiv.org/abs/2605.27090

Abbott, B. P., Abbott, R., Abbott, T. D., et al. 2016a, PhRvL, 116, 061102, doi: 10.1103/PhysRevLett.116.061102

—. 2016b, PhRvL, 116, 221101, doi: 10.1103/PhysRevLett.116.221101

—. 2016c, PhRvX, 6, 041015, doi: 10.1103/PhysRevX.6.041015

—. 2016d, CQGra, 33, 134001, doi: 10.1088/0264-9381/33/13/134001

—. 2016e, PhRvL, 116, 241102, doi: 10.1103/PhysRevLett.116.241102

—. 2017a, PhRvL, 119, 161101, doi: 10.1103/PhysRevLett.119.161101

—. 2017b, PhRvL, 119, 141101, doi: 10.1103/PhysRevLett.119.141101

—. 2017c, PhRvL, 118, 221101, doi: 10.1103/PhysRevLett.118.221101

—. 2019a, PhRvL, 123, 011102, doi: 10.1103/PhysRevLett.123.011102

—. 2019b, PhRvD, 100, 104036, doi: 10.1103/PhysRevD.100.104036

—. 2019c, PhRvX, 9, 031040, doi: 10.1103/PhysRevX.9.031040

—. 2020a, CQGra, 37, 055002, doi: 10.1088/1361-6382/ab685e

—. 2020b, LRR, 23, 3, doi: 10.1007/s41114-020-00026-9

Abbott, R., Abbott, T. D., Abraham, S., et al. 2020c, PhRvD, 102, 043015, doi: 10.1103/PhysRevD.102.043015

—. 2021a, ApJL, 915, L5, doi: 10.3847/2041-8213/ac082e

—. 2021b, PhRvD, 103, 122002, doi: 10.1103/PhysRevD.103.122002

—. 2021c, PhRvX, 11, 021053, doi: 10.1103/PhysRevX.11.021053

Abbott, R., Abbott, T. D., Acernese, F., et al. 2023, PhRvX, 13, 041039, doi: 10.1103/PhysRevX.13.041039

—. 2024, PhRvD, 109, 022001, doi: 10.1103/PhysRevD.109.022001

Abbott, R., Abe, H., Acernese, F., et al. 2025, PhRvD, 112, 084080, doi: 10.1103/PhysRevD.112.084080

Abd El Dayem, K., Abuter, R., Aimar, N., et al. 2025, A&A, 698, L15, doi: 10.1051/0004-6361/202554676

Abuter, R., Amorim, A., Anugu, N., et al. 2018, A&A, 615, L15, doi: 10.1051/0004-6361/201833718

Acernese, F., Agathos, M., Agatsuma, K., et al. 2015, CQGra, 32, 024001, doi: 10.1088/0264-9381/32/2/024001

Acernese, F., Agathos, M., Ain, A., et al. 2023, CQGra, 40, 185006, doi: 10.1088/1361-6382/acd92d

Adams, T., Buskulic, D., Germain, V., et al. 2016, CQGra, 33, 175012, doi: 10.1088/0264-9381/33/17/175012

Agathos, M., Del Pozzo, W., Li, T. G. F., et al. 2014, PhRvD, 89, 082001, doi: 10.1103/PhysRevD.89.082001

Akiyama, K., Alberdi, A., Alef, W., et al. 2019, ApJL, 875, L1, doi: 10.3847/2041-8213/ab0ec7

—. 2022a, ApJL, 930, L12, doi: 10.3847/2041-8213/ac6674

—. 2022b, ApJL, 930, L17, doi: 10.3847/2041-8213/ac6756

Albanesi, S., Bernuzzi, S., Damour, T., Nagar, A., & Placidi, A. 2023, PhRvD, 108, 084037, doi: 10.1103/PhysRevD.108.084037

Alexander, S., & Yunes, N. 2009, PhR, 480, 1, doi: 10.1016/j.physrep.2009.07.002

Allen, B., Anderson, W. G., Brady, P. R., Brown, D. A., & Creighton, J. D. E. 2012, PhRvD, 85, 122006, doi: 10.1103/PhysRevD.85.122006

Alvarez-Lopez, S., Liyanage, A., Ding, J., Ng, R., & McIver, J. 2024, CQGra, 41, 085007, doi: 10.1088/1361-6382/ad2194

Areeda, J. S., Smith, J. R., Lundgren, A. P., et al. 2017, A&C, 18, 27, doi: 10.1016/j.ascom.2017.01.003

Arnaudo, P., Carballo, J., & Withers, B. 2026, PhRvD, 114, 024025, doi: 10.1103/1bcl-q79x

Arun, K. G., Iyer, B. R., Qusailah, M. S. S., & Sathyaprakash, B. S. 2006a, PhRvD, 74, 024006, doi: 10.1103/PhysRevD.74.024006

—. 2006b, CQGra, 23, L37, doi: 10.1088/0264-9381/23/9/L01

Ashton, G., Hübner, M., Laskyothers, P. D., et al. 2019, ApJS, 241, 27, doi: 10.3847/1538-4365/ab06fc

Ashton, G., Udall, R., & Yarbrough, Z. 2025a, CBC Workflow. https://github.com/Rhiannon-Udall/cbcflow

Ashton, G., Talbot, C., Roy, S., et al. 2025b, Bilby TGR, v0.2, Zenodo, doi: 10.5281/zenodo.15676285

Aubin, F., Brighenti, F., Chierici, R., et al. 2021, CQGra, 38, 095004, doi: 10.1088/1361-6382/abe913

Baibhav, V., Cheung, M. H.-Y., Berti, E., et al. 2023, PhRvD, 108, 104020, doi: 10.1103/PhysRevD.108.104020

Baka, T., Cirok, B., Haris, K., Noller, J., & Krishnendu, N. V. 2026, PhRvD, 113, 064064, doi: 10.1103/bgvh-c1wj

Bernard, L., Blanchet, L., & Trestini, D. 2022, JCAP, 08, 008, doi: 10.1088/1475-7516/2022/08/008
Berti, E., & Cardoso, V. 2006, PhRvD, 74, 104020, doi: 10.1103/PhysRevD.74.104020
Berti, E., Cardoso, V., Carullo, G., et al. 2026, CQGra, 43, 123001, doi: 10.1088/1361-6382/ae59e2
Berti, E., Cardoso, V., Gonzalez, J. A., et al. 2007, PhRvD, 76, 064034, doi: 10.1103/PhysRevD.76.064034
Berti, E., Cardoso, V., & Starinets, A. O. 2009, CQGra, 26, 163001, doi: 10.1088/0264-9381/26/16/163001
Berti, E., Cardoso, V., & Will, C. M. 2006, PhRvD, 73, 064030, doi: 10.1103/PhysRevD.73.064030
Berti, E., & Klein, A. 2014, PhRvD, 90, 064012, doi: 10.1103/PhysRevD.90.064012
Bhagwat, S., Okounkova, M., Ballmer, S. W., et al. 2018, PhRvD, 97, 104065, doi: 10.1103/PhysRevD.97.104065
Blanchet, L. 2024, LRR, 27, 4, doi: 10.1007/s41114-024-00050-z
Blanchet, L., & Sathyaprakash, B. S. 1994, CQGra, 11, 2807, doi: 10.1088/0264-9381/11/11/020
—. 1995, PhRvL, 74, 1067, doi: 10.1103/PhysRevLett.74.1067
Blandford, R. D., & Znajek, R. L. 1977, MNRAS, 179, 433, doi: 10.1093/mnras/179.3.433
Błaut, A. 2012, PhRvD, 85, 043005, doi: 10.1103/PhysRevD.85.043005
Bourg, P., Panosso Macedo, R., Spiers, A., et al. 2025, PhRvL, 134, 061401, doi: 10.1103/PhysRevLett.134.061401
Boyle, M., Kidder, L. E., Ossokine, S., & Pfeiffer, H. P. 2014. https://arxiv.org/abs/1409.4431
Brito, R., Buonanno, A., & Raymond, V. 2018, PhRvD, 98, 084038, doi: 10.1103/PhysRevD.98.084038
Bucciotti, B., Juliano, L., Kuntz, A., & Trincherini, E. 2024, PhRvD, 110, 104048, doi: 10.1103/PhysRevD.110.104048
Bucciotti, B., Kuntz, A., Serra, F., & Trincherini, E. 2023, JHEP, 12, 048, doi: 10.1007/JHEP12(2023)048
Buonanno, A., Cook, G. B., & Pretorius, F. 2007, PhRvD, 75, 124018, doi: 10.1103/PhysRevD.75.124018
Cannon, K., Caudill, S., Chan, C., et al. 2021, SoftX, 14, 100680, doi: 10.1016/j.softx.2021.100680
Capuano, L., Santoni, L., & Barausse, E. 2024, PhRvD, 110, 084081, doi: 10.1103/PhysRevD.110.084081
Carter, B. 1971, PhRvL, 26, 331, doi: 10.1103/PhysRevLett.26.331
Carullo, G., Cotesta, R., Berti, E., & Cardoso, V. 2023, PhRvL, 131, 169002, doi: 10.1103/PhysRevLett.131.169002
Carullo, G., Del Pozzo, W., & Veitch, J. 2019, PhRvD, 99, 123029, doi: 10.1103/PhysRevD.99.123029
Carullo, G., Laghi, D., Johnson-McDaniel, N. K., et al. 2022, PhRvD, 105, 062009, doi: 10.1103/PhysRevD.105.062009
Chandra, K., Gamba, R., & Chiaramello, D. 2025. https://arxiv.org/abs/2512.04593
Chandrasekhar, S. 1983, The mathematical theory of black holes (Oxford, UK: Oxford University Press)
Chavda, A., Lagos, M., & Hui, L. 2025, JCAP, 07, 084, doi: 10.1088/1475-7516/2025/07/084
Cheung, M. H.-Y., Baibhav, V., Berti, E., et al. 2023, PhRvL, 130, 081401, doi: 10.1103/PhysRevLett.130.081401
Chiaramello, D., Cibrario, N., Lange, J., et al. 2025. https://arxiv.org/abs/2511.19593
Chung, A. K.-W., Lam, K. K.-H., & Yunes, N. 2025, PhRvD, 111, 124052, doi: 10.1103/g83n-rrlj
Chung, A. K.-W., & Yunes, N. 2024, PhRvL, 133, 181401, doi: 10.1103/PhysRevLett.133.181401
—. 2025. https://arxiv.org/abs/2506.14695
Colleoni, M., Vidal, F. A. R., García-Quirós, C., Akçay, S., & Bera, S. 2025, PhRvD, 111, 104019, doi: 10.1103/PhysRevD.111.104019
Cornish, N., Sampson, L., Yunes, N., & Pretorius, F. 2011, PhRvD, 84, 062003, doi: 10.1103/PhysRevD.84.062003
Cornish, N. J., & Littenberg, T. B. 2015, CQGra, 32, 135012, doi: 10.1088/0264-9381/32/13/135012
Cornish, N. J., Littenberg, T. B., Bécsy, B., et al. 2021, PhRvD, 103, 044006, doi: 10.1103/PhysRevD.103.044006
Cornish, N. J., Littenberg, T. B., Millhouse, M., et al. 2026, BayesWave software. https://git.ligo.org/lscsoft/bayeswave
Correia, A., Wang, Y.-F., Westerweck, J., & Capano, C. D. 2024, PhRvD, 110, L041501, doi: 10.1103/PhysRevD.110.L041501
Cotesta, R., Carullo, G., Berti, E., & Cardoso, V. 2022, PhRvL, 129, 111102, doi: 10.1103/PhysRevLett.129.111102
Creighton, J. D. E., & Anderson, W. G. 2011, Gravitational-wave physics and astronomy: An introduction to theory, experiment and data analysis (Weinheim: John Wiley & Sons, Ltd), doi: 10.1002/9783527636037
Cunningham, C. T., Price, R. H., & Moncrief, V. 1978, ApJ, 224, 643, doi: 10.1086/156413
Davies, G. S., Dent, T., Tápai, M., et al. 2020, PhRvD, 102, 022004, doi: 10.1103/PhysRevD.102.022004
Davis, D., Littenberg, T. B., Romero-Shaw, I. M., et al. 2022, CQGra, 39, 245013, doi: 10.1088/1361-6382/aca238
Davis, D., Yarbrough, Z., Areeda, J., et al. 2026. https://arxiv.org/abs/2605.16183
De Amicis, M., Cannizzaro, E., Carullo, G., & Sberna, L. 2026, PhRvD, 113, 024048, doi: 10.1103/bgjv-lvyd
Detweiler, S. L. 1980, ApJ, 239, 292, doi: 10.1086/158109
Dhani, A., Völkel, S. H., Buonanno, A., et al. 2025, PhRvX, 15, 031036, doi: 10.1103/5pks-qz6b
Di Renzo, F., Fidecaro, F., Razzano, M., & Sorrentino, N. 2024, in 55th Rencontres de Moriond on Gravitation. https://arxiv.org/abs/2401.15392
Divyajyoti, Krishnendu, N. V., Saleem, M., et al. 2024, PhRvD, 109, 023016, doi: 10.1103/PhysRevD.109.023016

Do, T., Hees, A., Ghez, A., et al. 2019, Sci, 365, 664, doi: 10.1126/science.aav8137
Dreyer, O., Kelly, B. J., Krishnan, B., et al. 2004, CQGra, 21, 787, doi: 10.1088/0264-9381/21/4/003
Essick, R., Godwin, P., Hanna, C., Blackburn, L., & Katsavounidis, E. 2020, MLS&T, 2, 015004, doi: 10.1088/2632-2153/abab5f
Estellés, H., Buonanno, A., Enficiaud, R., Foo, C., & Pompili, L. 2026, PhRvD, 113, 044049, doi: 10.1103/pjbd-pjxn
Ferreira, P. G. 2019, ARA&A, 57, 335, doi: 10.1146/annurev-astro-091918-104423
Finch, E., & Moore, C. J. 2022, PhRvD, 106, 043005, doi: 10.1103/PhysRevD.106.043005
Fisher, R. P., Hemming, G., Bizouard, M.-A., et al. 2021, SoftX, 14, 100677, doi: 10.1016/j.softx.2021.100677
Forward, R. L. 1978, PhRvD, 17, 379, doi: 10.1103/PhysRevD.17.379
Gao, B., Tang, S.-P., Wang, H.-T., Yan, J., & Fan, Y.-Z. 2024, PhRvD, 110, 044022, doi: 10.1103/PhysRevD.110.044022
Gelfand, I. M., Minlos, R. A., & Shapiro, Z. J. 1958, Predstavleniya gruppy vrashcheni i gruppy Lorentsa, ikh primeneniya [Representations of the rotation and Lorentz groups and their applications] (Gosudarstv. Izdat. Fiz.-Mat. Lit., Moscow, MR 0114876)
Ghosh, A., Brito, R., & Buonanno, A. 2021, PhRvD, 103, 124041, doi: 10.1103/PhysRevD.103.124041
Ghosh, A., Ghosh, A., Johnson-McDaniel, N. K., et al. 2016, PhRvD, 94, 021101(R), doi: 10.1103/PhysRevD.94.021101
Ghosh, A., Johnson-McDaniel, N. K., Ghosh, A., et al. 2018, CQGra, 35, 014002, doi: 10.1088/1361-6382/aa972e
Gibbons, G. W. 1975, CMaPh, 44, 245, doi: 10.1007/BF01609829
Giesler, M., Isi, M., Scheel, M. A., & Teukolsky, S. 2019, PhRvX, 9, 041060, doi: 10.1103/PhysRevX.9.041060
Gleiser, R. J., Nicasio, C. O., Price, R. H., & Pullin, J. 1996, PhRvL, 77, 4483, doi: 10.1103/PhysRevLett.77.4483
Grimaldi, L., Maggio, E., Pompili, L., & Buonanno, A. 2026, PhRvD, 113, L061506, doi: 10.1103/g19c-734l
Gross, D. J., & Sloan, J. H. 1987, NuPhB, 291, 41, doi: 10.1016/0550-3213(87)90465-2
Gu, H.-P., Wang, H.-T., & Shao, L. 2024, PhRvD, 109, 024058, doi: 10.1103/PhysRevD.109.024058
Gupta, I., Narayan, P., London, L., Tiwari, S., & Sathyaprakash, B. 2025. https://arxiv.org/abs/2511.11886
Gürsel, Y., & Tinto, M. 1989, PhRvD, 40, 3884, doi: 10.1103/PhysRevD.40.3884
Haegel, L., O'Neal-Ault, K., Bailey, Q. G., et al. 2023, PhRvD, 107, 064031, doi: 10.1103/PhysRevD.107.064031
Hamilton, E., London, L., & Hannam, M. 2023, PhRvD, 107, 104035, doi: 10.1103/PhysRevD.107.104035
Hanna, C., Caudill, S., Messick, C., et al. 2020, PhRvD, 101, 022003, doi: 10.1103/PhysRevD.101.022003
Harris, C. R., Millman, K. J., van der Walt, S. J., et al. 2020, Nature, 585, 357, doi: 10.1038/s41586-020-2649-2
Hawking, S. W. 1972, CMaPh, 25, 152, doi: 10.1007/BF01877517
Helmling-Cornell, A., Nguyen, P., Schofield, R., & Frey, R. 2024, CQGra, 41, 145003, doi: 10.1088/1361-6382/ad5139
Hofmann, F., Barausse, E., & Rezzolla, L. 2016, ApJL, 825, L19, doi: 10.3847/2041-8205/825/2/L19
Horbatsch, M. W., & Burgess, C. P. 2012, JCAP, 05, 010, doi: 10.1088/1475-7516/2012/05/010
Hoy, C., & Raymond, V. 2021, SoftX, 15, 100765, doi: 10.1016/j.softx.2021.100765
Hughes, S. A., & Menou, K. 2005, ApJ, 623, 689, doi: 10.1086/428826
Hunter, J. D. 2007, CSE, 9, 90, doi: 10.1109/MCSE.2007.55
Ishak, M. 2019, LRR, 22, 1, doi: 10.1007/s41114-018-0017-4
Isi, M. 2023, CQGra, 40, 203001, doi: 10.1088/1361-6382/acf28c
Isi, M., Chatziioannou, K., & Farr, W. M. 2019a, PhRvL, 123, 121101, doi: 10.1103/PhysRevLett.123.121101
Isi, M., & Farr, W. M. 2021. https://arxiv.org/abs/2107.05609
—. 2023, PhRvL, 131, 169001, doi: 10.1103/PhysRevLett.131.169001
—. 2024, RINGDOWN package, 1.0.0, ringdown.readthedocs.io, Zenodo, doi: 10.5281/zenodo.5094067
Isi, M., Giesler, M., Farr, W. M., Scheel, M. A., & Teukolsky, S. A. 2019b, PhRvL, 123, 111102, doi: 10.1103/PhysRevLett.123.111102
Isi, M., & Weinstein, A. J. 2017, Tech. Note, LIGO-P1700276. https://arxiv.org/abs/1710.03794
Isi, M., Weinstein, A. J., Mead, C., & Pitkin, M. 2015, PhRvD, 91, 082002, doi: 10.1103/PhysRevD.91.082002
Jacobson, T. 1999, PhRvL, 83, 2699, doi: 10.1103/PhysRevLett.83.2699
Jiménez Forteza, X., Bhagwat, S., Pani, P., & Ferrari, V. 2020, PhRvD, 102, 044053, doi: 10.1103/PhysRevD.102.044053
Jiménez-Forteza, X., Keitel, D., Husa, S., et al. 2017, PhRvD, 95, 064024, doi: 10.1103/PhysRevD.95.064024
Johnson-McDaniel, N. K., Ghosh, A., Ghonge, S., et al. 2022, PhRvD, 105, 044020, doi: 10.1103/PhysRevD.105.044020
Kanti, P., Mavromatos, N. E., Rizos, J., Tamvakis, K., & Winstanley, E. 1996, PhRvD, 54, 5049, doi: 10.1103/PhysRevD.54.5049
Kelly, B. J., & Baker, J. G. 2013, PhRvD, 87, 084004, doi: 10.1103/PhysRevD.87.084004
Kerr, R. P. 1963, PhRvL, 11, 237, doi: 10.1103/PhysRevLett.11.237
Khalil, M., Buonanno, A., Estelles, H., et al. 2023, PhRvD, 108, 124036, doi: 10.1103/PhysRevD.108.124036
Khalil, M., Sennett, N., Steinhoff, J., Vines, J., & Buonanno, A. 2018, PhRvD, 98, 104010, doi: 10.1103/PhysRevD.98.104010
Khusid, N., Farr, W., & Isi, M. 2026. https://arxiv.org/abs/2605.15271

Kidder, L. E., Will, C. M., & Wiseman, A. G. 1993, PhRvD, D47, 4183, doi: 10.1103/PhysRevD.47.R4183

Klimenko, S., Vedovato, G., Drago, M., et al. 2016, PhRvD, 93, 042004, doi: 10.1103/PhysRevD.93.042004

Klimenko, S., Yakushin, I., Rakhmanov, M., & Mitselmakher, G. 2004, CQGra, 21, S1685, doi: 10.1088/0264-9381/21/20/011

Klimenko, S., Vedovato, G., Drago, M., et al. 2011, PhRvD, 83, 102001, doi: 10.1103/PhysRevD.83.102001

Kramer, M., Stairs, I. H., Manchester, R. N., et al. 2021, PhRvX, 11, 041050, doi: 10.1103/PhysRevX.11.041050

Krishnendu, N. V., Arun, K. G., & Mishra, C. K. 2017, PhRvL, 119, 091101, doi: 10.1103/PhysRevLett.119.091101

Krishnendu, N. V., Saleem, M., Samajdar, A., et al. 2019, PhRvD, 100, 104019, doi: 10.1103/PhysRevD.100.104019

Küchler, L., Compère, G., & Pound, A. 2026, CQGra, 43, 015018, doi: 10.1088/1361-6382/ae2b44

Kwok, J. Y. L., Lo, R. K. L., Weinstein, A. J., & Li, T. G. F. 2022, PhRvD, 105, 024066, doi: 10.1103/PhysRevD.105.024066

Lagos, M., & Hui, L. 2023, PhRvD, 107, 044040, doi: 10.1103/PhysRevD.107.044040

Lee, Y. S. C., Doshi, S., Millhouse, M., & Melatos, A. 2025, PhRvD, 111, 082002, doi: 10.1103/PhysRevD.111.082002

Lestingi, J., D'Addario, G., & Sotiriou, T. P. 2025, PhRvD, 112, 064070, doi: 10.1103/vdk4-tg61

Li, T. G. F., Del Pozzo, W., Vitale, S., et al. 2012a, PhRvD, 85, 082003, doi: 10.1103/PhysRevD.85.082003

—. 2012b, JPhCS, 363, 012028, doi: 10.1088/1742-6596/363/1/012028

LIGO Scientific, Virgo, and KAGRA Collaboration. 2025, LVK Algorithm Library - LALSuite, doi: 10.7935/GT1W-FZ16

Littenberg, T. B., Kanner, J. B., Cornish, N. J., & Millhouse, M. 2016, PhRvD, 94, 044050, doi: 10.1103/PhysRevD.94.044050

Lo, R. K. L., Li, T. G. F., & Weinstein, A. J. 2019, PhRvD, 99, 084052, doi: 10.1103/PhysRevD.99.084052

London, L., Shoemaker, D., & Healy, J. 2014, PhRvD, 90, 124032, doi: 10.1103/PhysRevD.90.124032

Lu, N., Ma, S., Piccinni, O. J., Sun, L., & Finch, E. 2025, PhRvD, 112, 064047, doi: 10.1103/h6jy-sd94

Ma, S. 2026. https://arxiv.org/abs/2604.08680

Ma, S., Lu, N., Finch, E., et al. 2026, QNMRF software, https://github.com/Sizheng-Ma/qnm_filter, GitHub

Ma, S., Sun, L., & Chen, Y. 2023a, PhRvD, 107, 084010, doi: 10.1103/PhysRevD.107.084010

—. 2023b, PhRvL, 130, 141401, doi: 10.1103/PhysRevLett.130.141401

Ma, S., & Yang, H. 2024, PhRvD, 109, 104070, doi: 10.1103/PhysRevD.109.104070

Ma, S., Mitman, K., Sun, L., et al. 2022, PhRvD, 106, 084036, doi: 10.1103/PhysRevD.106.084036

Macas, R., & Lundgren, A. 2023, PhRvD, 108, 063016, doi: 10.1103/PhysRevD.108.063016

Macleod, D., Goetz, E., Davis, D., et al. 2021a, gwdetchar/gwdetchar, v2.3.1, Zenodo, doi: 10.5281/zenodo.2575786

Macleod, D. M., Areeda, J. S., Coughlin, S. B., Massinger, T. J., & Urban, A. L. 2021b, SoftX, 13, 100657, doi: 10.1016/j.softx.2021.100657

Maggio, E., Silva, H. O., Buonanno, A., & Ghosh, A. 2023, PhRvD, 108, 024043, doi: 10.1103/PhysRevD.108.024043

Mahapatra, P., Datta, S., Gupta, I., et al. 2025, PhRvD, 112, 104007, doi: 10.1103/c1sj-jc4v

Mandel, I., Farr, W. M., & Gair, J. R. 2019, MNRAS, 486, 1086, doi: 10.1093/mnras/stz896

May, T., Ma, S., Ripley, J. L., & East, W. E. 2024, PhRvD, 110, 084034, doi: 10.1103/PhysRevD.110.084034

Mehta, A. K., Buonanno, A., Cotesta, R., et al. 2023, PhRvD, 107, 044020, doi: 10.1103/PhysRevD.107.044020

Meidam, J., Tsang, K. W., Goldstein, J., et al. 2018, PhRvD, 97, 044033, doi: 10.1103/PhysRevD.97.044033

Messick, C., Blackburn, K., Brady, P., et al. 2017, PhRvD, 95, 042001, doi: 10.1103/PhysRevD.95.042001

Miani, A., Lazzaro, C., Prodi, G. A., et al. 2023, PhRvD, 108, 064018, doi: 10.1103/PhysRevD.108.064018

Mihaylov, D. P., Ossokine, S., Buonanno, A., et al. 2025, SoftX, 30, 102080, doi: 10.1016/j.softx.2025.102080

Mirshekari, S., Yunes, N., & Will, C. M. 2012, PhRvD, 85, 024041, doi: 10.1103/PhysRevD.85.024041

Mishra, C. K., Arun, K. G., Iyer, B. R., & Sathyaprakash, B. S. 2010, PhRvD, 82, 064010, doi: 10.1103/PhysRevD.82.064010

Mitman, K., Lagos, M., Stein, L. C., et al. 2023, PhRvL, 130, 081402, doi: 10.1103/PhysRevLett.130.081402

Mozzon, S., Nuttall, L. K., Lundgren, A., et al. 2020, CQGra, 37, 215014, doi: 10.1088/1361-6382/abac6c

Nair, R., & Yunes, N. 2020, PhRvD, 101, 104011, doi: 10.1103/PhysRevD.101.104011

Narayan, P., Johnson-McDaniel, N. K., & Gupta, A. 2023, PhRvD, 108, 064003, doi: 10.1103/PhysRevD.108.064003

—. 2026, PhRvD, 113, 044007, doi: 10.1103/5hbl-hr2x

Necula, V., Klimenko, S., & Mitselmakher, G. 2012, JPhCS, 363, 012032, doi: 10.1088/1742-6596/363/1/012032

Newman, E. T., & Penrose, R. 1966, JMP, 7, 863, doi: 10.1063/1.1931221

Ng, T. C. K., Wong, I. C. F., & Cirok, B. 2026, nullpol software, GitHub. https://github.com/isaac-cf-wong/nullpol

Nishizawa, A., Taruya, A., Hayama, K., Kawamura, S., & Sakagami, M.-a. 2009, PhRvD, 79, 082002, doi: 10.1103/PhysRevD.79.082002

Nitz, A. H., Dent, T., Dal Canton, T., Fairhurst, S., & Brown, D. A. 2017, ApJ, 849, 118, doi: 10.3847/1538-4357/aa8f50

Pacilio, C., Gerosa, D., & Bhagwat, S. 2024, PhRvD, 109, L081302, doi: 10.1103/PhysRevD.109.L081302
Payne, E., Isi, M., Chatziioannou, K., & Farr, W. M. 2023, PhRvD, 108, 124060, doi: 10.1103/PhysRevD.108.124060
Penrose, R. 1969, NCimR, 1, 252
—. 2002, GReGr, 34, 1141, doi: 10.1023/A:1016578408204
Perrone, D., Barreira, T., Kehagias, A., & Riotto, A. 2024, NuPhB, 999, 116432, doi: 10.1016/j.nuclphysb.2023.116432
Pompili, L., Buonanno, A., Estellés, H., et al. 2023, PhRvD, 108, 124035, doi: 10.1103/PhysRevD.108.124035
Pompili, L., Buonanno, A., & Pürrer, M. 2025a, PhRvD, 112, 044062, doi: 10.1103/5kbh-83k7
Pompili, L., Maggio, E., Silva, H. O., & Buonanno, A. 2025b, PhRvD, 111, 124040, doi: 10.1103/ng8w-98sz
Pratten, G., García-Quirós, C., Colleoni, M., et al. 2021, PhRvD, 103, 104056, doi: 10.1103/PhysRevD.103.104056
Press, W. H., & Teukolsky, S. A. 1973, ApJ, 185, 649, doi: 10.1086/152445
Puecher, A., Kalaghatgi, C., Roy, S., et al. 2022, PhRvD, 106, 082003, doi: 10.1103/PhysRevD.106.082003
Ramos-Buades, A., Buonanno, A., Estellés, H., et al. 2023, PhRvD, 108, 124037, doi: 10.1103/PhysRevD.108.124037
Redondo-Yuste, J., Carullo, G., Ripley, J. L., Berti, E., & Cardoso, V. 2024a, PhRvD, 109, L101503, doi: 10.1103/PhysRevD.109.L101503
Redondo-Yuste, J., Pereñiguez, D., & Cardoso, V. 2024b, PhRvD, 109, 044048, doi: 10.1103/PhysRevD.109.044048
Robinet, F., Arnaud, N., Leroy, N., et al. 2020, SoftX, 12, 100620, doi: 10.1016/j.softx.2020.100620
Robinson, D. C. 1975, PhRvL, 34, 905, doi: 10.1103/PhysRevLett.34.905
Romero-Shaw, I. M., Talbot, C., Biscoveanu, S., et al. 2020, MNRAS, 499, 3295, doi: 10.1093/mnras/staa2850
Roy, S., Haney, M., Pratten, G., Pang, P. T. H., & Van Den Broeck, C. 2026, PhRvD, 113, 024016, doi: 10.1103/855k-sys5
Sachdev, S., Caudill, S., Fong, H., et al. 2019. https://arxiv.org/abs/1901.08580
Saleem, M., Datta, S., Arun, K. G., & Sathyaprakash, B. S. 2022, PhRvD, 105, 084062, doi: 10.1103/PhysRevD.105.084062
Sänger, E. M., Roy, S., Agathos, M., et al. 2026, PhRvD, 113, 084070, doi: 10.1103/r43k-51yq
Sberna, L., Bosch, P., East, W. E., Green, S. R., & Lehner, L. 2022, PhRvD, 105, 064046, doi: 10.1103/PhysRevD.105.064046
Siegel, H., Isi, M., & Farr, W. M. 2025, PhRvD, 111, 044070, doi: 10.1103/PhysRevD.111.044070
Silva, H. O., Ghosh, A., & Buonanno, A. 2023, PhRvD, 107, 044030, doi: 10.1103/PhysRevD.107.044030
Silva, H. O., Holgado, A. M., Cárdenas-Avendaño, A., & Yunes, N. 2021, PhRvL, 126, 181101, doi: 10.1103/PhysRevLett.126.181101
Skilling, J. 2006, BayAn, 1, 833, doi: 10.1214/06-BA127
Smith, J. R., Abbott, T., Hirose, E., et al. 2011, CQGra, 28, 235005, doi: 10.1088/0264-9381/28/23/235005
Speagle, J. S. 2020, MNRAS, 493, 3132, doi: 10.1093/mnras/staa278
Stein, L. C. 2019, JOSS, 4, 1683, doi: 10.21105/joss.01683
Su, J., Khera, N., Casals, M., et al. 2026, PhRvD, 113, 104013, doi: 10.1103/3n8r-8kbb
Sullivan, J. M., Ranjan, S., & Millhouse, M. 2025, IGWN Wave Compare. https://git.ligo.org/bayeswave/igwn-wave-compare
Teukolsky, S. A. 1973, ApJ, 185, 635, doi: 10.1086/152444
Tiwari, A., Vijaykumar, A., Kapadia, S. J., Ghosh, S., & Nielsen, A. B. 2026, PhRvD, 113, 104072, doi: 10.1103/qtw8-3g56
Toubiana, A., Pompili, L., Buonanno, A., Gair, J. R., & Katz, M. L. 2024, PhRvD, 109, 104019, doi: 10.1103/PhysRevD.109.104019
Tsang, K. W., Rollier, M., Ghosh, A., et al. 2018, PhRvD, 98, 024023, doi: 10.1103/PhysRevD.98.024023
Uchikata, N., Narikawa, T., Nakano, H., et al. 2023, PhRvD, 108, 104040, doi: 10.1103/PhysRevD.108.104040
Usman, S. A., Nitz, A. H., Harry, I. W., et al. 2016, CQGra, 33, 215004, doi: 10.1088/0264-9381/33/21/215004
van de Meent, M., Buonanno, A., Mihaylov, D. P., et al. 2023, PhRvD, 108, 124038, doi: 10.1103/PhysRevD.108.124038
Varma, V., Field, S. E., Scheel, M. A., et al. 2019, PhRvR, 1, 033015, doi: 10.1103/PhysRevResearch.1.033015
Vazsonyi, L., & Davis, D. 2023, CQGra, 40, 035008, doi: 10.1088/1361-6382/acafd2
Viets, A. D., Wade, M., Urban, A. L., et al. 2018, CQGra, 35, 095015, doi: 10.1088/1361-6382/aab658
Vijaykumar, A., Tiwari, A., Kapadia, S. J., Arun, K. G., & Ajith, P. 2023, ApJ, 954, 105, doi: 10.3847/1538-4357/acd77d
Virgo Collaboration. 2021, PythonVirgoTools, v5.1.1, git.ligo.org/virgo/virgoapp/PythonVirgoTools
Virtanen, P., Gommers, R., Oliphant, T. E., et al. 2020, NatMe, 17, 261, doi: 10.1038/s41592-019-0686-2
Vishveshwara, C. V. 1970, PhRvD, 1, 2870, doi: 10.1103/PhysRevD.1.2870
Wald, R. M. 1974, PhRvD, 10, 1680, doi: 10.1103/PhysRevD.10.1680
Waskom, M. L. 2021, JOSS, 6, 3021, doi: 10.21105/joss.03021
Wette, K. 2020, SoftX, 12, 100634, doi: 10.1016/j.softx.2020.100634
Whiting, B. F. 1989, JMP, 30, 1301, doi: 10.1063/1.528308
Will, C. M. 2014, LRR, 17, 4, doi: 10.12942/lrr-2014-4
Williams, D., Veitch, J., Chiofalo, M. L., et al. 2023, JOSS, 8, 4170, doi: 10.21105/joss.04170
Wong, I. C. F., Pang, P. T. H., Lo, R. K. L., Li, T. G. F., & Van Den Broeck, C. 2021. https://arxiv.org/abs/2105.09485
Yagi, K., Stein, L. C., Yunes, N., & Tanaka, T. 2012a, PhRvD, 85, 064022, doi: 10.1103/PhysRevD.85.064022

Yagi, K., Yunes, N., & Tanaka, T. 2012b, PhRvL, 109, 251105, doi: 10.1103/PhysRevLett.116.169902

Yunes, N., & Hughes, S. A. 2010, PhRvD, 82, 082002, doi: 10.1103/PhysRevD.82.082002

Yunes, N., & Pretorius, F. 2009, PhRvD, 80, 122003, doi: 10.1103/PhysRevD.80.122003

Yunes, N., Yagi, K., & Pretorius, F. 2016, PhRvD, 94, 084002, doi: 10.1103/PhysRevD.94.084002

Zackay, B., Venumadhav, T., Roulet, J., Dai, L., & Zaldarriaga, M. 2021, PhRvD, 104, 063034, doi: 10.1103/PhysRevD.104.063034

Zhong, H., Isi, M., Chatziioannou, K., & Farr, W. M. 2024, PhRvD, 110, 044053, doi: 10.1103/PhysRevD.110.044053

Zhong, H., Isi, M., Farr, W. M., & Chatziioannou, K. 2026, HIERFIT, ND Hierarchical Analysis. https://git.ligo.org/haowen.zhong/2d-nd-hierarchical-analysis

Zhu, H., Pretorius, F., Ma, S., et al. 2024, PhRvD, 110, 124028, doi: 10.1103/PhysRevD.110.124028

Zhu, H., et al. 2025, PhRvD, 111, 064052, doi: 10.1103/PhysRevD.111.064052

Zimmerman, A., Haster, C.-J., & Chatziioannou, K. 2019, PhRvD, 99, 124044, doi: 10.1103/PhysRevD.99.124044

Zweizig, J. 2006, The Data Monitor Tool Project, labcit.ligo.caltech.edu/~jzweizig/DMT-Project.html

The LIGO Scientific Collaboration, the Virgo Collaboration, and the KAGRA Collaboration, A. G. Abac,[1] A. Abe,[2] I. Abouelfettouh,[3] F. Acernese,[4,5] K. Ackley,[6] A. Adam,[7] S. Adhicary,[8] D. Adhikari,[9,10] R. X. Adhikari,[11] V. K. Adkins,[12] S. Afroz,[13] A. Agapito,[14] D. Agarwal,[15] M. Agathos,[16] N. Aggarwal,[17] S. Aggarwal,[18] O. D. Aguiar,[19] I.-L. Ahrend,[20] L. Aiello,[21,22] A. Ain,[23] P. Ajith,[24] T. Akutsu,[25,26] L. Albers,[27] W. Ali,[28,29] S. Al-Kershi,[9,10] C. Allene,[30] A. Allocca,[31,5] S. Al-Shammari,[32] J. A. Alvarez,[33] S. Alvarez-Lopez,[34] W. Amar,[35] O. Amarasinghe,[32] A. Amato,[36,37] F. Amicucci,[38,39] C. Amra,[40] A. B. Anand,[33] C. Anand,[41] A. Ananyeva,[11] S. B. Anderson,[11] W. G. Anderson,[11] M. Andia,[42] M. Ando,[43,44] F. Andrade-Oliveira,[45] M. Andrés-Carcasona,[34] J. L. Andrey,[46] T. Andrić,[47,48] J. Anglin,[49] J. Anna,[50] J. M. Antelis,[51] S. Antier,[42] T. Aoki,[52] M. Aoumi,[53] E. Z. Appavuravther,[9,10] E. A. Appelt,[54] S. Appert,[11] S. K. Apple,[55] K. Arai,[11] A. Araya,[56] M. C. Araya,[11] M. Arca Sedda,[47,48] F. Arciprete,[21,22] J. S. Areeda,[57] N. Aritomi,[58] F. Armato,[28,29] S. Armstrong,[59] N. Arnaud,[60] M. Arogeti,[61] S. M. Aronson,[49] G. Ashton,[62] Y. Aso,[53,63] L. Asprea,[64] M. Assiduo,[65,66] S. Assis de Souza Melo,[67] S. M. Aston,[68] P. Astone,[38] P. S. Aswathi,[69] F. Attadio,[39,38] F. Aubin,[70] K. AultONeal,[50] G. Avallone,[4] N. Avdeev,[64] E. A. Avila,[51] S. Babak,[20] C. Badger,[71] S. Bae,[72] S. Bagnasco,[64] S. Baimukhametova,[73,74] L. Baiotti,[75] T. Baka,[76,37] K. A. Baker,[7] T. Baker,[77] G. Balbi,[78] G. Baldi,[79,80] N. Baldicchi,[81,82] M. Ball,[83] G. Ballardin,[67] M. Ballelli,[47,48] S. W. Ballmer,[84] S. Banagiri,[41] B. Banerjee,[47] D. Bankar,[85] T. M. Baptiste,[12] P. Baral,[86] M. Baratti,[87,88] J. C. Barayoga,[11] K. Baric,[11] B. C. Barish,[11] D. Barker,[3] D. Barman,[89] N. Barman,[85] F. Barone,[90,5] B. Barr,[91] M. Barrios,[33] L. Barsotti,[34] M. Barsuglia,[20] D. Barta,[92] M. A. Barton,[91] I. Bartos,[49] A. Basalaev,[9,10] R. Bassiri,[93] A. Basti,[88,87] M. Bawaj,[81,82] J. C. Bayley,[91] A. C. Baylor,[86] P. A. Baynard II,[61] M. Bazzan,[94,95] V. M. Bedakihale,[96] F. Beirnaert,[97] M. Bejger,[98] A. S. Bell,[91] C. Bellani,[99] D. S. Bellie,[100] D. Beltran-Martinez,[101] E. Benedetti,[38] W. Benoit,[18] I. Bentara,[60] M. Ben Yaala,[59] S. Bera,[102] F. Bergamin,[32] B. K. Berger,[93] M. Beroiz,[11] C. P. L. Berry,[91] I. Berry,[103] D. Bersanetti,[28] T. Bertheas,[104] A. Bertolini,[37,36] J. Betzwieser,[68] D. Beveridge,[7] N. Bevins,[105] J. Bezerra-Sobrinho,[106] R. Bhandare,[107] R. Bhatt,[11] A. Bhattacharjee,[108] D. Bhattacharjee,[109,110] S. Bhattacharyya,[111] S. Bhaumik,[112] V. Biancalana,[113] F. Bianchi,[82] I. A. Bilenko,[114] M. Bilicki,[115] G. Billingsley,[11] A. Binetti,[99] S. Bini,[11] S. Biot,[116] O. Birnholtz,[117] S. Biscoveanu,[118] A. Bisht,[10] M. Bitossi,[67,87] M.-A. Bizouard,[119] S. Blaber,[120] J. K. Blackburn,[11] L. A. Blagg,[121] C. D. Blair,[7,68] D. G. Blair,[7] M. Bloch,[122] N. Bode,[9,10] N. Boettner,[27] P. Bogdan,[123] G. Boileau,[119] M. Boldrini,[67] G. N. Bolingbroke,[124] L. D. Bonavena,[49] V. A. Bonhomme,[34] E. Bonilla,[93] M. S. Bonilla,[57] A. Bonino,[83] R. Bonnand,[35,125] A. Borchers,[9,10] N. Borghi,[126,78] V. Boschi,[87] S. Bose,[127] V. Bossilkov,[68] Y. Bothra,[37,128] A. Boudon,[60] T. D. Boybeyi,[18] M. Boyle,[129] A. Bozzi,[67] C. Bradaschia,[87] M. J. Brady,[130] P. R. Brady,[86] A. Branch,[68] M. Branchesi,[47,48] T. Briant,[131] A. Brillet,[119,*] M. Brinkmann,[9,10] P. Brockill,[86] E. Brockmueller,[9,10] A. F. Brooks,[11] D. D. Brown,[124] M. L. Brozzetti,[81,82] S. Brunett,[11] G. Bruno,[15] R. Bruntz,[123] J. Bryant,[132] Y. Bu,[133] F. Bucci,[66] A. Buchicchio,[39] A. Buggiani,[67] O. Bulashenko,[134,135] T. Bulik,[136] H. J. Bulten,[37] A. Buonanno,[137,1] K. Burtnyk,[3] R. Buscicchio,[138,139] N. Busdon,[94] D. Buskulic,[35] R. L. Byer,[93] R. Cabrita,[15] V. A. Cáceres-Barbosa,[8] L. Cadonati,[61] G. Cagnoli,[94] C. Cahillane,[84] A. Calafat,[83] J. Calderón Bustillo,[140] J. D. Callaghan,[91] T. A. Callister,[141] E. Calloni,[31,5] S. R. Callos,[121] K. Cannon,[44] V. Cantory,[18] H. Cao,[34] L. A. Capistran,[142] E. Capocasa,[20] G. Capoccia,[82] E. Capote,[3] C. Capuano,[84] G. Capurri,[88,87] F. Carbognani,[67] K. J. Cardona-Martínez,[12] M. Carlassara,[9,10] M. Carpinelli,[138,67] G. Carrillo,[121] G. Carullo,[132] A. Casallas-Lagos,[143] J. Casanueva Diaz,[67] C. Casentini,[144,22] S. Caudill,[145] M. Cavaglià,[110] R. Cavalieri,[67] A. Ceja,[100] G. Cella,[87] P. Cerdá-Durán,[146,147] E. Cesarini,[22] N. Chabbra,[69] W. Chaibi,[119] A. Chakraborty,[13] P. Chakraborty,[9,10] S. Chakraborty,[107] S. Chalathadka Subrahmanya,[27] C. Chan,[148] J. C. L. Chan,[149] M. Chan,[120] K. Chandra,[1] C.-Y. Chang,[150] K. Chang,[151] S. Chao,[151] P. Charlton,[152] E. Chassande-Mottin,[20] C. Chatterjee,[54] Debarati Chatterjee,[85] Deep Chatterjee,[34] M. Chaturvedi,[107] S. Chaty,[20] K. Chatziioannou,[11] A. Chen,[153] A. H.-Y. Chen,[154] D. Chen,[155] H. Chen,[150] H. Y. Chen,[156] S. Chen,[54] Yanbei Chen,[157] Yiwen Chen,[18] G. Cheng,[153] H. P. Cheng,[103] P. Chessa,[81,82] T. Cheunchitra,[133] H. T. Cheung,[158] S. Y. Cheung,[41] F. Chiadini,[159,160] G. Chiarini,[9,10] A. Chiba,[161] A. Chincarini,[28] D. Chintala,[109] A. Chiummo,[5,67] A. Chopra,[47] C. Chou,[162] S. Choudhary,[7] N. Christensen,[119,163] Y. K. Chu,[86] S. S. Y. Chua,[69] G. Ciani,[79,80] P. Ciecielag,[98] M. Cieślar,[136] M. Cifaldi,[22] B. Cirok,[164] F. Clara,[3] J. A. Clark,[11,61] T. A. Clarke,[118] A. Claveus,[165] M. R. Claypool,[121] S. Clesse,[116] F. Cleva,[119] S. M. Clyne,[130] E. Coccia,[47,48,166] E. Codazzo,[167] P.-F. Cohadon,[131] D. E. Cohen,[9,10] E. Colangeli,[77] O. Cole,[148] M. Colleoni,[83] C. G. Collette,[168] J. Collins,[68] S. Colloms,[91] A. Colombo,[38,169] G. Compère,[116] C. M. Compton,[3] G. Connolly,[121] L. Conti,[95] T. R. Corbitt,[12] I. Cordero-Carrión,[170] S. Corezzi,[81,82] M. Corman,[1] N. J. Cornish,[171] A. Corsi,[172] S. Cortese,[67] L. A. Corubolo,[21,22] L. Cotnoir,[123] R. Cottingham,[68] J. A. Cotturone,[100] M. W. Coughlin,[18] P. Couvares,[11,61] R. Coyne,[130] A. Cozzumbo,[47] J. D. E. Creighton,[86] T. D. Creighton,[173] S. Crook,[68] R. Crouch,[3] J. Csizmazia,[3] K. Csukás,[92] T. J. Cullen,[11] A. Cumming,[91] E. Cuoco,[126,78] M. Cusinato,[146] R. R. Cuzinatto,[174] L. V. da Conceição,[175] T. Dal Canton,[42] S. Dall'Osso,[78,126] S. Dal Pra,[176] G. Dálya,[104] Y. Dang,[8] B. D'Angelo,[28] S. Danilishin,[36,37] O. Danner,[108] S. D'Antonio,[38] K. Danzmann,[9,10] K. E. Darroch,[123] L. P. Dartez,[68] R. Das,[111] S. Das,[85] A. Dasgupta,[96] V. Dattilo,[67] A. Daumas,[20] I. Dave,[107] A. Davenport,[177] T. F. Davies,[7] D. Davis,[130] M. C. Davis,[18] P. Davis,[178,179] E. J. Daw,[180] M. Dax,[1] J. De Bolle,[97] E. deBruin,[18] M. Deenadayalan,[85] J. Degallaix,[181] M. De Laurentis,[31,5] C. J. Delgado Mendez,[101] F. De Lillo,[23] S. Della Torre,[139] W. Del Pozzo,[88,87] O. M. del Rio,[182] A. Demagny,[35] F. De Marco,[39,38] G. Demasi,[183,66] F. De Matteis,[21,22] C. de Melo,[174] N. Demos,[34] T. Dent,[140] A. Depasse,[15] N. DePergola,[105] R. De Pietri,[184,185] R. De Rosa,[31,5] C. De Rossi,[67] E. K. Derrick,[186] M. Desai,[34] D. DeSantis,[34] S. Deshmukh,[54] V. Deshmukh,[91] R. De Simone,[159,160] S. Determan,[187] S. Dhage,[15] A. Dhani,[1] R. Dhatri,[46] R. Dhurkunde,[77] R. Diab,[49] C. Diaz,[101] M. C. Díaz,[173] F. Diaz Guerra,[188,189] M. Di Cesare,[31,5] M. A. Dicorato,[82,190] T. Dietrich,[1] C. Di Fronzo,[7]

M. Di Giovanni,[191,87] D. Diksha,[37,36] J. Ding,[20,192] S. Di Pace,[39,38] I. Di Palma,[39,38] D. Di Piero,[188,189] F. Di Renzo,[66,183] Divyajyoti,[32] A. Dmitriev,[132] J. P. Docherty,[91] Z. Doctor,[100] N. Doerksen,[175] E. Dohmen,[3] A. Doke,[145] A. Domiciano De Souza,[193] L. D'Onofrio,[5] F. Donovan,[34] K. L. Dooley,[32] S. Doravari,[85] O. Dorosh,[194] F. Dosopoulou,[32] M. Drago,[39,38] J. C. Driggers,[3] M. Dubois,[104] R. S. Dumbreck,[32] U. Dupletsa,[47] D. D'Urso,[195,167] P. Dutta Roy,[49] H. Duval,[196] S. Dwivedi,[197] S. E. Dwyer,[3] C. Eassa,[3] W. East,[198] M. Eberhardt,[187] M. Ebersold,[45] M. Ebiri,[199] G. Eddolls,[84] A. Effler,[68] J. Eichholz,[132] H. Einsle,[119] M. Eisenmann,[25] M. Emma,[62] K. Endo,[161] R. Enficiaud,[1] V. Ernst,[15,200] L. Errico,[31,5] R. Espinosa,[173] M. Esposito,[5,31] R. C. Essick,[201] H. Estellés,[83] T. Etzel,[11] M. Evans,[34] T. Evstafyeva,[198] J. M. Ezquiaga,[149] F. Fabrizi,[65,66] V. Fafone,[21,22] S. Fairhurst,[32] X. Fan,[153] A. M. Farah,[201] B. Farr,[121] W. M. Farr,[202,203] M. Favata,[204] M. Fays,[200] M. Fazio,[59] J. Feicht,[11] M. M. Fejer,[93] J.-N. Feldhusen,[27] E. Fenyvesi,[92,205] A. Feo,[184,185] J. Fernandes,[112] T. Fernandes,[206,146] G. Fernández Rodríguez,[170] D. Fernando,[199] S. Ferraiuolo,[207,39,38] T. A. Ferreira,[19] M. Ferrer-Martinez,[83] F. Fidecaro,[88,87] P. Figura,[98] E. Finch,[11] I. Fiori,[67] M. Fishbach,[201] R. P. Fisher,[123] S. K. Fitzgerald,[91] V. Fiumara,[208,160] R. Flaminio,[35] B. Flanagan,[32] S. M. Fleischer,[182] L. S. Fleming,[209] F. Flocco,[94] E. Floden,[18] H. Fong,[120] J. A. Font,[146,147] F. Fontinele-Nunes,[18] C. Foo,[1] B. Fornal,[210] P. W. F. Forsyth,[69] A. Fragkos,[211,74] N. Franchini,[212] A. Franco-Ordovas,[11] F. Frappez,[35] F. Frasconi,[87] J. P. Freed,[50] Z. Frei,[213] A. Freise,[37,128] O. Freitas,[206,146] R. Frey,[121] W. Frischhertz,[68] P. Fritschel,[34] V. V. Frolov,[68] M. Fuentes-Garcia,[11] R. Fujii,[161] T. Fujimori,[2] Y. Fujiwara,[214] P. Fulda,[49] M. Fyffe,[68] J. R. Gair,[1] S. Galaudage,[193] V. Galdi,[215] M. Galimberti,[67] A. Gamboa,[1] S. Gamoji,[216] A. Ganguly,[85] B. Garaventa,[28] P. García Abia,[101] J. García-Bellido,[217] C. García-Quirós,[83] J. W. Gardner,[69] S. Garg,[44] J. Gargiulo,[67] X. Garrido,[42] A. Garron,[83] F. Garufi,[31,5] P. A. Garver,[93] C. Gasbarra,[218,22] F. Gautier,[219] V. Gayathri,[86] T. Gayer,[84] G. Gemme,[28] A. Gennai,[87] V. Gennari,[104] J. George,[107] R. George,[156] O. Gerberding,[27] L. Gergely,[164] A. Ghinassi,[126,78] Archisman Ghosh,[97] Sayantan Ghosh,[112] Shaon Ghosh,[204] Shrobana Ghosh,[9,10] Suprovo Ghosh,[220] Tathagata Ghosh,[85,221] J. A. Giaime,[12,68] K. D. Giardina,[68] D. R. Gibson,[209] C. Gier,[59] F. Gittins,[76] J. Glanzer,[11] F. Glotin,[42] E. Glowacki,[222] J. Godfrey,[121] R. V. Godley,[9,10] O. Godwin,[11] A. S. Goettel,[223] E. Goetz,[120] J. Golomb,[11] S. Gomez Lopez,[39,38] G. González,[12] P. Goodarzi,[46] S. R. Goode,[41] A. Goodwin-Jones,[15] M. Gosselin,[67] S. M. Goss-Grubbs,[18] C. Gostiaux,[70] R. Gouaty,[35] D. W. Gould,[69] D. Goupilliere,[179,178] K. Govorkova,[34] A. Grado,[81,82] V. Graham,[91] A. E. Granados,[18] M. Granata,[181] V. Granata,[224,160] S. Gras,[34] P. Grassia,[11] C. Gray,[3] R. Gray,[91] G. Greco,[82] A. C. Green,[37,36] L. Green,[225] S. R. Green,[223] A. M. Gretarsson,[50] E. M. Gretarsson,[50] D. Griffith,[11] L. Grimaldi,[9] C. Grimaud,[35] H. Grote,[32] S. Grunewald,[1] A. G. Guerrero,[226] G. M. Guidi,[65,66] T. Guidry,[3] H. K. Gulati,[96] F. Gulminelli,[178,179] H. Guo,[153] W. Guo,[7] Y. Guo,[37] A. Gupta,[89] I. Gupta,[100] N. C. Gupta,[96] S. K. Gupta,[49] V. Gupta,[18] N. Gupte,[1] N. Guttman,[41] F. Guzman,[142] M. Haberland,[1] S. Haino,[227] E. D. Hall,[34] E. Z. Hamilton,[83] G. Hammond,[91] W.-B. Han,[228] M. Haney,[37] J. Hanks,[3] C. Hanna,[8] M. D. Hannam,[32] O. A. Hannuksela,[229] H. Hansen,[3] J. Hanson,[68] R. Harada,[44] A. R. Hardison,[187] S. Harikumar,[98] K. Haris,[230] I. Harley-Trochimczyk,[142] J. Harms,[47,48] G. M. Harry,[231] I. W. Harry,[77] M. T. Hartman,[40,232,20] B. Haskell,[233,234] C.-J. Haster,[225] K. Haughian,[91] H. Hayakawa,[53] K. Hayama,[235] J. Hedberg,[50] A. Heffernan,[83] D. Hegde,[15] M. C. Heintze,[68] J. Heinzel,[34] H. Heitmann,[119] F. Hellman,[33] A. F. Helmling-Cornell,[186] G. Hemming,[67] O. Henderson-Sapir,[124] M. Hendry,[91] I. S. Heng,[91] M. H. Hennig,[91] C. Henshaw,[61] A. Heranval,[8] M. Heurs,[9,10] A. L. Hewitt,[236,237] J. Heynen,[15] J. Heyns,[34] S. Hido,[221] S. Hild,[36,37] M. Hill,[123] S. Hill,[91] Y. Himemoto,[238] C. Hirose,[53] D. Hofman,[181] N. A. Holland,[11] K. Holley-Bockelmann,[54] I. J. Hollows,[180] D. E. Holz,[226] L. Honet,[116] K. M. Hoops,[216] M. E. Hoque,[239] D. J. Horton-Bailey,[33] J. Hough,[91] S. Hourihane,[11] N. T. Howard,[54] E. J. Howell,[7] C. G. Hoy,[77] P. Hsi,[34] H.-Y. Hsieh,[240] C. Hsiung,[241] S.-H. Hsu,[242] W.-F. Hsu,[99] H. Y. Huang,[151] Y. Huang,[8] A. D. Huddart,[243] B. Hughey,[50] D. C. Y. Hui,[244] S. Husa,[83] L. Iampieri,[39,38] G. A. Iandolo,[36] M. Ianni,[22,21] Y. Ichinose,[221] K. Ide,[214] R. Iden,[245] A. Ierardi,[47,48] S. Ikeda,[155] H. Imafuku,[44] K. Imai,[221] Y. Inoue,[151] P. Iosif,[188,189] J. Irwin,[91,76] K. Ishida,[2] R. Ishikawa,[214] T. Ishikawa,[52] H. Ishino,[2] M. Isi,[246,203] K. S. Isleif,[247] Y. Itoh,[2,248] S. Iwaguchi,[52] M. M. Iwaya,[32,221] B. R. Iyer,[24] C. Jacquet,[104] T. Jacquot,[42] S. J. Jadhav,[249] S. P. Jadhav,[148] K. Jain,[32] A. L. James,[11] K. Jani,[54] S. Jani,[18] J. Janquart,[15,250] N. N. Janthalur,[249] S. Jaraba,[251] P. Jaranowski,[222] R. Jaume,[83] W. Javed,[32] M. Jensen,[3] W. Jia,[34] J. Jiang,[103] H.-B. Jin,[252,253] S.-J. Jin,[7] G. R. Johns,[123] N. A. Johnson,[49] N. K. Johnson-McDaniel,[89] M. C. Johnston,[225] R. Johnston,[91] N. Johny,[9,10] D. H. Jones,[69] D. I. Jones,[220] R. Jones,[91] P. Joshi,[61] S. K. Joshi,[85] G. Joubert,[60] J. Ju,[254] L. Ju,[7] I. L. Juarez-Reyes,[121] K. Jung,[255] H. B. Kabagoz,[34] B. Kacskovics,[92] T. Kajita,[221] I. Kaku,[2] V. Kalogera,[100] M. Kalomenopoulos,[225] M. Kamiizumi,[53] N. Kanda,[248,2] S. Kandhasamy,[85] G. Kang,[256] J. B. Kanner,[11] S. J. Kapadia,[85] D. P. Kapasi,[57] A. Karia,[37,128] A. S. Karia,[257] R. Kashyap,[112] M. Kasprzack,[11] H. Kato,[161] T. Kato,[221] E. Katsavounidis,[34] W. Katzman,[68] R. Kaushik,[107] K. Kawabe,[3] S. Kawamura,[52] D. Keitel,[83] S. A. Kemper,[55] L. J. Kemperman,[124] J. Kennington,[8] R. Kesharwani,[85] J. S. Key,[258] R. Khadela,[9,10] S. S. Khadkikar,[8] F. Y. Khalili,[114] C. Khamar,[201] F. Khan,[9,10] M. Khursheed,[107] N. M. Khusid,[202,203] W. Kiendrebeogo,[259] C. Kim,[260] G. Kim,[261] J. C. Kim,[262] K. Kim,[263] M. H. Kim,[254] S. Kim,[244] Y.-M. Kim,[263] C. Kimball,[100] K. Kimes,[57] M. Kinnear,[32] J. S. Kissel,[3] S. Klimenko,[49] A. M. Knee,[158] N. Knust,[9,10] K. Kobayashi,[221] S. M. Koehlenbeck,[93] K. Kohri,[264] K. Kokeyama,[32,52] S. Koley,[47,200] P. Kolitsidou,[83] A. E. Koloniari,[265] K. Komori,[25,43] K. Kompanets,[18] A. K. H. Kong,[266] A. Kontos,[186] K. Kopczuk,[109] L. M. Koponen,[132] M. Korobko,[27] X. Kou,[18] N. Kouvatsos,[71] T. Koyama,[161] D. B. Kozak,[11] E. Kraja,[67] S. L. Kranzhoff,[36,37] V. Kringel,[9,10] N. V. Krishnendu,[132] S. Kroker,[267] A. Królak,[268,194] K. Kruska,[9,10] J. Kubisz,[269] K. Kubota,[221] G. Kuehn,[9,10] D. Kukla,[18] A. Kulur Ramamohan,[69] Achal Kumar,[49] Anil Kumar,[249] Dhruv Kumar,[8,91] Praveen Kumar,[140] Prayush Kumar,[24] Rahul Kumar,[3] Rakesh Kumar,[96] Ravi Kumar,[18] J. Kume,[270,44] K. Kuns,[34] N. Kuntimaddi,[32] S. Kuroyanagi,[217,271,272] K. Kwak,[255] K. Kwan,[69] S. Kwon,[44] G. Lacaille,[91] A. Laeuger,[157] D. Laghi,[45] A. H. Laity,[130] N. Lajili,[125,273] A. Lakhal,[131]

E. Lalande,[274] M. Lalleman,[23] S. Lalvani,[100] M. Landry,[3] R. N. Lang,[34] A. Lange,[18] J. A. Lange,[64] R. Langgin,[225] B. Lantz,[93] I. La Rosa,[83] O. Laske,[8] P. D. Lasky,[41] L. Lavezzi,[64] J. Lawrence,[173] M. Laxen,[68] A. Lazzarini,[11] C. Lazzaro,[275, 167] P. Leaci,[39, 38] L. Lealı,[18] Y. K. Lecoeuche,[120] H. W. Lee,[276] J. Lee,[84] K. Lee,[254] R.-K. Lee,[150] R. Lee,[34] Sungho Lee,[277] Sunjae Lee,[254] W. Lee,[255] Y. Lee,[151] F. Legger,[64] I. N. Legred,[11] J. Lehmann,[9, 10] L. Lehner,[198] M. Le Jean,[181, 125] A. Lemaître,[278] R. Lemrani Alaoui,[125, 273] M. Lenti,[66, 183] M. Leonardi,[79, 80, 279] M. Lequime,[40] M. Lesovsky,[11] N. Letendre,[35] M. Lethuillier,[60] Y. Levin,[41] S. Lexmond,[128] K. Leyde,[202, 203] A. K. Y. Li,[44] F. Li,[69] K. L. Li,[280] T. G. F. Li,[99] X. Li,[157] Y. Li,[100] Z. Li,[91] Q. Liang,[153] C-Y. Lin,[281] E. T. Lin,[282] F. Lin,[151] L. C.-C. Lin,[280] Y.-C. Lin,[282] C. Lindsay,[209] S. D. Linker,[216] A. Liu,[229] F. Liu,[42] G. C. Liu,[241] Jian Liu,[7] S. Liu,[153] F. Llamas Villarreal,[173] J. Llobera-Querol,[83] R. K. L. Lo,[149] J.-P. Locquet,[99] S. C. G. Loggins,[165] L. T. London,[71] A. Longo,[65, 66] M. Lopez Portilla,[76] A. Lorenzo-Medina,[140] V. Loriette,[42] M. Lormand,[68] M. Lorusso,[78] G. Losurdo,[191, 87] T. P. Lott IV,[229] J. D. Lough,[9, 10] H. A. Loughlin,[34] C. O. Lousto,[199] N. K. Y. Low,[133] N. Lu,[69] H. Lück,[9, 10] O. Lukina,[34] D. Lumaca,[22] A. P. Lundgren,[283, 284] L. Lunghini,[67] A. W. Lussier,[274] L.-T. Ma,[285] S. Ma,[198] X. Ma,[46] M. Ma'arif,[151] S. MacBride,[45] K. Machida,[161] D. M. Macleod,[32] I. A. O. MacMillan,[11] A. Macquet,[42] S. S. Madekar,[166] S. Maenaut,[99] S. S. Magare,[85] R. M. Magee,[11] E. Maggio,[1, 38] M. Magnozzi,[28, 29] P. Mahapatra,[32] M. Mahesh,[27] S. Majhi,[85] E. Majorana,[39, 38] C. N. Makarem,[11] E. Makelele,[109] N. Malagon,[199] D. Malakar,[110] J. A. Malaquias-Reis,[19] U. Mali,[201] S. Maliakal,[11] A. Malik,[107] L. Mallick,[175, 201] A.-K. Malz,[62] N. Man,[119] M. Mancarella,[102] V. Mandic,[18] V. Mangano,[195, 167] Z. Mangi,[199] B. Mannix,[121] G. L. Mansell,[84] M. Manske,[86] M. Mantovani,[67] M. Mapelli,[94, 95, 286] S. Marchetti,[94, 95] F. Marion,[35] J. Mark,[18] A. S. Markosyan,[93] J. Markus,[18] E. Maros,[11] S. Marsat,[104] F. Martelli,[65, 66] I. W. Martin,[91] R. M. Martin,[204] B. B. Martinez,[142] M. Martinez,[166, 287] V. Martinez,[288] A. Martini,[79, 80] Juan Carlos Martins,[289] Julio C. Martins,[19] D. V. Martynov,[132] E. J. Marx,[34] L. Massaro,[36, 37] A. Masserot,[35] M. Masso-Reid,[91] T. Masters,[109] S. Mastrogiovanni,[38] G. Mastropasqua,[78] M. Matiushechkina,[9, 10] A. Matte-Landry,[274] L. Maurin,[219] N. Mavalvala,[34] N. Maxwell,[3] A. McCann,[121] G. McCarrol,[68] R. McCarthy,[3] D. E. McClelland,[69] S. McCormick,[68] L. McCuller,[11] L. I. McDermott,[127] C. McElhenny,[123] G. I. McGhee,[91] K. B. M. McGowan,[54] J. McIver,[120] A. McLeod,[7] I. McMahon,[45] T. McRae,[69] R. McTeague,[91] K. McWhirter,[8] D. Meacher,[86] B. N. Meagher,[84] R. Mechum,[199] L. G. Medeiros,[106] R. M. Mehta,[18] A. Melatos,[133] C. S. Menoni,[177] R. A. Mercer,[86] L. Mereni,[181] K. Merfeld,[121] E. L. Merilh,[68] J. R. Mérou,[83] C. Messick,[86] M. Meyer-Conde,[30] F. Meylahn,[9, 10] H. Miao,[290] C. Michel,[181] Y. Michimura,[44] H. Middleton,[132] D. P. Mihaylov,[109] S. J. Miller,[11] M. Millhouse,[61] E. Milotti,[188, 189] V. Milotti,[94] E. Minakaki,[128] Y. Minenkov,[22] Ll. M. Mir,[166] L. Mirasola,[291] C.-A. Miritescu,[166] A. Mishra,[24] C. Mishra,[111] T. Mishra,[77] A. Mitchell,[93] J. G. Mitchell,[50] O. Mitchem,[121] K. Mitman,[129] S. Mitra,[85] V. P. Mitrofanov,[114] K. Mitsuhashi,[25] R. Mittleman,[34] O. Miyakawa,[53] S. Miyoki,[53] G. Mo,[11] L. Mobilia,[65, 66] S. R. P. Mohapatra,[11] M. Molina-Ruiz,[33] M. Mondin,[216] M. Montani,[65, 66] G. Montefusco,[179] C. J. Moore,[236] D. Moraru,[3] A. More,[85] S. More,[85] C. Moreno,[292] E. A. Moreno,[34] G. Moreno,[3] A. Moreso Serra,[134] C. Morgan,[32] S. Morisaki,[221] S. Moriwaki,[221] Y. Moriwaki,[161] G. Morras,[217] A. Moscatello,[94] M. Mould,[223] B. Mours,[70] C. M. Mow-Lowry,[37, 128] L. Muccillo,[183, 66] F. Muciaccia,[39, 38] Arunava Mukherjee,[239] D. Mukherjee,[132] Samanwaya Mukherjee,[24] Soma Mukherjee,[173] Subroto Mukherjee,[96] Suvodip Mukherjee,[13] N. Mukund,[34] A. Mullavey,[68] C. L. Mungioli,[7] Y. Murakami,[221] M. Murakoshi,[214] P. G. Murray,[91] D. Nabari,[79, 80] S. Nadji,[181] A. Nagar,[64, 293] N. Nagarajan,[1] K. Nakagaki,[53] A. Nakamura,[52] K. Nakamura,[25] H. Nakano,[294] M. Nakano,[11] D. Nanadoumgar-Lacroze,[166] D. Nandi,[12] V. Napolano,[67] S. U. Naqvi,[111] P. Narayan,[89] A. Nardecchia,[39, 38] I. Nardecchia,[22] T. Narikawa,[221] H. Narola,[76] R. S. Nathan,[69] L. Naticchioni,[38] R. K. Nayak,[295] J. Neeson,[32] L. Negri,[76] A. Nela,[91] C. Nelle,[121] A. Nelson,[142] T. J. N. Nelson,[68] A. Nemmani,[98] A. Neunzert,[3] M. Newell,[16] S. Ng,[57] T. C. K. Ng,[37, 76] L.-A. T. Nguyen,[296] T. T. Nguyen,[296] L. Nguyen Quynh,[296] A. B. Nielsen,[297] Y. Nishino,[25, 298] A. Nishizawa,[299] S. Nissanke,[300, 37] W. Niu,[8] F. Nocera,[67] J. Noller,[301] M. Norman,[32] C. North,[32] J. Novak,[251, 302] G. Nurbek,[173] L. K. Nuttall,[77] K. Obayashi,[214] J. Oberling,[3] C. E. Ochoa,[46] C. O'Connor,[84] J. O'Dell,[243] E. Oelker,[34] M. Oertel,[251, 302] G. Oganesyan,[47, 48] J. J. Oh,[262] T. O'Hanlon,[68] M. Ohashi,[53, 30] F. Ohme,[9, 10] Y. Okabe,[161] I. Oke,[59] R. Oliveira,[303] R. Omer,[18] N. O'Neill,[84] M. Onishi,[161] K. Oohara,[304, 305] P. Ophardt,[247] R. J. Oram,[68] B. O'Reilly,[68] R. O'Shaughnessy,[199] S. Oshino,[53] J. Ostrovska,[132] A. Osumi,[52] I. Ota,[12] G. Othman,[247] M. Otsuka,[25, 298] D. J. Ottaway,[124] A. Ouzriat,[60] H. Overmier,[68] B. J. Owen,[108] A. E. Pace,[8] M. A. Page,[25] A. Pai,[112] S. Pal,[295] M. A. Palaia,[87, 88] M. Pálfi,[213] C. Palomba,[38] H. Pan,[266] J. Pan,[7] K.-C. Pan,[150, 282] P. K. Panda,[249] Shiksha Pandey,[8] Swadha Pandey,[34] P. T. H. Pang,[37, 76] F. Pannarale,[39, 38] B. C. Pant,[107] F. H. Panther,[7] M. Panzeri,[65, 66] F. Paoletti,[87] A. Paoli,[67] A. Paolone,[38, 306] A. Papadopoulos,[91] E. E. Papalexakis,[46] L. Papalini,[87, 88] G. Papigkiotis,[265] A. Paquis,[42] A. Parisi,[81, 82] B.-J. Park,[277] Jihwan Park,[260] Junegyu Park,[261] W. Parker,[68] G. Pascale,[9, 10] D. Pascucci,[97] A. Pasqualetti,[67] L. Passenger,[41] D. Passuello,[87] O. Patane,[3] A. V. Patel,[151] L. Pathak,[85] A. Patra,[32] B. Patricelli,[88, 87] B. G. Patterson,[32] K. Paul,[111, 37] S. Paul,[121] E. Payne,[11] T. Pearce,[32] M. Pedraza,[11] A. Pele,[11] F. E. Peña Arellano,[216] X. Peng,[132] Y. Peng,[61] S. Penn,[84, 307] A. Perreca,[47, 48] J. Perret,[20] D. Pesios,[265] S. Petracca,[215] C. Petrillo,[81] H. P. Pfeiffer,[1] H. Pham,[68] K. A. Pham,[18] K. S. Phukon,[132] H. Phurailatpam,[229] M. Piarulli,[104] L. Piccari,[39, 38] O. J. Piccinni,[83] M. Pichot,[119] A. Pied,[91] M. Piendibene,[88, 87] F. Piergiovanni,[65, 66] L. Pierini,[38] G. Pierra,[38] V. Pierro,[308, 160] M. Pillas,[309, 42] B. Pillon,[50] L. Pinard,[181] I. M. Pinto,[308, 160, 310, 31] M. Pinto,[67] B. J. Piotrzkowski,[86] M. Pirello,[3] A. Pisarski,[222] M. D. Pitkin,[236, 91] E. Placidi,[39, 38] M. L. Planas,[1] C. Plunkett,[34] R. Poggiani,[88, 87] E. Polini,[119] M. Polo,[101] J. Pomper,[87, 88] L. Pompili,[223] J. Poon,[229] E. Porcelli,[37] A. S. Porter,[108] E. K. Porter,[20] C. Posnansky,[8] J. Powell,[148] G. S. Prabhu,[85] M. Pracchia,[200] A. K. Prajapati,[96] K. Prasai,[311] R. Prasanna,[249] P. Prasia,[312] G. Pratten,[132] G. Principe,[188, 189] G. A. Prodi,[79, 80] P. Prosperi,[87] P. Prosposito,[21, 22] A. Puecher,[1] J. Pullin,[12] M. Punturo,[82] P. Puppo,[38] M. Pürrer,[130]

H. Qi,[16] M. Qiao,[153] J. Qin,[69] G. Quéméner,[179,125] V. Quetschke,[173] P. J. Quinonez,[50] R. Rading,[247] I. Rainho,[146] S. Raja,[107] C. Rajan,[107] B. Rajbhandari,[108] M. R. Raj Sah,[13] K. E. Ramirez,[68] F. A. Ramis Vidal,[83] M. Ramos Arevalo,[173] A. Ramos-Buades,[83] S. Ranjan,[61] M. Ranjbar,[46] K. Ransom,[68] P. Rapagnani,[39,38] B. Ratto,[50] A. Ravichandran,[145] A. Ray,[100] V. Raymond,[32] M. Razzano,[88,87] J. Read,[57] J. Redepenning,[18] J. Regan,[225] T. Regimbau,[35] T. Reichardt,[148] S. Reid,[59] C. Reissel,[34] D. H. Reitze,[11] A. I. Renzini,[45,138,139] B. Revenu,[122,42] A. Revilla-Peña,[134] F. Ricci,[39,38] M. Ricci,[38,39] A. Ricciardone,[88,87] J. Rice,[84] J. W. Richardson,[46] M. L. Richardson,[34] K. Riles,[158] H. K. Riley,[32] A. Riminucci,[65,66] F. Robinet,[42] M. Robinson,[3] A. Rocchi,[22] J. Rodriguez,[84] R. Rodriguez Lopez,[177] L. Rolland,[35] J. G. Rollins,[11] A. E. Romano,[313] R. Romano,[4,5] A. Romero-Rodríguez,[35] I. M. Romero-Shaw,[32] J. H. Romie,[68] S. Ronchini,[8,47,48] T. J. Roocke,[124] T. J. Rosauer,[46] C. A. Rose,[61] D. Rosińska,[136] M. P. Ross,[55] M. Rossello-Sastre,[83] B. I. Rotimi,[84] S. Rowan,[91] K. Rowlands,[187] S. K. Roy,[202,203] S. Roy,[15,250] T. RoyChowdhury,[86] D. Rozza,[138,139] P. Ruggi,[67] G. H. Ruiz,[165] E. Ruiz Morales,[314,217] K. Ruiz-Rocha,[54] V. Russ,[182] S. M. S,[230] S. Sachdev,[61] T. Sadecki,[3] F. Safai Tehrani,[38] P. Saffarieh,[37,128] S. Safi-Harb,[175] S. Saha,[282] T. Sainrat,[20] S. Sajith Menon,[315,39,38] K. Sakai,[316] Y. Sakai,[30] M. Sakellariadou,[71] S. Sakon,[8] F. Salces-Carcoba,[11] L. Salconi,[67] M. Saleem,[156] F. Salemi,[39,38] M. Sallé,[37] M. Salomé,[60] S. U. Salunkhe,[85] S. Salvador,[179,178] A. Salvarese,[156] A. Samajdar,[76,37] P. M. Samir,[186] A. Sanchez,[3] E. J. Sanchez,[11] J. Sanchez,[68] D. Sanchez-Cid,[45] N. Sanchis-Gual,[146] J. R. Sanders,[187] E. M. Sänger,[1] F. Santoliquido,[47,48] E. Sapkin,[41] F. Sarandrea,[64] T. R. Saravanan,[85] P. Sarkar,[9,10] A. Sasli,[18] P. Sassi,[82,81] B. Sassolas,[181] B. S. Sathyaprakash,[8,32] O. Sauter,[49] R. L. Savage,[3] T. Savicheva,[177] T. Sawada,[53] H. L. Sawant,[85] D. Schaetzl,[11] M. Scheel,[157] A. Schiebelbein,[201] M. G. Schiworski,[84] K. Schluterman,[50] P. Schmidt,[132] R. Schnabel,[27] M. Schneewind,[9,10] R. M. S. Schofield,[121,3] M. Schoor,[35] K. Schouteden,[99] B. W. Schulte,[9,10] M. Schulz,[47,48] B. F. Schutz,[32,9,10] E. Schwartz,[197] M. Scialpi,[317] J. Scott,[91] S. M. Scott,[69] R. M. Sedas,[68] T. C. Seetharamu,[91] M. Seglar-Arroyo,[166] Y. Sekiguchi,[318] D. Sellers,[68] N. Sembo,[2] E. G. Seo,[91] J. W. Seo,[99] G. Seong,[260] V. Sequino,[31,5] M. Serra,[38] C. K. Sethi,[145] A. Sevrin,[196] T. Shaffer,[3] U. S. Shah,[61] M. A. Shaikh,[319] L. Shao,[320] J. Sharkey,[91] A. K. Sharma,[83] Preeti Sharma,[12] Priyanka Sharma,[107] Sushant Sharma-Chaudhary,[18] P. Shawhan,[137] T. Shen,[69] Z.-H. Shi,[150] K. Shimode,[53] H. Shinkai,[321] S. Shirke,[85] D. H. Shoemaker,[34] D. M. Shoemaker,[156] R. W. Short,[3] S. ShyamSundar,[107] H. Siegel,[198] V. Sierra,[292] D. Sigg,[3] L. Silenzi,[36,37] P. J. S. Silva,[289] L. Silvestri,[39,176] M. Simmonds,[124] L. P. Singer,[322] A. Singh,[89] D. Singh,[33] M. K. Singh,[32] N. Singh,[83] S. Singh,[245,25] M. R. Sinha,[41] A. M. Sintes,[83] V. Skliris,[32] B. J. J. Slagmolen,[69] T. J. Slaven-Blair,[7] J. Smetana,[132] D. A. Smith,[68] J. R. Smith,[57] J. Smith,[32] L. Smith,[188,189] W. J. Smith,[54] S. Soares de Albuquerque Filho,[65,66] M. Soares-Santos,[45] K. Somiya,[245] I. Song,[282] S. Soni,[46] V. Sordini,[60] F. Sorrentino,[28] H. Sotani,[323] N. E. Sovitzky,[324] F. Spada,[87] V. Spagnuolo,[37] A. P. Spencer,[91] M. Spera,[189,325] P. Spinicelli,[67] A. K. Srivastava,[96] F. Stachurski,[91] V. V. Stanford,[108] A. Stanton,[32] D. A. Steer,[326] J. Steinhoff,[1] N. Steinle,[175] J. Steinlechner,[36,37] S. Steinlechner,[36,37] C. Stephens,[32] N. Stergioulas,[265] M. StPierre,[130] J. Stremiz,[57] M. D. Strong,[12] A. Strunk,[3] M. Suchenek,[98] S. Sudhagar,[98] R. Sugimoto,[43] L. Suleiman,[57] K. D. Sullivan,[12] J. Sun,[262,79] L. Sun,[69] S. Sunil,[96] J. Suresh,[119] P. J. Sutton,[32] K. Suzuki,[245] M. Suzuki,[221] A. Svizzeretto,[81] B. L. Swinkels,[37] A. Syx,[125] M. J. Szczepańczyk,[143] M. Tacca,[37] M. Tagliazucchi,[126,78] H. Tagoshi,[221] S. C. Tait,[11] H. Takaba,[155] K. Takada,[221] H. Takahashi,[30] R. Takahashi,[25] A. Takamori,[56] S. Takano,[9,10] H. Takeda,[327,328] I. Takimoto Schmiegelow,[47,48] C. Talbot,[118] M. Tamaki,[221] N. Tamanini,[104] D. Tanabe,[151] K. Tanaka,[245] S. J. Tanaka,[214] S. Tanioka,[32] D. B. Tanner,[49] W. Tanner,[9,10] L. Tao,[46,329] R. D. Tapia,[8] E. N. Tapia San Martín,[37] A. Taruya,[330] J. D. Tasson,[163] J. G. Tau,[199] A. Tejera,[172] J. G. Temple,[109] Y. Teng,[86] H. Themann,[216] A. Theodoropoulos,[146] M. P. Thirugnanasambandam,[85] L. M. Thomas,[11] M. Thomas,[68] P. Thomas,[3] J. E. Thompson,[220] S. R. Thondapu,[107] E. Thrane,[41] J. Tissino,[47,48] A. Tiwari,[85] Pawan Tiwari,[47] Praveer Tiwari,[331] S. Tiwari,[45] V. Tiwari,[132] M. R. Todd,[84] E. Tofani,[38] M. Toffano,[94] A. M. Toivonen,[18] K. Toland,[91] T. Tomaru,[25] V. Tommasini,[11] H. Tong,[41] C. I. Torrie,[11] I. Tosta e Melo,[332] E. Tournefier,[35] A. Trapananti,[190,82] R. Travaglini,[78] F. Travasso,[190,82] G. Traylor,[68] L. Traylor,[57] M. Trevor,[137] M. C. Tringali,[67] A. Tripathee,[158] G. Troian,[188,189] A. Trovato,[188,189] L. Trozzo,[5] R. J. Trudeau,[11] T. Tsang,[333] S. Tsuchida,[334] K. Tsuji,[52] L. Tsukada,[225] A. Tuci,[50] M. Turconi,[119] C. Turski,[97] H. Ubach,[134,135] A. S. Ubhi,[132] N. Uchikata,[221] T. Uchiyama,[53] R. P. Udall,[120] T. Uehara,[335] V. Undheim,[297] V. Upadhyaya,[145] L. E. Uronen,[229] T. Ushiba,[53] M. Vacatello,[87,88] H. Vahlbruch,[9,10] G. Vajente,[11] J. Valencia,[83] M. Valentini,[128,37] E. Vallejo-Pagès,[166] S. A. Vallejo-Peña,[313] S. Vallero,[64] M. van Dael,[37,336] E. Van den Bossche,[196] J. F. J. van den Brand,[36,128,37] C. Van Den Broeck,[76,37] M. van der Kolk,[128] M. van der Sluys,[76,37] A. Van de Walle,[42] J. van Dongen,[37] K. Vandra,[105] M. VanDyke,[127] H. van Haevermaet,[23] J. V. van Heijningen,[37] P. Van Hove,[70] J. Vanier,[274] J. Vanosky,[3] N. van Remortel,[23] M. Vardaro,[36,37] A. F. Vargas,[133] V. Varma,[145] A. Vecchio,[132] G. Vedovato,[95] J. Veitch,[91] P. J. Veitch,[124] S. Venikoudis,[15] P. Verdier,[60] M. Vereecken,[97] D. Verkindt,[35] B. Verma,[145] S. Verma,[116] Y. Verma,[107] S. M. Vermeulen,[11] F. Vetrano,[65] A. Veutro,[38,39] A. Viceré,[65,66] S. Vidyant,[84] A. D. Viets,[324] A. Vijaykumar,[201] A. Vilkha,[199] N. Villanueva Espinosa,[146] E. T. Vincent,[61] J.-Y. Vinet,[119] S. Viret,[60] S. Vitale,[34] A. Vives,[121] L. Vizmeg,[182] B. Vizzone,[61] H. Vocca,[81,82] D. Voigt,[27] E. R. G. von Reis,[3] J. S. A. von Wrangel,[9,10] W. E. Vossius,[247] L. Vujeva,[149] S. P. Vyatchanin,[114] J. Wack,[11] L. E. Wade,[109] M. Wade,[109] K. J. Wagner,[199] R. M. Wald,[226] L. Wallace,[11] R.-Z. Wan,[337] H. Wang,[245] P. Wang,[150] W. H. Wang,[173] Y. F. Wang,[1] Z. Wang,[153] R. L. Ward,[69] J. Warner,[3] M. Was,[35] T. Washimi,[25] N. Y. Washington,[11] D. Watarai,[44] B. Weaver,[3] S. A. Webster,[91] N. L. Weickhardt,[27] M. Weinert,[9,10] A. J. Weinstein,[11] R. Weiss,[34,†] L. Wen,[7] K. Wette,[69] C. Wheeler,[68] J. T. Whelan,[199] B. F. Whiting,[49] E. G. Wickens,[77] D. Wilken,[9,10] B. M. Williams,[127] D. Williams,[91] M. J. Williams,[77] N. S. Williams,[1] J. L. Willis,[11] B. Willke,[9,10] M. Wils,[99] L. Wimmer,[221] C. W. Winborn,[110] A. Wingfield,[123] J. Winterflood,[7] C. C. Wipf,[11] G. Woan,[91] N. E. Wolfe,[34]

H. T. Wong,[151] I. C. F. Wong,[99] T. Wouters,[76,37] J. L. Wright,[3] M. Wright,[76] B. Wu,[84] C. Wu,[150] D. S. Wu,[9,10] H. Wu,[150] K. Wu,[127] Z. Wu,[104] E. Wuchner,[57] D. M. Wysocki,[86] V. A. Xu,[33] Y. Xu,[83] N. Yadav,[64] H. Yamamoto,[11] K. Yamamoto,[161] T. S. Yamamoto,[44] T. Yamamoto,[53] R. Yamazaki,[214] T. Yan,[132] H. Yang,[290] K. Z. Yang,[18] Y. Yang,[162] Z. Yarbrough,[12] J. Yébana Carrilero,[83] A. B. Yelikar,[54] X. Yin,[34] J. Yokoyama,[338,44,43] T. Yokozawa,[53] M. Yoshihara,[52] S. Yuan,[7] H. Yuzurihara,[53] M. Zanatta,[79] M. Zanolin,[50] M. Zeeshan,[199] T. Zelenova,[67] J.-P. Zendri,[95] M. Zeoli,[15] M. Zerrad,[40] M. Zevin,[100] H. Zhang,[153] J. Zhang,[15] L. Zhang,[11] N. Zhang,[61] R. Zhang,[103] T. Zhang,[132] C. Zhao,[7] J. Zhao,[339] Yue Zhao,[340] Yuhang Zhao,[20] L.-M. Zheng,[32] Y. Zheng,[110] L. Zhizhong,[82] H. Zhong,[18] H. Zhou,[84] H. O. Zhu,[7] X.-J. Zhu,[339] Z.-H. Zhu,[339,337] Z. Zhu,[199] D. Z. Zieba,[91] A. B. Zimmerman,[156] L. Zimmermann,[60] and M. E. Zucker[34,11]

[1]*Max Planck Institute for Gravitational Physics (Albert Einstein Institute), D-14476 Potsdam, Germany*

[2]*Department of Physics, Graduate School of Science, Osaka Metropolitan University, 3-3-138 Sugimoto-cho, Sumiyoshi-ku, Osaka City, Osaka 558-8585, Japan*

[3]*LIGO Hanford Observatory, Richland, WA 99352, USA*

[4]*Dipartimento di Fisica "E.R. Caianiello", Università di Salerno, I-84084 Fisciano, Salerno, Italy*

[5]*INFN, Sezione di Napoli, I-80126 Napoli, Italy*

[6]*University of Warwick, Coventry CV4 7AL, United Kingdom*

[7]*OzGrav, University of Western Australia, Crawley, Western Australia 6009, Australia*

[8]*The Pennsylvania State University, University Park, PA 16802, USA*

[9]*Max Planck Institute for Gravitational Physics (Albert Einstein Institute), D-30167 Hannover, Germany*

[10]*Leibniz Universität Hannover, D-30167 Hannover, Germany*

[11]*LIGO Laboratory, California Institute of Technology, Pasadena, CA 91125, USA*

[12]*Louisiana State University, Baton Rouge, LA 70803, USA*

[13]*Tata Institute of Fundamental Research, Mumbai 400005, India*

[14]*Centre de Physique Théorique, Aix-Marseille Université, Campus de Luminy, 163 Av. de Luminy, 13009 Marseille, France*

[15]*Université catholique de Louvain, B-1348 Louvain-la-Neuve, Belgium*

[16]*Queen Mary University of London, London E1 4NS, United Kingdom*

[17]*University of California, Davis, Davis, CA 95616, USA*

[18]*University of Minnesota, Minneapolis, MN 55455, USA*

[19]*Instituto Nacional de Pesquisas Espaciais, 12227-010 São José dos Campos, São Paulo, Brazil*

[20]*Université Paris Cité, CNRS, Astroparticule et Cosmologie, F-75013 Paris, France*

[21]*Università di Roma Tor Vergata, I-00133 Roma, Italy*

[22]*INFN, Sezione di Roma Tor Vergata, I-00133 Roma, Italy*

[23]*Universiteit Antwerpen, 2000 Antwerpen, Belgium*

[24]*International Centre for Theoretical Sciences, Tata Institute of Fundamental Research, Bengaluru 560089, India*

[25]*Gravitational Wave Science Project, National Astronomical Observatory of Japan, 2-21-1 Osawa, Mitaka City, Tokyo 181-8588, Japan*

[26]*Advanced Technology Center, National Astronomical Observatory of Japan, 2-21-1 Osawa, Mitaka City, Tokyo 181-8588, Japan*

[27]*Universität Hamburg, D-22761 Hamburg, Germany*

[28]*INFN, Sezione di Genova, I-16146 Genova, Italy*

[29]*Dipartimento di Fisica, Università degli Studi di Genova, I-16146 Genova, Italy*

[30]*Research Center for Space Science, Advanced Research Laboratories, Tokyo City University, 3-3-1 Ushikubo-Nishi, Tsuzuki-Ku, Yokohama, Kanagawa 224-8551, Japan*

[31]*Università di Napoli "Federico II", I-80126 Napoli, Italy*

[32]*Cardiff University, Cardiff CF24 3AA, United Kingdom*

[33]*University of California, Berkeley, CA 94720, USA*

[34]*LIGO Laboratory, Massachusetts Institute of Technology, Cambridge, MA 02139, USA*

[35]*Univ. Savoie Mont Blanc, CNRS, Laboratoire d'Annecy de Physique des Particules - IN2P3, F-74000 Annecy, France*

[36]*Maastricht University, 6200 MD Maastricht, Netherlands*

[37]*Nikhef, 1098 XG Amsterdam, Netherlands*

[38]*INFN, Sezione di Roma, I-00185 Roma, Italy*

[39]*Università di Roma "La Sapienza", I-00185 Roma, Italy*

[40]*Aix Marseille Univ, CNRS, Centrale Med, Institut Fresnel, F-13013 Marseille, France*

[41]*OzGrav, School of Physics & Astronomy, Monash University, Clayton 3800, Victoria, Australia*

[42]*Université Paris-Saclay, CNRS/IN2P3, IJCLab, 91405 Orsay, France*

[43]*Department of Physics, The University of Tokyo, 7-3-1 Hongo, Bunkyo-ku, Tokyo 113-0033, Japan*

[44]*Research Center for the Early Universe (RESCEU), The University of Tokyo, 7-3-1 Hongo, Bunkyo-ku, Tokyo 113-0033, Japan*

[45]*University of Zurich, Winterthurerstrasse 190, 8057 Zurich, Switzerland*

[46]*University of California, Riverside, Riverside, CA 92521, USA*

[47]*Gran Sasso Science Institute (GSSI), I-67100 L'Aquila, Italy*

[48]*INFN, Laboratori Nazionali del Gran Sasso, I-67100 Assergi, Italy*

[49]*University of Florida, Gainesville, FL 32611, USA*
[50]*Embry-Riddle Aeronautical University, Prescott, AZ 86301, USA*
[51]*Tecnologico de Monterrey, Escuela de Ingeniería y Ciencias, 64849 Monterrey, Nuevo León, Mexico*
[52]*Nagoya University, Nagoya, 464-8601, Japan*
[53]*KAGRA Observatory, Institute for Cosmic Ray Research, The University of Tokyo, 238 Higashi-Mozumi, Kamioka-cho, Hida City, Gifu 506-1205, Japan*
[54]*Vanderbilt University, Nashville, TN 37235, USA*
[55]*University of Washington, Seattle, WA 98195, USA*
[56]*Earthquake Research Institute, The University of Tokyo, 1-1-1 Yayoi, Bunkyo-ku, Tokyo 113-0032, Japan*
[57]*California State University Fullerton, Fullerton, CA 92831, USA*
[58]*Department of Applied Physics, Graduate School of Engineering, The University of Tokyo, 7-3-1 Hongo, Bunkyo-ku, Tokyo 113-8656, Japan*
[59]*SUPA, University of Strathclyde, Glasgow G1 1XQ, United Kingdom*
[60]*Université Claude Bernard Lyon 1, CNRS, IP2I Lyon / IN2P3, UMR 5822, F-69622 Villeurbanne, France*
[61]*Georgia Institute of Technology, Atlanta, GA 30332, USA*
[62]*Royal Holloway, University of London, London TW20 0EX, United Kingdom*
[63]*Department of Astronomical Science, The Graduate University for Advanced Studies (SOKENDAI), 2-21-1 Osawa, Mitaka City, Tokyo 181-8588, Japan*
[64]*INFN Sezione di Torino, I-10125 Torino, Italy*
[65]*Università degli Studi di Urbino "Carlo Bo", I-61029 Urbino, Italy*
[66]*INFN, Sezione di Firenze, I-50019 Sesto Fiorentino, Firenze, Italy*
[67]*European Gravitational Observatory (EGO), I-56021 Cascina, Pisa, Italy*
[68]*LIGO Livingston Observatory, Livingston, LA 70754, USA*
[69]*OzGrav, Australian National University, Canberra, Australian Capital Territory 0200, Australia*
[70]*Université de Strasbourg, CNRS, IPHC UMR 7178, F-67000 Strasbourg, France*
[71]*King's College London, University of London, London WC2R 2LS, United Kingdom*
[72]*Korea Institute of Science and Technology Information, Daejeon 34141, Republic of Korea*
[73]*Département de Physique Nucléaire et Corpusculaire, Université de Genève, 24 quai E. Ansermet, CH-1211 Geneva, Switzerland*
[74]*Gravitational Wave Science Center, UniGe, -, Switzerland*
[75]*International College, The University of Osaka, 1-1 Machikaneyama-cho, Toyonaka City, Osaka 560-0043, Japan*
[76]*Institute for Gravitational and Subatomic Physics (GRASP), Utrecht University, 3584 CC Utrecht, Netherlands*
[77]*University of Portsmouth, Portsmouth, PO1 3FX, United Kingdom*
[78]*Istituto Nazionale Di Fisica Nucleare - Sezione di Bologna, viale Carlo Berti Pichat 6/2 - 40127 Bologna, Italy*
[79]*Università di Trento, Dipartimento di Fisica, I-38123 Povo, Trento, Italy*
[80]*INFN, Trento Institute for Fundamental Physics and Applications, I-38123 Povo, Trento, Italy*
[81]*Università di Perugia, I-06123 Perugia, Italy*
[82]*INFN, Sezione di Perugia, I-06123 Perugia, Italy*
[83]*IAC3–IEEC, Universitat de les Illes Balears, E-07122 Palma de Mallorca, Spain*
[84]*Syracuse University, Syracuse, NY 13244, USA*
[85]*Inter-University Centre for Astronomy and Astrophysics, Pune 411007, India*
[86]*University of Wisconsin-Milwaukee, Milwaukee, WI 53201, USA*
[87]*INFN, Sezione di Pisa, I-56127 Pisa, Italy*
[88]*Università di Pisa, I-56127 Pisa, Italy*
[89]*The University of Mississippi, University, MS 38677, USA*
[90]*Dipartimento di Medicina, Chirurgia e Odontoiatria "Scuola Medica Salernitana", Università di Salerno, I-84081 Baronissi, Salerno, Italy*
[91]*IGR, University of Glasgow, Glasgow G12 8QQ, United Kingdom*
[92]*HUN-REN Wigner Research Centre for Physics, H-1121 Budapest, Hungary*
[93]*Stanford University, Stanford, CA 94305, USA*
[94]*Università di Padova, Dipartimento di Fisica e Astronomia, I-35131 Padova, Italy*
[95]*INFN, Sezione di Padova, I-35131 Padova, Italy*
[96]*Institute for Plasma Research, Bhat, Gandhinagar 382428, India*
[97]*Universiteit Gent, B-9000 Gent, Belgium*
[98]*Nicolaus Copernicus Astronomical Center, Polish Academy of Sciences, 00-716, Warsaw, Poland*
[99]*Katholieke Universiteit Leuven, Oude Markt 13, 3000 Leuven, Belgium*
[100]*Northwestern University, Evanston, IL 60208, USA*
[101]*Centro de Investigaciones Energéticas Medioambientales y Tecnológicas, Avda. Complutense 40, 28040, Madrid, Spain*
[102]*Aix-Marseille Université, Université de Toulon, CNRS, CPT, Marseille, France*
[103]*Northeastern University, Boston, MA 02115, USA*
[104]*Laboratoire des 2 infinis - Toulouse, Université de Toulouse, CNRS/IN2P3, Toulouse, France, Toulouse, France*
[105]*Villanova University, Villanova, PA 19085, USA*

[106]*Federal University of Rio Grande do Norte, Campus Universitário - Lagoa Nova, Natal - RN, 59078-970, Brazil*
[107]*RRCAT, Indore, Madhya Pradesh 452013, India*
[108]*University of Maryland, Baltimore County, Baltimore, MD 21250, USA*
[109]*Kenyon College, Gambier, OH 43022, USA*
[110]*Missouri University of Science and Technology, Rolla, MO 65409, USA*
[111]*Indian Institute of Technology Madras, Chennai 600036, India*
[112]*Indian Institute of Technology Bombay, Powai, Mumbai 400 076, India*
[113]*Università di Siena, Dipartimento di Scienze Fisiche, della Terra e dell'Ambiente, I-53100 Siena, Italy*
[114]*Lomonosov Moscow State University, Moscow 119991, Russia*
[115]*Center for Theoretical Physics, Polish Academy of Sciences, 02-668, Warsaw, Poland*
[116]*Université libre de Bruxelles, 1050 Bruxelles, Belgium*
[117]*Bar-Ilan University, Ramat Gan, 5290002, Israel*
[118]*Princeton University, Princeton, NJ 08544 USA*
[119]*Université Côte d'Azur, Observatoire de la Côte d'Azur, CNRS, Artemis, F-06304 Nice, France*
[120]*University of British Columbia, Vancouver, BC V6T 1Z4, Canada*
[121]*University of Oregon, Eugene, OR 97403, USA*
[122]*Subatech, CNRS/IN2P3 - IMT Atlantique - Nantes Université, 4 rue Alfred Kastler BP 20722 44307 Nantes CÉDEX 03, France*
[123]*Christopher Newport University, Newport News, VA 23606, USA*
[124]*OzGrav, University of Adelaide, Adelaide, South Australia 5005, Australia*
[125]*Centre national de la recherche scientifique, 75016 Paris, France*
[126]*DIFA- Alma Mater Studiorum Università di Bologna, Via Zamboni, 33 - 40126 Bologna, Italy*
[127]*Washington State University, Pullman, WA 99164, USA*
[128]*Department of Physics and Astronomy, Vrije Universiteit Amsterdam, 1081 HV Amsterdam, Netherlands*
[129]*Cornell University, Ithaca, NY 14850, USA*
[130]*University of Rhode Island, Kingston, RI 02881, USA*
[131]*Laboratoire Kastler Brossel, Sorbonne Université, CNRS, ENS-Université PSL, Collège de France, F-75005 Paris, France*
[132]*University of Birmingham, Birmingham B15 2TT, United Kingdom*
[133]*OzGrav, University of Melbourne, Parkville, Victoria 3010, Australia*
[134]*Institut de Ciències del Cosmos (ICCUB), Universitat de Barcelona (UB), c. Martí i Franquès, 1, 08028 Barcelona, Spain*
[135]*Departament de Física Quàntica i Astrofísica (FQA), Universitat de Barcelona (UB), c. Martí i Franqués, 1, 08028 Barcelona, Spain*
[136]*Astronomical Observatory, University of Warsaw, 00-478 Warsaw, Poland*
[137]*University of Maryland, College Park, MD 20742, USA*
[138]*Università degli Studi di Milano-Bicocca, I-20126 Milano, Italy*
[139]*INFN, Sezione di Milano-Bicocca, I-20126 Milano, Italy*
[140]*IGFAE, Universidade de Santiago de Compostela, E-15782 Santiago de Compostela, Spain*
[141]*Williams College, Williamstown, MA 01267 USA*
[142]*University of Arizona, Tucson, AZ 85721, USA*
[143]*Faculty of Physics, University of Warsaw, Ludwika Pasteura 5, 02-093 Warszawa, Poland*
[144]*Istituto di Astrofisica e Planetologia Spaziali di Roma, 00133 Roma, Italy*
[145]*University of Massachusetts Dartmouth, North Dartmouth, MA 02747, USA*
[146]*Departamento de Astronomía y Astrofísica, Universitat de València, E-46100 Burjassot, València, Spain*
[147]*Observatori Astronòmic, Universitat de València, E-46980 Paterna, València, Spain*
[148]*OzGrav, Swinburne University of Technology, Hawthorn VIC 3122, Australia*
[149]*Niels Bohr Institute, University of Copenhagen, 2100 Kóbenhavn, Denmark*
[150]*Department of Physics, National Tsing Hua University, No. 101 Section 2, Kuang-Fu Road, Hsinchu 30013, Taiwan*
[151]*National Central University, Taoyuan City 320317, Taiwan*
[152]*OzGrav, Charles Sturt University, Wagga Wagga, New South Wales 2678, Australia*
[153]*University of Chinese Academy of Sciences / International Centre for Theoretical Physics Asia-Pacific, Beijing 100190, China*
[154]*Institute of Physics, National Yang Ming Chiao Tung University, 101 Univ. Street, Hsinchu, Taiwan*
[155]*Kamioka Branch, National Astronomical Observatory of Japan, 238 Higashi-Mozumi, Kamioka-cho, Hida City, Gifu 506-1205, Japan*
[156]*University of Texas, Austin, TX 78712, USA*
[157]*CaRT, California Institute of Technology, Pasadena, CA 91125, USA*
[158]*University of Michigan, Ann Arbor, MI 48109, USA*
[159]*Dipartimento di Ingegneria Industriale (DIIN), Università di Salerno, I-84084 Fisciano, Salerno, Italy*
[160]*INFN, Sezione di Napoli, Gruppo Collegato di Salerno, I-80126 Napoli, Italy*
[161]*Faculty of Science, University of Toyama, 3190 Gofuku, Toyama City, Toyama 930-8555, Japan*
[162]*School of Physical Science and Technology, ShanghaiTech University, 393 Middle Huaxia Road, Pudong, Shanghai, 201210, China*

[163] *Carleton College, Northfield, MN 55057, USA*
[164] *University of Szeged, Dóm tér 9, Szeged 6720, Hungary*
[165] *St. Thomas University, Miami Gardens, FL 33054, USA*
[166] *Institut de Física d'Altes Energies (IFAE), The Barcelona Institute of Science and Technology, Campus UAB, E-08193 Bellaterra (Barcelona), Spain*
[167] *INFN Cagliari, Physics Department, Università degli Studi di Cagliari, Cagliari 09042, Italy*
[168] *Université Libre de Bruxelles, Brussels 1050, Belgium*
[169] *INAF, Osservatorio Astronomico di Brera sede di Merate, I-23807 Merate, Lecco, Italy*
[170] *Departamento de Matemáticas, Universitat de València, E-46100 Burjassot, València, Spain*
[171] *Montana State University, Bozeman, MT 59717, USA*
[172] *Johns Hopkins University, Baltimore, MD 21218, USA*
[173] *The University of Texas Rio Grande Valley, Brownsville, TX 78520, USA*
[174] *Instituto de Ciências e Tecnologia - Universidade Federal de Alfenas, BR 267 - Rodovia José Aurélio Vilela, nº11.999, Km 533 37715-400 Cidade Universitária - Poços de Caldas - MG - Brasil, Brazil*
[175] *University of Manitoba, Winnipeg, MB R3T 2N2, Canada*
[176] *INFN-CNAF - Bologna, Viale Carlo Berti Pichat, 6/2, 40127 Bologna BO, Italy*
[177] *Colorado State University, Fort Collins, CO 80523, USA*
[178] *Université de Normandie, ENSICAEN, UNICAEN, CNRS/IN2P3, LPC Caen, F-14000 Caen, France*
[179] *Laboratoire de Physique Corpusculaire Caen, 6 boulevard du maréchal Juin, F-14050 Caen, France*
[180] *The University of Sheffield, Sheffield S10 2TN, United Kingdom*
[181] *Université Claude Bernard Lyon 1, CNRS, Laboratoire des Matériaux Avancés (LMA), IP2I Lyon / IN2P3, UMR 5822, F-69622 Villeurbanne, France*
[182] *Western Washington University, Bellingham, WA 98225, USA*
[183] *Università di Firenze, Sesto Fiorentino I-50019, Italy*
[184] *Università di Parma, I-43124 Parma, Italy*
[185] *INFN, Sezione di Milano Bicocca, Gruppo Collegato di Parma, I-43124 Parma, Italy*
[186] *Bard College, Annandale-On-Hudson, NY 12504, USA*
[187] *Marquette University, Milwaukee, WI 53233, USA*
[188] *Dipartimento di Fisica, Università di Trieste, I-34127 Trieste, Italy*
[189] *INFN, Sezione di Trieste, I-34127 Trieste, Italy*
[190] *Università di Camerino, I-62032 Camerino, Italy*
[191] *Scuola Normale Superiore, I-56126 Pisa, Italy*
[192] *Corps des Mines, Mines Paris, Université PSL, 60 Bd Saint-Michel, 75272 Paris, France*
[193] *Université Côte d'Azur, Observatoire de la Côte d'Azur, CNRS, Lagrange, F-06304 Nice, France*
[194] *National Center for Nuclear Research, 05-400 Świerk-Otwock, Poland*
[195] *Università degli Studi di Sassari, I-07100 Sassari, Italy*
[196] *Vrije Universiteit Brussel, 1050 Brussel, Belgium*
[197] *Trinity College, Hartford, CT 06106, USA*
[198] *Perimeter Institute, Waterloo, ON N2L 2Y5, Canada*
[199] *Rochester Institute of Technology, Rochester, NY 14623, USA*
[200] *Université de Liège, B-4000 Liège, Belgium*
[201] *Canadian Institute for Theoretical Astrophysics, University of Toronto, Toronto, ON M5S 3H8, Canada*
[202] *Stony Brook University, Stony Brook, NY 11794, USA*
[203] *Center for Computational Astrophysics, Flatiron Institute, New York, NY 10010, USA*
[204] *Montclair State University, Montclair, NJ 07043, USA*
[205] *HUN-REN Institute for Nuclear Research, H-4026 Debrecen, Hungary*
[206] *Centro de Física das Universidades do Minho e do Porto, Universidade do Minho, PT-4710-057 Braga, Portugal*
[207] *Aix Marseille Univ, CNRS/IN2P3, CPPM, Marseille, France*
[208] *Dipartimento di Ingegneria, Università della Basilicata, I-85100 Potenza, Italy*
[209] *SUPA, University of the West of Scotland, Paisley PA1 2BE, United Kingdom*
[210] *Barry University, Miami Shores, FL 33168, USA*
[211] *Department of Astronomy, University of Geneva, Chemin Pegasi 51, 1290 Versoix, Switzerland*
[212] *Centro de Astrofísica e Gravitação, Departamento de Física, Instituto Superior Técnico - IST, Universidade de Lisboa - UL, Av. Rovisco Pais 1, 1049-001 Lisboa, Portugal*
[213] *Eötvös University, Budapest 1117, Hungary*
[214] *Department of Physical Sciences, Aoyama Gakuin University, 5-10-1 Fuchinobe, Sagamihara City, Kanagawa 252-5258, Japan*
[215] *University of Sannio at Benevento, I-82100 Benevento, Italy and INFN, Sezione di Napoli, I-80100 Napoli, Italy*
[216] *California State University, Los Angeles, Los Angeles, CA 90032, USA*
[217] *Instituto de Fisica Teorica UAM-CSIC, Universidad Autonoma de Madrid, 28049 Madrid, Spain*

[218]*Istituto Nazionale di Astrofisica - Osservatorio di Roma, Viale del Parco Mellini 84 - 00136 Roma, Italy*
[219]*Laboratoire d'Acoustique de l'Université du Mans, UMR CNRS 6613, F-72085 Le Mans, France*
[220]*University of Southampton, Southampton SO17 1BJ, United Kingdom*
[221]*KAGRA Observatory, Institute for Cosmic Ray Research, The University of Tokyo, 5-1-5 Kashiwa-no-Ha, Kashiwa City, Chiba 277-8582, Japan*
[222]*Faculty of Physics, University of Białystok, 15-245 Białystok, Poland*
[223]*University of Nottingham NG7 2RD, UK*
[224]*Dipartimento di Ingegneria Industriale, Elettronica e Meccanica, Università degli Studi Roma Tre, I-00146 Roma, Italy*
[225]*University of Nevada, Las Vegas, Las Vegas, NV 89154, USA*
[226]*University of Chicago, Chicago, IL 60637, USA*
[227]*Institute of Physics, Academia Sinica, 128 Sec. 2, Academia Rd., Nankang, Taipei 11529, Taiwan*
[228]*Shanghai Astronomical Observatory, Chinese Academy of Sciences, 80 Nandan Road, Shanghai 200030, China*
[229]*The Chinese University of Hong Kong, Shatin, NT, Hong Kong*
[230]*Nirula Institute of Technology, Kolkata, West Bengal 700109, India*
[231]*American University, Washington, DC 20016, USA*
[232]*Aix Marseille Université, Jardin du Pharo, 58 Boulevard Charles Livon, 13007 Marseille, France*
[233]*Dipartimento di Fisica, Università degli studi di Milano, Via Celoria 16, I-20133, Milano, Italy*
[234]*INFN, sezione di Milano, Via Celoria 16, I-20133, Milano, Italy*
[235]*Department of Applied Physics, Fukuoka University, 8-19-1 Nanakuma, Jonan, Fukuoka City, Fukuoka 814-0180, Japan*
[236]*University of Cambridge, Cambridge CB2 1TN, United Kingdom*
[237]*University of Lancaster, Lancaster LA1 4YW, United Kingdom*
[238]*College of Industrial Technology, Nihon University, 1-2-1 Izumi, Narashino City, Chiba 275-8575, Japan*
[239]*Saha Institute of Nuclear Physics, Bidhannagar, West Bengal 700064, India*
[240]*Institute of Photonics Technologies, National Tsing Hua University, No. 101 Section 2, Kuang-Fu Road, Hsinchu 30013, Taiwan*
[241]*Department of Physics, Tamkang University, No. 151, Yingzhuan Rd., Danshui Dist., New Taipei City 25137, Taiwan*
[242]*Department of Electrophysics, National Yang Ming Chiao Tung University, 101 Univ. Street, Hsinchu, Taiwan*
[243]*Rutherford Appleton Laboratory, Didcot OX11 0DE, United Kingdom*
[244]*Department of Astronomy and Space Science, Chungnam National University, 9 Daehak-ro, Yuseong-gu, Daejeon 34134, Republic of Korea*
[245]*Graduate School of Science, Institute of Science Tokyo, 2-12-1 Ookayama, Meguro-ku, Tokyo 152-8551, Japan*
[246]*Columbia University, New York, NY 10027, USA*
[247]*Helmut Schmidt University, D-22043 Hamburg, Germany*
[248]*Nambu Yoichiro Institute of Theoretical and Experimental Physics (NITEP), Osaka Metropolitan University, 3-3-138 Sugimoto-cho, Sumiyoshi-ku, Osaka City, Osaka 558-8585, Japan*
[249]*Directorate of Construction, Services & Estate Management, Mumbai 400094, India*
[250]*Royal Observatory of Belgium, Avenue Circulaire, 3, 1180 Uccle, Belgium*
[251]*Observatoire Astronomique de Strasbourg, Université de Strasbourg, CNRS, 11 rue de l'Université, 67000 Strasbourg, France*
[252]*National Astronomical Observatories, Chinese Academy of Sciences, 20A Datun Road, Chaoyang District, Beijing, China*
[253]*School of Astronomy and Space Science, University of Chinese Academy of Sciences, 20A Datun Road, Chaoyang District, Beijing, China*
[254]*Sungkyunkwan University, Seoul 03063, Republic of Korea*
[255]*Department of Physics, Ulsan National Institute of Science and Technology (UNIST), 50 UNIST-gil, Ulju-gun, Ulsan 44919, Republic of Korea*
[256]*Chung-Ang University, Seoul 06974, Republic of Korea*
[257]*Vrije Universiteit Amsterdam, 1081 HV, Amsterdam, Netherlands*
[258]*University of Washington Bothell, Bothell, WA 98011, USA*
[259]*Université Paris-Saclay, Université Paris Cité, CEA, CNRS, AIM, 91191, Gif-sur-Yvette, France*
[260]*Ewha Womans University, Seoul 03760, Republic of Korea*
[261]*Department of Astronomy, Yonsei University, 50 Yonsei-Ro, Seodaemun-Gu, Seoul 03722, Republic of Korea*
[262]*National Institute for Mathematical Sciences, Daejeon 34047, Republic of Korea*
[263]*Korea Astronomy and Space Science Institute, Daejeon 34055, Republic of Korea*
[264]*Division of Science, National Astronomical Observatory of Japan, 2-21-1 Osawa, Mitaka City, Tokyo 181-8588, Japan*
[265]*Department of Physics, Aristotle University of Thessaloniki, 54124 Thessaloniki, Greece*
[266]*National Tsing Hua University, Hsinchu City 30013, Taiwan*
[267]*Technical University of Braunschweig, D-38106 Braunschweig, Germany*
[268]*Institute of Mathematics, Polish Academy of Sciences, 00656 Warsaw, Poland*
[269]*Astronomical Observatory, Jagiellonian University, 31-007 Cracow, Poland*
[270]*Department of Physics and Helsinki Institute of Physics, University of Helsinki, Gustaf Hallstromin katu 2,, FI-00014, Finland*
[271]*Instituto de Fisica Teorica UAM-CSIC, Universidad Autonoma de Madrid, 28049 Madrid, Spain*
[272]*Department of Physics, Nagoya University, ES building, Furocho, Chikusa-ku, Nagoya, Aichi 464-8602, Japan*
[273]*Centre de Calcul IN2P3, 21 avenue Pierre de Coubertin, Campus de la Doua, 69100 Villeurbanne, France*

[274] *Université de Montréal/Polytechnique, Montreal, Quebec H3T 1J4, Canada*
[275] *Università degli Studi di Cagliari, Via Università 40, 09124 Cagliari, Italy*
[276] *Department of Computer Simulation, Inje University, 197 Inje-ro, Gimhae, Gyeongsangnam-do 50834, Republic of Korea*
[277] *Korea Astronomy and Space Science Institute (KASI), 776 Daedeokdae-ro, Yuseong-gu, Daejeon 34055, Republic of Korea*
[278] *NAVIER, École des Ponts, Univ Gustave Eiffel, CNRS, Marne-la-Vallée, France*
[279] *Gravitational Wave Science Project, National Astronomical Observatory of Japan (NAOJ), Mitaka City, Tokyo 181-8588, Japan*
[280] *Department of Physics, National Cheng Kung University, No.1, University Road, Tainan City 701, Taiwan*
[281] *National Center for High-performance Computing, National Institutes of Applied Research, No. 7, R&D 6th Rd., Hsinchu Science Park, Hsinchu City 30076, Taiwan*
[282] *Institute of Astronomy, National Tsing Hua University, No. 101 Section 2, Kuang-Fu Road, Hsinchu 30013, Taiwan*
[283] *Institució Catalana de Recerca i Estudis Avançats, E-08010 Barcelona, Spain*
[284] *Institut de Física d'Altes Energies, E-08193 Barcelona, Spain*
[285] *Institute of Astronomy, National Tsing Hua University, No. 101 Section 2, Kuang-Fu Road, Hsinchu 30013, Taiwan*
[286] *Institut fuer Theoretische Astrophysik, Zentrum fuer Astronomie Heidelberg, Universitaet Heidelberg, Albert Ueberle Str. 2, 69120 Heidelberg, Germany*
[287] *Institucio Catalana de Recerca i Estudis Avançats (ICREA), Passeig de Lluís Companys, 23, 08010 Barcelona, Spain*
[288] *Université de Lyon, Université Claude Bernard Lyon 1, CNRS, Institut Lumière Matière, F-69622 Villeurbanne, France*
[289] *Universidade Estadual Paulista, R. Dr. José Barbosa de Barros, 1780 - Jardim Paraiso, Botucatu - SP, 18610-307, Brazil*
[290] *Tsinghua University, Beijing 100084, China*
[291] *Departament de Física, Universitat de les Illes Balears, IAC3 –IEEC, Crta. Valldemossa km 7.5, E-07122 Palma, Spain*
[292] *Universidad de Guadalajara, 44430 Guadalajara, Jalisco, Mexico*
[293] *Institut des Hautes Etudes Scientifiques, F-91440 Bures-sur-Yvette, France*
[294] *Faculty of Law, Ryukoku University, 67 Fukakusa Tsukamoto-cho, Fushimi-ku, Kyoto City, Kyoto 612-8577, Japan*
[295] *Indian Institute of Science Education and Research, Kolkata, Mohanpur, West Bengal 741252, India*
[296] *Phenikaa University, Nguyen Trac Street, Duong Noi, Hanoi, Vietnam*
[297] *University of Stavanger, 4021 Stavanger, Norway*
[298] *Department of Astronomy, The University of Tokyo, 7-3-1 Hongo, Bunkyo-ku, Tokyo 113-0033, Japan*
[299] *Physics Program, Graduate School of Advanced Science and Engineering, Hiroshima University, 1-3-1 Kagamiyama, Higashihiroshima City, Hiroshima 739-8526, Japan*
[300] *GRAPPA, Anton Pannekoek Institute for Astronomy and Institute for High-Energy Physics, University of Amsterdam, 1098 XH Amsterdam, Netherlands*
[301] *University College London, London WC1E 6BT, United Kingdom*
[302] *Observatoire de Paris, 75014 Paris, France*
[303] *Instituto Tecnológico de Aeronáutica, Praça Marechal Eduardo Gomes, 50 - Vila das Acacias, São José dos Campos - SP, 12228-900, Brazil*
[304] *Graduate School of Science and Technology, Niigata University, 8050 Ikarashi-2-no-cho, Nishi-ku, Niigata City, Niigata 950-2181, Japan*
[305] *Niigata Study Center, The Open University of Japan, 754 Ichibancho, Asahimachi-dori, Chuo-ku, Niigata City, Niigata 951-8122, Japan*
[306] *Consiglio Nazionale delle Ricerche - Istituto dei Sistemi Complessi, I-00185 Roma, Italy*
[307] *Hobart and William Smith Colleges, Geneva, NY 14456, USA*
[308] *Dipartimento di Ingegneria, Università del Sannio, I-82100 Benevento, Italy*
[309] *Institut d'Astrophysique de Paris, Sorbonne Université, CNRS, UMR 7095, 75014 Paris, France*
[310] *Museo Storico della Fisica e Centro Studi e Ricerche "Enrico Fermi", I-00184 Roma, Italy*
[311] *Kennesaw State University, Kennesaw, GA 30144, USA*
[312] *Government Victoria College, Palakkad, Kerala 678001, India*
[313] *Universidad de Antioquia, Medellín, Colombia*
[314] *Departamento de Física - ETSIDI, Universidad Politécnica de Madrid, 28012 Madrid, Spain*
[315] *Ariel University, Ramat HaGolan St 65, Ari'el, Israel*
[316] *Department of Electronic Control Engineering, National Institute of Technology, Nagaoka College, 888 Nishikatakai, Nagaoka City, Niigata 940-8532, Japan*
[317] *Dipartimento di Fisica e Scienze della Terra, Università Degli Studi di Ferrara, Via Saragat, 1, 44121 Ferrara FE, Italy*
[318] *Faculty of Science, Toho University, 2-2-1 Miyama, Funabashi City, Chiba 274-8510, Japan*
[319] *Seoul National University, Seoul 08826, Republic of Korea*
[320] *Kavli Institute for Astronomy and Astrophysics, Peking University, Yiheyuan Road 5, Haidian District, Beijing 100871, China*
[321] *Faculty of Information Science and Technology, Osaka Institute of Technology, 1-79-1 Kitayama, Hirakata City, Osaka 573-0196, Japan*
[322] *NASA Goddard Space Flight Center, Greenbelt, MD 20771, USA*
[323] *Faculty of Science and Technology, Kochi University, 2-5-1 Akebono-cho, Kochi-shi, Kochi 780-8520, Japan*
[324] *Concordia University Wisconsin, Mequon, WI 53097, USA*
[325] *Scuola Internazionale Superiore di Studi Avanzati, Via Bonomea, 265, I-34136, Trieste TS, Italy*
[326] *Laboratoire de Physique de l'ENS, Université Paris Cité, Ecole Normale Supérieure, Université PSL, Sorbonne Université, CNRS, 75005 Paris, France*
[327] *The Hakubi Center for Advanced Research, Kyoto University, Yoshida-honmachi, Sakyou-ku, Kyoto City, Kyoto 606-8501, Japan*
[328] *Department of Physics, Kyoto University, Kita-Shirakawa Oiwake-cho, Sakyou-ku, Kyoto City, Kyoto 606-8502, Japan*

[329]

[330] *Yukawa Institute for Theoretical Physics (YITP), Kyoto University, Kita-Shirakawa Oiwake-cho, Sakyou-ku, Kyoto City, Kyoto 606-8502, Japan*

[331] *Chennai Mathematical Institute, Chennai 603103, India*

[332] *University of Catania, Department of Physics and Astronomy, Via S. Sofia, 64, 95123 Catania CT, Italy*

[333] *Southeastern Louisiana University, Hammond, LA 70402, USA*

[334] *National Institute of Technology, Fukui College, Geshi-cho, Sabae-shi, Fukui 916-8507, Japan*

[335] *Department of Communications Engineering, National Defense Academy of Japan, 1-10-20 Hashirimizu, Yokosuka City, Kanagawa 239-8686, Japan*

[336] *Eindhoven University of Technology, 5600 MB Eindhoven, Netherlands*

[337] *School of Physics and Technology, Wuhan University, Bayi Road 299, Wuchang District, Wuhan, Hubei, 430072, China*

[338] *Kavli Institute for the Physics and Mathematics of the Universe (Kavli IPMU), WPI, The University of Tokyo, 5-1-5 Kashiwa-no-Ha, Kashiwa City, Chiba 277-8583, Japan*

[339] *Department of Astronomy, Beijing Normal University, Xinjiekouwai Street 19, Haidian District, Beijing 100875, China*

[340] *Hong Kong University of Science and Technology, Clear Water Bay, HK, Hong Kong*